

PREVENTIVE PEDIATRIC HEALTH CARE 2018

NINTH EDITION



OFFICIAL PUBLICATION OF THE
PHILIPPINE PEDIATRIC SOCIETY

PREVENTIVE PEDIATRIC HEALTH CARE HANDBOOK

2018



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MESSAGE

A clearer understanding of the evolution of health and disease in adulthood has led to the recognition of the critical importance of the individual's health status in the early years. Longitudinal studies of pulmonary function have shown that deficits in infancy are carried on to later years and adulthood, many times only made worse by illness and environmental insults. The First 1000 Days program is founded on the concept that insults to the health of the fetus and up to age two years have long term consequences. Assuring the optimal growth and development of the infant, the child and the adolescent has become an urgent responsibility.

In response to this call, the Committee on Preventive Health Care of the Philippine Pediatric Society has updated the PPS Preventive Health Care Handbook with its 8th edition. As a resource, the handbook is replete with information and guidance on newborn screening, immunization, developmental surveillance and screening, blood pressure measurement, child maltreatment and adolescent health care, among others.

Franklin D. Roosevelt once stated, "We cannot always build the future for our youth, but we can build our youth for the future". This short but meaningful statement is embodied by this Handbook.

We would like to express our gratitude to the Committee led by the chair, Dr. Maria Rosario S. Cruz, the vice-chairs, Drs. Naomi S. Nocheseda, Christian T. Caligagan and Mary Antonette C. Madrid and their advisers, Drs. Miguel L. Noche, Jr. and Loida T. Villanueva. The output of their dedication and hard work will serve as an important resource to pediatricians.



ALEXANDER O. TUAZON, MD, MDE
President
Philippine Pediatric Society

MESSAGE

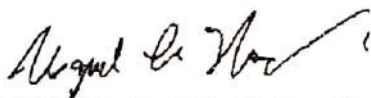
*T*he advancement of child health care is a never-ending process. This is facilitated by our colleagues in research and technology. Their research findings have been excellently transformed to the clinician's day to day care of patients from primary prevention to secondary and tertiary preventive strategies.

The Committee on Preventive Pediatric Health Care has meticulously monitored and assessed these advances in primary health care of common conditions that create a burden on the health status of Filipino children. This is evident from the first edition of the handbook to this present 8th edition. The Committee's recommendations started with simple beginnings utilizing a number of foreign data in its 1st edition in 2000 to the present edition of mostly local-based studies. Moreover, in the choice of what to recommend for use in clinical practice for each subject area, the Committee exercised extreme caution to see to it that each strategy is effective and will not result to adverse outcome to the Filipino child. Hence, optimum child health and safety are the highest priorities strongly considered in the choice of the recommendations.

Most clinicians including seasoned pediatricians have now gained more and more confidence in incorporating the recommendations of this handbook in their clinical practice. It can be stated that, through the Committee, PPS has contributed significantly to the health care of Filipino children by providing an effective, safe and inexpensive reference in promoting optimum health and safety for our children. I am deeply honored to be part of this innovative project.

Congratulations once again to the PPS Committee on Pediatric Health Care led by Dr. Maria Rosario S. Cruz and its indefatigable members for this commendable effort. Your group has continuously provided us clinicians updated and safe strategies for primary prevention of common illnesses through this handbook.

More power to you!!!



MIGUEL L. NOCHE, JR., MD, FPPS, FPSAAI

Adviser

Committee of Pediatric Preventive Health Care

FOREWORD

The concept of health has dramatically changed through the years. From the control of disease to its prevention, this transformational paradigm of wellness has redefined global strides to improve health across the world's population.

As the leading advocate of childhood health and well-being, the Philippine Pediatric Society Inc. has consistently engineered its programs to ensure that these are aligned with the global tempo of contemporary health directives and more importantly attuned to the prevailing needs of Filipino children. These consisted of continuing professional education done face-to-face and virtually, clinical and community-based researches as well as the dissemination of timely and relevant guidelines that help provide evidence in day-to-day decision-making.

One such time-tested publication is the "Preventive Pediatric Health Care Handbook" which has through the years, provided relevant and pragmatic recommendations towards optimum pediatric healthcare despite resource-limited and cost-constrained settings.

The idea of a preventive healthcare handbook was spearheaded by Dr. Miguel L. Noche, Jr. in 2002, who as Chairperson of the then Committee on Preventive Health envisioned the creation of a pamphlet to serve as a guide for pediatricians in delivering holistic ambulatory care to their patients. His committee later on expanded the coverage of pediatric wellness visits to include immunization schedules, anticipatory guidance, health screening and risk identification in order to promote a well-rounded approach to child health. This idea has remained an indelible imprint in the succeeding editions of the pamphlet which later became a set of recommendations now widely used by medical learners, residents-in-training, and practicing clinicians as a handy reference material in their daily patient encounter. It is in loving memory that this latest offering is being published in his honor.

In this 2018 edition, the handbook contains updates on blood pressure measurement, newborn screening, adolescent health care, child maltreatment, developmental surveillance and screening, and the latest on childhood immunization.

It is our hope that this document be of use to all interested stakeholders in childhood health – medical, paramedical and allied health professionals who have the best interest of the Filipino child in mind.

Allow me to thank all the dedicated and committed pediatricians who have served as chair, member or adviser of this Committee for fostering genuine awareness, championing productive collaboration and promoting contributive discussions on health issues which are universal for child welfare and fundamental for child healthcare givers.

Rest assured, we shall continue to work towards this template of progressive action hinged on principles that espouse "prevention over cure, screening more than treating and surveillance beyond compliance" as we build a future ready and able to nurture the next generation of Filipinos.



Maria Rosario S. Cruz, MD, FPPS
Chair, Committee on Preventive Pediatric Health Care Handbook

LIST OF ABBREVIATIONS

5TU PPD	- 5 Tuberculin Units Purified Protein Derivative
AAP	- American Academy of Pediatrics
AAPD	- American Academy of Pediatric Dentistry
AMA	- American Medical Association
BCPC	- (Local) Barangay Council for the Protection of Children
BMI	- Body Mass Index
CIU	- Crisis Intervention Unit
CP	- Cerebral Palsy
DOH	- Department of Health
DBP	- Diastolic Blood Pressure
DSWD	- Department of Social Welfare and Development
EINC	- Essential Intrapartum and Newborn Care
GAPS	- Guidelines for Adolescent Preventive Services
HEEADSSS	- Home, Education/Employment, Eating, Activities, Drugs, Sexuality, Suicidality, Safety/Spirituality
LVH	- Left Ventricular Hypertrophy
MR	- Mental Retardation
PASOO	- Philippine Association for the Study of Overweight and Obesity
PEP	- Pre-Exposure Prophylaxis
PIDSP	- Pediatric Infectious Disease Society of the Philippines
PFV	- Philippine Foundation for Vaccination
PPS	- Philippine Pediatric Society
ROP	- Retinopathy of Prematurity
ROR	- Red Orange Reflex
SBP	- Systolic Blood Pressure
TIPP	- The Injury Prevention Program
UPCHK	- University of the Philippines - College of Human Kinetics
UP PGH	- University of the Philippines - Philippine General Hospital
US CDC-NCHS	- United States Center for Disease Control and Prevention National Center for Health Statistics
USPSTF	- United States Preventive Service Task Force
WCPDs	- Philippine National Police Women and Child Protection Desks
WHO	- World Health Organization

DISCLAIMER

"The recommendations contained in this document are intended to GUIDE practitioners in the conduct of anticipatory care/guidance and periodic health examinations of infants, children and adolescents. In no way should the recommendations be regarded as absolute rules, since nuances and peculiarities in individual cases or particular communities may entail differences in the specific approach. In the end, the recommendations should supplement and not replace sound clinical judgment made on a case to case basis."

ANNOTATIONS

1. Prenatal education may be done through a structured mothers' class or face to face counseling with a health care professional or worker. Education and counseling must include the following areas of concern (Appendix 1):

- Breastfeeding (Appendix 7)
- Newborn Care and Procedures at Birth
- Anticipatory Guidance to decrease the risk of injury and identify risk factors for child maltreatment
- Prevention of smoking, alcohol intake and exposure to teratogens
- Tetanus Toxoid Immunization for the mother
- Maternal nutrition (to include folic acid supplementation)

2. Every infant must be totally appraised at birth and monitored daily until discharge.

Colostrum is the perfect first food for the newborn. Latching-on and breastfeeding must be initiated during the first 30 minutes to one hour after delivery of the infant (1, 9, 11, 12)

3. The optimal time of discharge of a healthy term newborn is decided by the physicians caring for both mother and child. For newborns discharged <48 hours after delivery, a definitive appointment must be made for the infant to be examined within 48 hours of discharge (61, 63). (Appendix 2)

4. Developmental, psychosocial, and chronic disease issues for children and adolescents may require frequent counseling and treatment visits separate from routine preventive care visits.

5. If a child comes under care for the first time at any point on the schedule, or if any items are not accomplished at the suggested age, the preventive care services should be brought up to date at the earliest possible time.

6. Risk assessment and screening using the HEEADSSS format is part of a complete history-taking of adolescent patients (184). (Appendix 3)

7. Every well child visit must be an opportunity for the health care professional to evaluate the overall development of a child. History taking, observing the child and doing a thorough physical examination remain to be the most powerful instruments available to the pediatrician in identifying concerns that may need monitoring or referral (Appendix 4).

The WHO Child Growth Standards include 'Windows of Achievement' which describe the range and time lines for six key motor development milestones (Figure 1). These motor development milestones must be interpreted in the light of other neurodevelopmental findings in a child.

"The Philippine Ambulatory Pediatrics Association, Inc. strongly recommends that pediatricians advise parents about the importance of reading aloud to their children during the first years of life. Research shows that this helps them develop language and literacy skills, thus making children ready to learn and read in school. The Developmental Milestones of Early Literacy from Reach Out and Read, Inc. in Boston, Massachusetts, describes the motor and cognitive skills of children from 6 months to 5 years of age, and inform parents what

they can do to support their children's emergent literacy behaviors (Figure 2)."

8. Red Flag signs for Atopy from the Philippine Society of Allergy, Asthma, and Immunology: Any child with a family history of atopy (asthma, atopic dermatitis, allergic rhinitis, drug / food allergy) who presents with recurrent/persistent symptoms of 1 or more of the following should be closely monitored, investigated or referred to the subspecialist(s) when warranted:

- Respiratory symptoms: chronic cough with or without wheezing, shortness of breath, chest tightness, trouble sleeping due to coughing, fatigue, problems with feeding or grunting during infancy
- Nasal symptoms: frequent sneezing, rhinorrhea, itchiness, nasal congestion
- Ocular symptoms: bluish, brownish discoloration around both eyes, puffiness under the eyes, redness and tearing, itchiness
- Skin symptoms: dryness and itchiness
- Gastrointestinal symptoms: itchiness of the roof of the mouth and throat, colic, vomiting, stomach cramps, diarrhea and bloody stools

9. Approach to a thorough physical examination and interpretation of findings must be age - appropriate. Respect for an older child's privacy and minimizing the child's discomfort are basic in pediatric physical examination. Additional procedures to be performed for adolescent patients are mentioned in Appendix 3.

10. The WHO Child Growth Standards are used as reference standard for weight, height and head circumference. Interpretation of growth points are based on Z-scores (standard deviation scores) and not on percentile scores.

Figures 3-27 show the following: Z-score interpretation, Head Circumference for Age for Girls, Weight for Age for Girls (Birth to 2 years), Length for Age for Girls (Birth to 2 years), Weight for Length for Girls (Birth to 2 years), BMI for Age for Girls (Birth to 2 years), Weight for Age for Girls (2 to 5 years), Height for Age for Girls (2 to 5 years), Weight for Height for Girls (2 to 5 years), BMI for Age for Girls (2 to 5 years), Weight for Age for Girls (5 to 10 years), Height for Age for Girls (5 to 19 years), BMI for Age for Girls (5 to 19 years), Head Circumference for Age for Boys, Weight for Age for Boys (Birth to 2 years), Length for Age for Boys (Birth to 2 years), Weight for Length for Boys (Birth to 2 years), BMI for Age for Boys (Birth to 2 years), Weight for Age for Boys (2 to 5 years), Height for Age for Boys (2 to 5 years), Weight for Height for Boys (2 to 5 years), BMI for Age for Boys (2 to 5 years), Weight for Age for Boys (5 to 10 years), Height for Age for Boys (5 to 19 years), BMI for Age for Boys (5 to 19 years).

The following excerpts were lifted from the WHO Child Growth Standards recommendations (67):

"If a child is less than 2 years old, measure the recumbent length. If a child is age 2 years or more and able to stand, measure the standing height. In general, standing height is ~ 0.7 cm less than recumbent length. If a child less than 2 years old will not lie down

for measurement of length, measure standing height and add 0.7 cm to convert it to length. If a child 2 years old or more cannot stand, measure recumbent length and subtract 0.7 cm to convert it to height.

Weight-for-length/height is a reliable growth indicator even when the age is not known.

Body Mass Index (BMI) measurement standards enable early detection and prevention of overweight and obesity problems."

The WHO cautions the health care workers about edema associated with kwashiorkor which can hide the fact that a child has very low weight. When plotting the weight of the child with edema it is important to note on the growth chart that the child has edema.

To be consistent with the use of the blood pressure percentile tables (Annotation 11), the US CDC-NCHS grow charts are also included (Figures 28-31).

11. The Pediatric Nephrology Society of the Philippines recommends routine blood pressure (BP) measurement annually for all children > 3 years of age and adolescents. BP however should also be checked in every encounter on all ill patients and all patients at risk such as those with obesity, are taking medications known to increase BP, have renal disease, a history of aortic arch obstruction or coarctation, or diabetes.

An updated Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents developed by the American Academy of Pediatrics and was endorsed by the American

Heart Association was recently published. New normative BP tables based on normal-weight children have been recommended as to age, gender and height percentile (73). Figures 30 and 31 are tables that show the blood pressure levels for boys and girls respectively. Until we have our own population-based blood pressure levels of Filipino children, these tables may be used to interpret the blood pressure levels of our patients.

How to use the BP Tables:

- a. Determine the height percentile of the patient using the US CDC-NCHS growth charts (Figure 28 and 29)
- b. Measure and record the patient's SBP and DBP.
- c. Use the correct gender table for SBP and DBP (Figure 30 and 31)
- d. Find the child's age on the column at the left side of the table. Follow the age row horizontally across the table to intersect with the vertical column of the child's height percentile.
- e. Find the SBP on the left columns of the table and the DBP on the right columns.
- f. Find the corresponding BP percentile on the vertical column to the right of the age column.

A new feature in these tables is that the BP values are presented according to the updated definitions of BP categories and Stages. A normotensive child aged 1-13 years old has a BP reading less than the 90th percentile for age, gender and height percentile whereas for children age >13 years, normal BP <120/<80 mm Hg. For children with normal BP, encourage healthy diet, sleep and physical work.

Elevated BP in children is defined as >90th percentile to <95th percentile or 120/80 mm Hg to <95th percentile (whichever is lower) for ages 1 to 13 years old and BP 120/<80 to 129/<80 mm Hg for > 13 years old. Counseling on physical activity and diet management, if obese, must be done. Medical investigations for the presence of factors that might need pharmacologic therapy (chronic kidney disease, diabetes mellitus, heart failure) must likewise be done. If in doubt, a referral to the subspecialist will be prudent.

Stage 1 Hypertension for aged 1-13 years is defined as BP > 95th percentile to <95th percentile +12 mm Hg or 130/80 to 139/89 mm Hg (whichever is lower) and for children aged > 13 years, a BP 130/80 to 139/89 mm Hg. For **Stage 2 Hypertension** 1-13 y/o: BP > 95th percentile +12 mm Hg, or > 140/90 mm Hg (whichever is lower) and for children > 13 y/o, a BP > 140/90 mm Hg. Hypertensive patients must be referred to the subspecialist for further investigation and management.

12. General Procedures may be modified depending upon entry point into schedule and individual need.

13. Article 3 Section 54 of the Newborn Screening Act of 2004 (Republic Act No. 9288) states "*Obligation to Inform*. Any health practitioner who delivers, or assists in the delivery of a newborn in the Philippines shall, prior to delivery, inform the parents or legal guardian of the newborn of the availability, nature and benefits of newborn screening.

Newborn screening aims for the early detection of disorders to ensure early intervention preventing life-threatening complications. Newborn

Screening Reference Center (NSRC) Memorandum 2014-015 regarding the protocol on collecting blood samples states that "newborn screening should be ideally done immediately after 24 hours from birth" regardless of gestational age and clinical status.

NSRC Memorandum 2014-028 states that preterm (<37 weeks), LBW (<2000g) and sick neonates (receiving intensive care) may have a sample taken before 24 hours from birth if blood transfusion will be done but otherwise a sample should be taken immediately after 24 hours from birth and a repeat screening should be collected at the 28th day of life (53).

Expanded newborn screening has been made available since December 2014.(51) It increases the number of screened disorders from six to twenty-eight. All health professionals are encouraged to organize and participate in continuing medical education and to advocate for newborn screening (48).

14. The PPS Policy Statement on "Neonatal Hearing Screening" recommends screening for all Newborns whether high risk or non-high risk (56). Pertinent sections of Republic Act No. 9709 (The Universal Newborn Hearing Screening and Intervention Act of 2009) include the following:

Section 5: "*Obligation to Inform* - Any health practitioner who delivers, or assists in the delivery, of a newborn in the Philippines shall, prior to delivery, inform the parents or legal guardian of the newborn of the availability, nature and benefits of hearing loss screening among newborns or children three months old and below."

Section 6: "*Obligation to Perform Newborn Hearing Loss Screening*

and Audiologic Diagnostic Evaluation
– All infants born in hospitals in the Philippines shall be made to undergo newborn hearing loss screening before discharge, unless the parents or legal guardians of the newborn object to the screening subject to Section 7 of this Act. **Infants who are not born in hospitals** should be **screened within the first three (3) months after birth**.

In the event of a **positive newborn hearing loss screening** result, the newborn shall **undergo audiologic diagnostic evaluation** in a timely manner to allow appropriate follow-up, recall and referral for intervention before the age of six (6) months: provided, that audiologic diagnostic evaluation shall be performed by Newborn Hearing Screening Centers duly certified by DOH.” (58)

15. The Philippine Society of Pediatric Ophthalmology and Strabismus (PSPOS) and the Philippine Academy of Ophthalmology (PAO) recommend the regular eye and vision screening examination of children from infancy until maturity of their visual system. The **single most effective** way of determining a child's eye health is through a **non-invasive and simple Eye and Vision Screening Test** (Appendix 5).

The Philippine Pediatric Society Policy Statement on “Pediatric Blindness Prevention and Vision Screening” asserts that proper dietary supplementation, measles immunization, routine pediatric eye evaluation for all patients, and subsequent referral of children at high risk for blindness are key steps in the prevention of blindness in Filipino children (79). The Clinical Practice Guideline on “The Routine Eye Examination as a Screening Tool for

Retinoblastoma” recommends routine eye examination of infants and children for early detection of **leukocoria** and **strabismus**, the most presenting signs of retinoblastoma (60).

16. Every visit should be an opportunity to update a child's immunization.

Figure 33 and 34 summarize the recommendations for immunization of infants, children and adolescents from the Philippine Pediatric Society, the Pediatric Infectious Disease Society of the Philippines and the Philippine Foundation for Vaccination.

- Immunization recommendations for adolescents are summarized in Figure 34.
- Republic Act no. 9482 the Anti Rabies Act of 2007 mandates the creation of a National Rabies Prevention and Control Program. One of the proposed activities of the program is the provision of free routine immunization or Pre Exposure Prophylaxis (PEP) for schoolchildren aged five (5) to fourteen (14) years. (84)

17. **Iron Supplementation** as recommended by the DOH (87):

Targets	Preparation	Dose/ Duration
Low birth weight	Drops: 15mg elemental iron/0.6ml	0.3ml once a day to start at two months of age until 6 months when complementary foods are given

A N N O T A T I O N S

Targets	Preparation	Dose/ Duration
Infants 6-11 months	Drops containing 15ml elemental iron/0.6ml	0.6 ml once a day for 3 months
Children 1-5 yrs old	Syrup containing 30mg elemental iron/5ml	1 tsp once a day for 3 months or 30mg once a week for 6 months with supervised administration
Adolescent Girls (10-19 yrs.)	Tablet containing 60 mg elemental iron with 400mcg folic acid (coated)	One tablet once a day

18. Vitamin A supplementation as recommended by the DOH (87):

Targets	Preparation	Dose/ Duration
Infants 6-11 months	100,000 I.U.	1 dose only (One capsule is given any- time between 6-11 months but usually given at 9 months of age during the measles immunization)
Children 12-59 months	200,000 I.U.	1 capsule every 6 months

The PPS Policy Statement on "Zinc Supplements in Children" cites the beneficial role of zinc supplementation

in the prevention of pneumonia and diarrhea. The recommended dose and dose interval though have yet to be set. (93)

19. The Department of Health Administrative Order 2015-0054: Revised Guidelines on Mass Drug Administration and the Management of Adverse Events Following Deworming (AEFD) and Serious Adverse Events (SAE) recommends deworming for all children aged 1 to 12 years (96).

The WHO and the DOH both recommend the use of either albendazole or mebendazole in the following doses and schedule (96, 97, 100):

- **Albendazole**
12 months to 23 months: 200 mg, single dose every 6 months
24 months and above: 400 mg, single dose every 6 months

- **Mebendazole**
12 months and above: 500 mg, single dose every 6 months
Either drug shall be taken ON FULL STOMACH.

Deworming must not be done in the children with (97):

- severe malnutrition
- high-grade fever
- profuse diarrhea
- abdominal pain
- serious illness
- previous hypersensitivity to antihelminthic drug

The DOH has a National Filariasis Elimination Program implemented in municipalities endemic for filariasis. Mass treatment with Diethylcarbamazine Citrate (DEC)

and Albendazole includes children from 2 years old and above (101).

20. Age-appropriate discussion and counseling should be an integral part of each visit.

21. The Philippine Pediatric Dental Society, Inc. endorses the recommendations of the American Academy of Pediatric Dentistry (AAPD) and the American Dental Association pertinent to preventive dental care (Appendix 6).

The first dental visit is recommended to be done at the time of eruption of the first tooth and no later than 12 months of age. During the first dental visit, the dentist will assess:

- the child's general health, growth and behavior
- the child's oral hygiene and periodontal health and,
- the risk for developing oral disease. The dentist will likewise provide education on infant oral health and evaluate and optimize fluoride exposure (168).

PPS Policy Statement on Fluorides in the Prevention of Dental Caries in Children cites ways on how to prevent early childhood caries such as involving the parents, dentists, physicians and the government in promoting good oral health, as well as recommendations on the use of fluoride varnish (167).

22. Counseling regarding breastfeeding that was started during the prenatal period must be continued during well child visits. Mothers must be encouraged to exclusively breastfeed up to six (6) months and continued on up to two (2) years and beyond. Safe, adequate, timely and properly fed complementary feeding using

fresh, natural and indigenous food shall begin at six (6) months to meet the evolving nutritional requirements of infants (14, 15, 17, 114). The Philippine Society of Pediatric Gastroenterology and Nutrition issued guidelines on breastfeeding and complementary feeding (Appendix 7).

Early on, children must be taught the value of eating healthy foods in a balanced diet as well as avoiding unhealthy foods (120). Giving of foods that are too sweet (sweetened beverages, candies), too salty (chips, curls) or too oily (gravies, dressings) should be avoided (114, 120, 121)

It is imperative that hand hygiene be practiced at all times.

The food pyramid crafted by the Philippine Society of Pediatric Gastroenterology and Nutrition is a guide for physicians and parents in providing daily healthy diets for children (Figure 32).

Health care professionals and parents must be aware that exposure of children to media food advertising especially commercials for convenient foods, processed foods or sweetened drinks may influence children's choices toward higher-fat or higher-energy foods (122)

Physicians may work with school administrators in promoting school-based nutrition programs to ensure that children are provided with healthy food in school (119).

23. Physical activity, along with a well-balanced healthy diet, is a major principle to healthy living (117, 130). A physically active lifestyle among children and adolescents will be carried into adulthood and reduce health problems related to inactivity

ANNOTATIONS

(131). Physical activity can be in the form of sports and games, dance, physical recreational activities, household chores and other lifestyle related physical activities (134)

Age-appropriate physical activities for children and adolescents for 60 minutes daily (PASOO) or on most days of the week (UPCHK) are the current recommendations (134). Health care professionals must educate parents and discourage children from prolonged periods of sedentary activity (TV viewing and computer games) for periods greater than two hours per day. These messages are similarly incorporated in the Philippine Pediatric Society Policy Statement on "Physical Activity for Schoolchildren" which lists recommendations for schools, physicians regular physical activity in children (130).

24. The following are Policy Statements of the Philippine Pediatric Society: Child Safety in Private Motor Vehicles, Child Safety in Public Motor Vehicles, Child Pedestrian Injury Prevention, Child Helmet Use, Drowning Prevention, Burn Injury Prevention, Household Products Poisoning, Medicinal Poisoning and Watusi Poisoning, Backpacks and Children, Noise in the Environment, Recreational Noise, Fetal and Neonatal Exposure to Noise and Fireworks Related Injuries. (137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150)

Anticipatory Guidance for adolescents is contained in Appendix 3.

The Injury Prevention Program (TIPP) of the American Academy of Pediatrics similarly provide guidelines for pediatricians to counsel parents

and children about adopting behaviors to prevent injuries from birth to adolescence (153).

25. The World Health Organization defines child maltreatment as "all forms of physical and/or emotional ill-treatment, sexual abuse, neglect or negligent treatment or commercial or other exploitation, resulting in actual or potential harm to the child's health, survival, development or dignity in the context of a relationship of responsibility, trust and power" (154).

There is **NO** one risk factor that is predictive of child maltreatment and there is **NO** one characteristic that defines resiliency of a child to traumatic experiences (154, 158, 161, 162).

The WHO listed factors that increase the susceptibility of a child to maltreatment as well as factors that may offer protective effect (Appendix 8).

The Child Protection Unit Network, Inc. enumerated the "5 Steps to Protect Children" (158, 162) (Appendix 9).

Children learn behaviors and have their value systems shaped by television, movies, music, music videos, video games, computer games and the internet. The Philippine Pediatric Society Policy Statement on "Effects of Media Sex and Violence on Children and adolescents" advocates minimization of media exposure for Filipino youth and urges parents to create a healthier and friendlier environment for their children to reduce the negative effects of media influence (169)

The Implementing Rules and Regulations of Republic Act 7610 (Anti Child Abuse Law) Section 4 states that "The head of any public or private hospital, medical clinic and similar institution, as well as the attending

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physician and nurse, shall report, either orally or in writing, to Department of Social Welfare and Development (DSWD) the examination and/or treatment of a child who appears to have suffered abuse within 48 hours from knowledge of the same." They may also report suspected cases to the Local Barangay Council for the Protection of Children (BCPC), Local Government Unit (City/ Municipal/ District) Social Welfare Office, Crisis Intervention Unit (CIU) - DSWD-NCR and Philippine National Police Women and Child Protection Desks (WCPDs). Clinicians should maintain a high index of suspicion for past and present incidence of domestic violence (163, 165, 166)

26. Lead is an ubiquitous environmental toxicant that can attack many different organ systems. Among children, the best studied effect of lead exposure is cognitive impairment. The Philippine Pediatric Society Policy Statement "Lead Poisoning in Children" presents background information on lead poisoning in children, several lead exposure prevention strategies and recommendations for the prevention of lead poisoning in children (103).

27. Iron deficiency anemia is associated with cognitive and psychomotor abnormalities in children. At risk are those with poor nutritional history and those with a past or family history of anemia.

The Philippine Society of Pediatric Hematology recommends a complete blood count at least once between the following time intervals for those at risk: 6-24 months, 2-6 years and 10-19 years. Special attention should be given to infants 6 months to less than 12 months, and 12 months to 23

months (94). Actively menstruating female adolescents and fad dieters are likewise at risk. The Philippine Society of Adolescent Medicine Specialists recommends a complete blood count at each stage of adolescence.

28. The Philippine Society of Adolescent Medicine Specialists recommends screening urinalysis on first encounter with an adolescent patient (Appendix 3). Urinalysis must likewise be done for all patients with signs and symptoms referable to a possible renal disease regardless of age.

29. As indicated in Appendix 3, annual health screening for sexually active females includes vaginal wet mount and PAP smear. Sexually-active males must undergo serologic test for syphilis while both male and female sexually active adolescents should have annual non-culture test for gonorrhea and Chlamydia.

30. Screening for tuberculosis is targeted among individuals who are at risk for developing the disease (206).

Using 5 TTU PPD or 2TU-RT23 test read at 48-72 hrs, regardless of BCG status, an induration (not erythema) of > 5 mm is considered positive in the presence of any or all of the following: history of close contact with a known or suspected case of TB, clinical findings suggestive of TB, chest x-ray suggestive of TB, and Immunosuppressed condition. In the absence of the above factors, an induration > 10 mm is considered positive (206).

Appendix 1. Prenatal Visit, Education and Counseling

Education and counseling on Essential Intrapartum and Newborn Care (EINC or *Unang Yakap*) and breastfeeding must begin during the prenatal period (2, 9, 14, 17). The recommended EINC practice for immediate care of the normal newborn are a series of time-bound interventions at the time of birth that can be enforced immediately in all health care settings. It emphasizes the step-by-step performance of a sequence of four core actions which are: (1)

- (1) immediate and thorough drying of the newborn,
- (2) early skin-to-skin contact between mother and newborn,
- (3) properly-timed cord clamping and cutting, and
- (4) non-separation of newborn and mother for early breastfeeding.

These time sensitive interventions should not be pre-empted nor undermined by other interventions. Unnecessary interventions in newborn care include routine suctioning, early bathing, routine separation from the mother, foot printing, application of various substances to the cord, and giving pre-lacteals or artificial milk formula or other breast milk substitutes.

Aside from this, to promote, protect, and support breastfeeding, mothers must be informed of the recommended period of early initiation of breastfeeding, exclusive breastfeeding up to 6 months and continued breastfeeding up to two years and beyond after introduction of complementary foods (12, 13, 14, 15).

Other recommended procedures done during the infant's birth may be explained during the prenatal visits. These include early rooming-in, Kangaroo Mother Care, newborn screening, hearing screening and immunizations with Hepatitis B vaccine and BCG (1, 80, 81).

It is during the prenatal visits when the health care professional may elicit information regarding the parent's education, profession, attitude regarding the pregnancy, planned disciplinary method or child rearing approach, financial security, family support system and such other factors that are vital in the assessment for the child's future exposure to or prevention of neglect, maltreatment or violence (154). Similarly, the prenatal visit is a good opportunity to inquire about a family history of genetic or chromosomal abnormality and development disability. The health care professional must monitor, counsel and refer whenever necessary to give the infant the best possible start in life (29).

Discussion regarding injury prevention and potential exposure of the mother and child to environmental toxicants such as lead may begin during the prenatal visits (103).

Pregnant women must be informed about the deleterious effects of smoking, alcohol intake and exposure to known teratogens during pregnancy (6, 36). They must likewise be advised and encouraged to take folic acid rich food and supplements on top of the recommended healthy diet for a pregnant woman (33, 34, 35). Tetanus Toxoid immunization must be started or continued during pregnancy (80).

Pediatricians, obstetricians, midwives, nurses and other health care professionals/workers must work together to promote the welfare of the mother and the unborn child both in normal and high-risk pregnancies.

Appendix 2. Discharge and Follow-up of Healthy Term Newborns

The Philippine Society of Newborn Medicine lists the following **minimum criteria** for **discharging newborns before 48 hours** (61, 63):

- **Uncomplicated** antepartum, intrapartum and postpartum courses for both mother and newborn
- **Vaginal delivery, singleton, completed 37 weeks, AGA**
- **Normal and stable vital signs** during the **preceding 12 hours** (RR <60/min, CR 100-160/min, axillary temperature 36.5°C - 37.4°C properly clothed in an open crib)
- Has **urinated and passed at least one stool**
- Has documented **proper latch, milk transfer, swallowing, infant satiety** and absence of nipple discomfort
- Normal physical examination
- **No evidence of significant jaundice** in the **first 24 hours of life**
- **Educability and ability of the parents to care for their child** (recognize signs of illness, care of the umbilical cord/ skin/ genitalia, maternal confidence in feeding her infant and parents' understanding of the importance of follow-up visit or emergency consultation)
- **Must follow-up within the next 48 hours**

The purpose of the follow-up visit is to (63):

- Assess the infant's general health, hydration, and presence/ degree of jaundice; weigh the patient; identify any new problems; and obtain historical evidence of adequate urination and defecation patterns for the infant.
- **Reinforce maternal or family education in infant care**, particularly regarding infant feeding.
- Review feeding pattern and technique, including observation of breastfeeding for adequacy of position, latch-on, and swallowing.
- Assess quality of mother-infant interaction and details of infant behavior.
- Review the outstanding results of laboratory tests performed before discharge.
- Perform screening tests (see section on General Procedures), if not yet done, and other tests that may be clinically indicated, such as serum bilirubin.
- Suggest and encourage compliance to recommended schedule of periodic follow-up and preventive care.

Appendix 3. Adolescent Health Care

The Philippine Society of Adolescent Medicine Specialists, cognizant of the rapid physical, cognitive and psychosocial changes occurring in each adolescent patient, recommends an annual health screening and preventive services for this special population. During the annual visit, the adolescent should undergo the following:

1. **Complete history-taking** to screen for risks and protective factors using the tool **HEEADSSS** (31, 32) which means: **H-home**, **E-education/employment**, **E-eating**, **A-activities**, including media and internet exposure, **D-drugs** (alcohol, tobacco, and other drugs), **S-sexuality**, **S-suicidality/depression**, **S-safety**. It is recommended to add another S, which screens for **strength/spirituality**.

2. **Physical Examination**

This should be done in privacy and preferably by a health care provider that is of the same gender as the teen patient. In addition to the routine physical examination, the following should be done:

- 2.1 **Tanner Staging/Sexual Maturity Rating**
- 2.2 **Breast examination**
- 2.3 Examination of the **spine and shoulders**. Check for **scoliosis** and **kyphosis**.
- 2.4 Inspection of the **genitals and anus**. A more thorough examination is warranted in symptomatic patients.

3. **Screening Procedures**

- 3.1 **Complete Blood Count** (or at least Hemoglobin/Hematocrit) at every stage of adolescence
- 3.2 **Urinalysis** on first encounter
- 3.3 **Vaginal wet mount, PAP smear** for **sexually active females**
- 3.4 Serologic test for **syphilis** for **sexually active males** Non-culture test for **gonorrhea** and **Chlamydia** for **both** male and females who are sexually active

4. **Immunization Update:**

Please refer to Figure 34. Summary Table: Immunization of Teens and Pre-teens 2018 (7 to 18 yrs old)

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5. Anticipatory Guidance and Counseling

- 5.1 Self breast examination for females
- 5.2 Healthy Lifestyle: physical activity, diet, avoidance of alcohol, smoking, drug use
- 5.3 Sexual behavior and the risk of acquiring STIs including HIV
- 5.4 Injury and accident prevention: use of sports protective gear, seat belts, no driving under the influence of alcohol, no smoking in bed, no hand gun use.

Confidentiality is a major issue in attending to adolescent patients. In addition, guidance and counseling is now directed to the patient with diminishing participation of the parent/guardian.

Guidelines for Adolescent Health Care (Updated as of October 2017)	
PROCEDURE	FREQUENCY
HISTORY	Annual
PHYSICAL EXAMINATION	Annual
	Do Sexual Maturation Rating (SMR) at each stage of adolescence (early, middle, late)
Breast Self Awareness ¹ (198)	Early adolescence
Testicular Examination ²	Older adolescence
Visual screening (191, 192)	Once during middle adolescence
Scoliosis (197)	For Females: age 10 -12 years For Males: age 13 -14 years
MEASUREMENTS	
Weight	Annual
Height	Annual
Body Mass Index	Annual Use WHO Z-score charts (Figures 15 and 27)
Blood Pressure	Annual
DEVELOPMENTAL/BEHAVIORAL HEALTH	
Psychosocial Assessment (31, 32)	Annual Use HEADSSS structured interview format
Alcohol, Tobacco, and Drug/ Substance Use Assessment (40, 164, 199, 200, 201) (No Local consensus available)	Annual Assess using CRAFFT (version 2.0) (201)
Depression Screening (202)	Annual Use PHQ-9 starting at age 12 years old (203)

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SCREENING PROCEDURES	
Anemia (Hemoglobin/Hematocrit)	Once during each stage of adolescence (early, middle, late)
Urinalysis	Once during adolescence, preferably on initial visit
Tuberculosis	For at risk
Sexually Transmitted Infections (191, 192, 196) (No local consensus available) Chlamydia and Gonorrhea (40, 164, 192, 193, 196) Trichomonas vaginalis (164) Syphilis (164, 196) HIV	Annual For sexually active adolescents Annual For sexually active adolescents With pretest and post test counseling
Cervical Dysplasia (191, 192, 196)	PAP smear at age 21 years, regardless of sexual activity
IMMUNIZATIONS (Figure 34)	Use all opportunities to give age-appropriate vaccines and do catch-up vaccinations
ANTICIPATORY GUIDANCE	
Injury prevention (153), nutrition, physical activity, alcohol, tobacco, drug use, mental health and responsible sexual behaviors	Annual
Men having sex with men (MSM) (No local consensus available)	Behavioral counseling, recommendation to delay sexual activity

1 - Defined as : Understanding how breasts normally look and feel to be able to detect changes and report then.

2 - Perform Visual inspection for SMR and observation for signs of STIs (warts, vesicles); Examine for hydrocele, hernia, varicocele, mass

Appendix 4. Developmental Surveillance and Screening

The Philippine Society for Developmental and Behavioral Pediatrics (PSDBP) recommends that developmental surveillance be done at every well child visit. Developmental surveillance is a process by which the health care professional recognizes the children who may be at risk of developmental and behavioral conditions. At every well child visit, developmental surveillance has the following components (207, 208):

1. Eliciting and attending to the parents' concerns about their child's development
2. Maintaining a developmental history
3. Making accurate and informed observations of the child

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4. Identifying the presence of risk and protective factors
5. Documenting the process and findings

Another component of developmental surveillance is the promotion of healthy development (213). This is done through providing parents with developmental promotion and behavior management strategies.

Furthermore the PSDBP recommends that **developmental screening** be done at specified ages particularly at **9, 18 and 30 months** (207, 208) and **yearly thereafter**. Developmental screening is the process of administering a standardized tool designed to identify children who are at risk for developmental and behavioral disorders. It is also done when surveillance activities detect risks and anytime when parents express concerns about their child's development (207, 208). At **18 and 24 months**, an **Autism Spectrum Disorder specific screening tool** (e.g. Modified Checklist for Autism in Toddlers) should also be routinely administered (214).

Developmental red flags can also be used. As children grow and develop, one needs to be mindful that each child develops at his/her own pace and the range of what is considered to be "normal" is quite wide. However, there are absolute indicators often referred to as RED FLAGS that identify developmental markers suggesting the need for further neurodevelopmental evaluation. The tables below outline some of the selected developmental indicators in the first five years presented in different functional domains (215).

Gross Motor Red Flags

AGE	RED FLAG
4 ½ months	Does not pull up to sit
5 months	Does not roll over
7-8 months	Does not sit without support
9-10 months	Does not stand while holding on
15 months	Not walking
2 years	Not climbing up or down the stairs
2 ½ years	Not jumping with both feet
3 years	Unable to stand on one foot momentarily
4 years	Not hopping
5 years	Unable to walk a straight line back and forth or balance on one foot for 5-10 seconds

Fine Motor Red Flags

AGE	RED FLAG
3 ½ months	Persistence of grasp reflex
4-5 months	Unable to hold a rattle
7 months	Unable to hold an object in each hand
10-11 months	Absence of pincer grasp
15 months	Unable to put in or take out
20 months	Unable to remove socks or gloves alone
2 years	Unable to stack five blocks, not scribbling

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2 ½ years	Not turning a single page of a book
3 years	Unable to stack eight blocks or draw a straight line
4 years	Unable to stack ten blocks or copy a circle
4 ½ years	Unable to copy a square
5 years	Unable to build a staircase of blocks or copy a cross

Language Red Flags

AGE	RED FLAG
5-6 months	Not babbling
8-9 months	Not saying "da" or "ba"
10-11 months	Not saying "dada" or "baba"
18 months	Has less than three words with meaning. Unable to achieve shared attention
2 years	No two-word phrases or repetition of phrases
2 ½ years	Not using at least one personal pronoun
3 ½ years	Speech only half understandable
4 years	Does not understand prepositions
5 years	Not using proper syntax in short sentences

Cognitive Red Flags

AGE	RED FLAG
2-3 months	Not alert to mother with special interest
6-7 months	Not searching for dropped object
8-9 months	No interest in peek-a-boo
12 months	Does not search for hidden objects
15-18 months	No interest in cause-and-effect games
2 years	Does not categorize similarities
3 years	Does not know own full name
4 years	Cannot pick shorter or longer of two lines
4 ½ years	Cannot count sequentially
5 years	Does not know color or any letters
5 ½ years	Does not know own birthday or address

Psychosocial Red Flags

AGE	RED FLAG
3 months	Not smiling socially
6-8 months	Not laughing in playful situations
1 year	Hard to console, stiffens when approached
2 years	Kicks, bites and screams easily and without provocation Rocks back and forth in crib
3-5 years	No eye contact or engagement with other children In constant motion Resists discipline Does not play with other children

Appendix 5

COMPONENTS OF EYE AND VISION SCREENING AND SUMMARY OF RECOMMENDATIONS

COMPONENTS OF EYE AND VISION SCREENING	RECOMMENDATIONS	
1. History	a. All high risk infants should be referred to an ophthalmologist for comprehensive eye examination. This include but not limited to the following:	1) Premature infants whose age of gestation at birth meets the criteria for screening Retinopathy of Prematurity (ROP) a) Thirty two (32) weeks or younger Age of Gestation or; b) One thousand five hundred grams (1,500 g) or lighter or; c) Older than 32 weeks and/or heavier than 1,500g) with a stormy medical course in the NICU 2) Infants and children with metabolic disorders 3) Infants and children with medical conditions known to have associated eye problems (i.e. Down's Syndrome, idiopathic arthritis, neurofibromatosis) 4) Infants and children with history of "squinting", "head tilt" or "head turn" 5) Children with history of visual difficulties and learning problems
	b. Family history of eye problems such as the following:	1) Strabismus, 2) Amblyopia, 3) Congenital cataract, 4) Congenital glaucoma, 5) Retinoblastoma, 6) Ocular and systemic genetic diseases
2. Visual Acuity Assessment	a. Fixes and Follows Test	For infants and pre-verbal children, the fixation behaviour can be determined by using a toy, picture, or other interesting objects as a target, with unoccluded eyes and with one eye occluded. If the child fails to fixate and follow an object

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		on each eye, the child should be referred to an Ophthalmologist with training and experience in the evaluation and treatment of children's eye diseases.
	b. Subjective/ Formal Visual Acuity Testing	For children who can participate in subjective/formal visual acuity testing, using LEA Pictures chart for preschoolers or children who are not familiar with letters, and the Sloan letters, HOTV chart for literate children. Other validated chart may be used for screening. a. Children age 3 to 3^{11/12} years old and below should be able to read at least 20/50 (10/25) or better. b. Children age 4 to 4^{11/12} years old should be able to read at least 20/40 (10/20) or better. c. Children 5 years old and older are expected to read 20/30 (10/15). Children who do not pass the minimum criteria should be referred for an additional ocular assessment to be performed by and Ophthalmologist with training and experience in the evaluation and treatment of children's eye diseases.
3. External Inspection of the Eye, Pupil and Adnexa	The external of the eye and its adnexa should be inspected for abnormalities like eyelid deformities, ptosis and hemangioma. The pupils should be centered, round and constrict when light is directed at it. Any abnormalities noted should warrant a referral.	
4. Ocular Motility Assessment	a. Corneal Light Reflex	Using a penlight as a target, the corneal light reflex should be seen simultaneously at the center of the pupil of each eye. A corneal light reflex not seen at the center of the pupil warrants a referral.
	b. Versions and Ductions	The ocular motility of the eyes should be examined using an

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		interesting object or a penlight as a target. The eyes should be able to move together in all directions of gaze. Any limitations or lag in the eye movements should warrant a referral.
	c. Cross Over Test	A cross over cover test or alternate cover test should be performed to check the presence of ocular misalignment . This is done by alternately covering the eyes while the patient is fixing at a target . Any movement noticed during this test warrants a referral .
5. Red Orange Reflex Examination	Using an ophthalmoscope in a darkened surrounding, the light is directed through the pupil. A red orange glow reflex should be observed through the ophthalmoscope signifying the light from the ophthalmoscope was able to enter the eye and reflect back from the choroidal blood vessels. An absence of the Red Orange Reflex or the presence of a white reflex pupil warrants a referral.	

The **vision testing using LEA Chart** or its equivalent is preferably done at **distance** and **near starting 3 years of age**. The vision testing procedure is as follows:

- Distance visual acuity should be tested at **6 meters (20 ft)** or **3 meters (10 ft)** and at **reading distance of 34-40 cm (14-16 inches)** from child under good illumination
- Establish a method of communication i.e. naming or pointing (matching). Decide with child which means will be used to identify symbols.
- "Let's play a game"
- Start with both eyes open to acquaint child with the "game".
- Point briefly at each of the 4 symbols (circle, house, apple, square) on the lowest line, observe the baseline responses for comprehension, speed and accuracy.
- Demonstrate to child that the "game" can be played with 1 eye occluded.
- Cover the top line with a white card and ask the child to identify only the 1st symbol on the line below the covering card.
- Repeat procedure for the each lower line moving quickly down (to avoid tiring the child) until the child hesitates or misidentifies a symbol.
- Upon reaching the misidentified symbol, move back up 1 line and ask the child to identify all the symbols on that line.
- If the child skips a symbol, ask the child to try again.

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- **Visual Acuity** is recorded as the **last line** on which **at least 4 of the 5 symbols are read correctly**. (Ex. If up to 10th line VA=20/40 or 0.5). Always test until this threshold line. If only 3 out of 5 are read correctly, record the result: visual acuity of the previous line (+3).
- Repeat test with the other eye occluded.
- If the child wears glasses, they should be worn.
- Watch out for peekers.
- Frustration or disinterest often indicates that the child no longer recognizes a symbol.
- Pointing at the line to be read is preferable over pointing at a specific symbol.
- If results are inconclusive, repeat or refer.

Appendix 6. Preventive Dental Care

- **Use of Fluoride Toothpaste.** **Twice daily** use of fluoride-containing toothpaste is recommended as a primary preventive measure. Parents and caregivers must ensure that the recommended amount of fluoride toothpaste is used. Young children must always be supervised while brushing and should be taught to spit out the toothpaste and to **avoid rinsing after brushing**.

Recommended Use of Fluoride Toothpaste in Children

Age Group	Fluoride Concentration (ppm)	Minimum Daily Use	Amount of Toothpaste to be Used	Amount of Fluoride (~mg)
6 months to less than 2 years old	1000 ppm	Twice daily	Smear 2.5mm 0.125g	2 x 0.125 = 0.25mg
2 to 6 years old	1000 ppm	Twice daily	Pea size 5mm 0.25g	2 x 0.25 = 0.50mg
6 months to less than 2 years old	1000 ppm	Twice daily 21	Smear 2.5mm 0.125g	2 x 0.125 = 0.25mg
6 years old and above	1500 ppm	Twice daily	Full length of bristle 10-20mm 0.5 1.0g	2 x 0.50 = 1.0mg

- **Topical Fluoride Treatment.** Professionally applied topical fluoride has been proven to prevent or reverse enamel demineralization. The American Academy of Pediatric Dentistry (AAPD) recommends that children at moderate caries risk should receive a professional fluoride treatment at least every 6 months; those with high caries risk should receive topically-applied fluoride more frequently (172, 176).
- **Other Anticipatory Measures.** Anticipatory care includes guidance on oral hygiene and proper diet. Cleansing the infant's teeth as soon as they erupt with either washcloth or soft brush will help reduce bacterial colonization. The use of dental floss is important to reduce interproximal caries.

The education of parents includes the cariogenicity of some food and beverages, dental caries and its relationship with prolonged bottle feeding or bottle feeding while asleep, and the maintenance of good oral hygiene in the mother that has a significant impact on the child's caries rate (171, 172, 179).

Appendix 7. Breastfeeding and Complementary Feeding Philippine Society of Pediatric Gastroenterology and Nutrition

A. Breastfeeding

1. Benefits of Breastmilk
 - Safe, sterile and always available
 - With perfect nutrients to fully sustain the growth and development of the baby from birth to six months of age; after 6 months, still a good source of nutrients when given with adequate complementary food
 - Easily digested and absorbed; efficiently used by the baby's immature system
 - Contains antibodies and substances which protect the baby against infection
 - Contains fats (DHA) which enhance brain development and intelligence of the baby
2. Advantages of Breastfeeding
 - Promotes emotional bonding between baby and mother
 - Protects the mother's health against cancer (breast, uterus, ovaries), obesity and post-partum hemorrhage
 - Promotes early return to pre-pregnancy weight
 - Gives the family big financial savings

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3. Correct Breastfeeding Techniques

- Support the baby's head and the entire body throughout the feeding; the head, back and hips should be facing the breast and aligned in a straight manner.
- Maintain the position of the baby in such a way he is "face to face", "chest to chest", and "tummy to tummy" with the mother.
- Support the breast with the hand of the opposite arm in a C-hold position: thumb above, 4 fingers under the breast.
- Stimulate the infant to open the mouth wide by stroking the corner of the baby's lips; check that the chin touches the breast and the lower lip is turned outward.
- Ensure that the baby grasps the entire nipple plus one inch of the surrounding areola.
- Allow the baby to suck 15 to 30 minutes per breast to extract both foremilk and hindmilk.
- Empty the breast around 8 to 10 times or more a day to ensure adequate milk supply.

4. Breastmilk Expression and Storage

- Express breastmilk by hand or by using a breast pump when breastmilk supply is abundant and when the mother is planning to go back to work.
- Store in sterile polypropylene (cloudy hard plastic) containers, properly labeled with the date and time of breastmilk collection.

5. Recommended Breastmilk Storage Period

- Room temperature ($<25^{\circ}\text{C}$): 4 hours
- Room temperature ($>25^{\circ}\text{C}$): 1 hour
- Refrigerator (4°C): 8 days
- Freezer compartment of a 1-door refrigerator: 2 weeks
- Freezer compartment of a 2-door refrigerator: 3 months
- Deep freezer with constant temperature (-20°C): 6 months

B. Diet of a Lactating Mother

- Rice 6 cups
- Fruits (vitamin C rich; different varieties) 4 pieces
- Vegetables (green leafy and yellow) 1 ½ cups
- Meat, fish, poultry, seafoods 5 pieces (matchbox size for meat), or 2 cups cut into small pieces
- Egg 4 pieces a week
- Mongo, beans, taho 1 ½ cups 3x a week

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- Milk 2 glasses
- Fats (olive oil, corn oil, butter) 7 teaspoons
- Fluids 7 glasses water;
1 glass fresh fruit juice

C. Complementary Food

Must be:

- **Timely** – introduced at 6 months of age
- **Adequate** – provides sufficient energy, protein and micronutrients to sustain growth: use PSPGN Dietary Prescription Food Guide Pyramid and FNRI Pinggang Pinoy as tools.
- **Safe** – hygienically prepared and stored; feed using clean utensils, NOT bottles and nipples
- **Properly fed** – meal frequency and feeding methods suitable for age (guide or self-feeding using clean hands, spoon and fork, cups and bowls; using locally fresh and natural foods)

How to introduce:

- Begin with **one new food** at a time to be given for **3 days**.
- Start with **lugaw or cereals, fruits or vegetables** in any order, giving **one to two teaspoons a day**.
- Start with **pureed food** at **6 months** of age. Introduce “**finger foods**” around **8 months** of age; **lumpy or chopped foods** at **10 months** of age; **table food** at **12 months** of age.
- Feed **6–8 month old infant 2–3 times a day**; **9–24 month old infants 3–4 times a day**. Give additional nutritious snacks once or twice a day.
- Offer a variety of food to improve the quality of food intake; avoid drinks with low nutrient value (sweet beverages).
- **Do not add salt to the infant's diet before one year of age**.
- Give **supplements of iron, zinc, calcium, and vitamin B12**, if diet is primarily plant-based
- Practice **responsive feeding**. Feed infants directly and assist older children. Feed slowly and patiently. DO NOT force-feed; make feeding a pleasurable experience.

Appendix 8. Child Maltreatment

The World Health Organization presents an ecological model describing the risk factors for child maltreatment. The complex interaction of factors in the individual, social relationships, community and society must be understood to effectively deal with problems of child maltreatment (154).

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The risk factors are not themselves diagnostic but in situations where resources are limited, children and families identified as having several of these factors should have priority for receiving services.

It is important for pediatricians to recognize and report on going child maltreatment (republic Act No. 7610). Suspected child abuse cases may be reported to the following:

- Local Barangay Council for the protection of Children (BCPC)
- Local Government Unit (City/Municipal District) Social Welfare Office
- Crisis Intervention Unit (CIU) – DSWD-NCR (24 hours):
Contact No. 734-8635 (NCR cases only)
- Philippine National Police Women and Child Protection Desks (WCPDs)

All hospitals should have a protocol in the handling of women and child abuse cases. The Department of Health Administrative Order No. 2013-0011 on the establishment of women and child protection in all government hospitals can be downloaded from:

<http://www.doh.gov.ph/content/women-and-children-protection-program.html>

Recommended Anticipatory Guidance Topics for Prevention of Violence Against Children (VAC)

TOPICS	Infancy	Toddler	Prechool	School Age	Adolescence
Domestic Violence					
Infant crying and dangers of shaking					
Positive discipline					
Safety on others' homes or with other people					
Normal sexual behavior					
Good touch/ bad touch					
Bullying					
Peer relationships					
Mobile phone and internet safety					
Alcohol and substance abuse					
Dating violence					

A P P E N D I X

Protective Factors

Factors that appear to facilitate resilience include:

- secure attachment of the infant to the adult family member
- high levels of parental care during childhood
- non-association with delinquent or substance-abusing peers
- a warm and supportive relationship with a non-offending parent
- a lack of abuse-related stress

Based on the current understanding of early child development, **it is clear that stable family units can be a powerful source of protection for children. Good parenting, strong attachment between parents and children, and positive non-physical disciplinary techniques are likely to be protective factors."**

The World Health Organization has taken the step in ending violence against children as there is evidence that this is preventable. The Seven INSPIRE evidence based strategies enumerated below unify many sectors to raise awareness and give policy makers the tools to be able to act and promote prevention of violence against all children (155).

- Implementation and enforcement of laws
- Norms and values
- Safe environments
- Parent and caregiver support
- Income and economic strengthening
- Response and support services
- Educational and life skills

APPENDIX

Appendix 9. "5 Steps to Protect Children (162)" Child Protection Unit Network, Inc.

Pediatricians and health care professionals may disseminate this guide and use it during well child counseling.

Step 1. *Learn the facts.* Majority of sexual offenders of children are family members, friends and neighbors people that the child and the child's family trust. Boys, in almost the same frequency as girls, are also being sexually abused. Few girls report the abuse but boys tend not to report at all.

Step 2. *Minimize the opportunity for sexual abuse by eliminating or reducing one-adult/one-child situations.* More than 80% of sexual abuse cases occur in situations where a child is left alone with an adult or an older youth.

School organizations, clubs, sports teams, faith groups must eliminate situations of one-adult/one-child.

Talk to your child when he/she returns from an outing. Notice the child's behavior and whether the child can tell you with confidence how the time was spent.

Tell the adults who care for your child that you and your child are educated about child abuse. Be that direct.

Step 3. *Talk about it.* Teach your children what parts of their bodies others should not touch. Do not be afraid that you are teaching them about "sex." You are protecting them. Mention that the abuser can be a family member, a friend or an older youth.

Children are afraid to "tell" an abuse. The abuser shames the child, tells the child that his/her parents will be angry, confuses the child about what is right or wrong, or threatens the child or a family member. *Break the barrier by talking openly about it.*

If a child seems uncomfortable or resistant to being with a particular adult (an uncle or a ninong) ask why.

Step 4. *Recognize the Signs.* Learn the signs of sexual abuse. Physical signs are not common. Emotional and behavioral signs are more common such as "too perfect" behavior, withdrawal, depression, unexplained anger or rebellion, running away, failing in school, unusual interest in or knowledge of sexual matters, fear of a person, intense dislike at being left somewhere or with someone. Know the textmates of your child.

Step 5. React Responsibly. The Future well-being of a child is at stake. Have the courage to report suspected abuse. Do not close your eyes and pretend that it will go away. It will not go away. If the child is not helped, the abuse will continue.

You can bring the child to the **Child Protection Unit of PGH, PCMC and East Avenue Medical Center**. It is the duty of the hospital administrators, doctors, nurses, government teachers and employees of government agencies to report abuse.

How to react to the knowledge of abuse. Offer support:

- **Believe the child** and make sure the child knows you believe him/her. Very few reports of child abuse are not true.
- **Thank the child for telling you** and for having the courage to do so.
- **Encourage the child to talk** but don't ask leading questions.
- **Seek professional help.**

Use your voice and your vote to make your community a safer place for children. Ask what schools or organizations in your community have child abuse prevention policies and help with their creation. Demand that the government put their resources into protecting children from sexual abuse and into responding to reports of sexual abuse.

You can download educational materials on child sexual abuse prevention for parents on <https://www.d2i.org/>

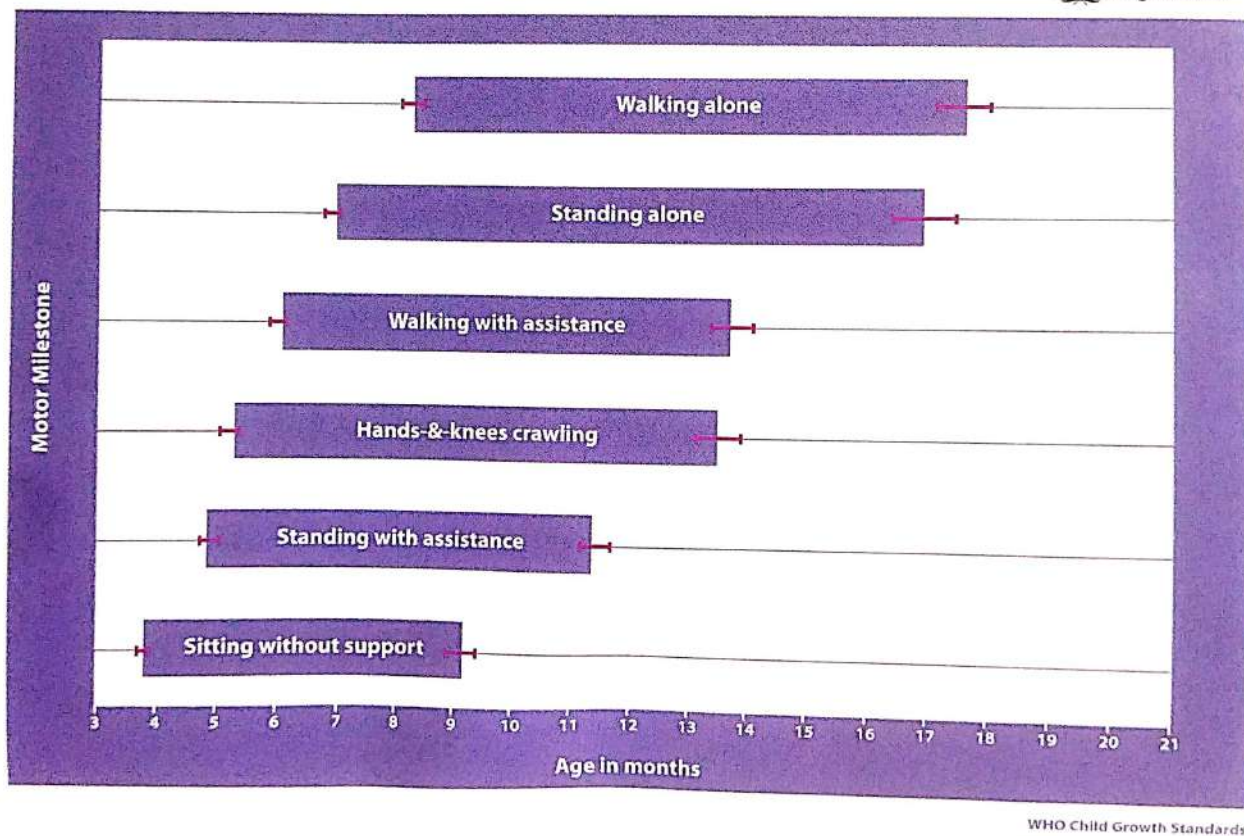
FIGURES

1. Windows of Achievement
2. Developmental Milestones of Early Literacy
3. Z Score Interpretation
4. Head Circumference for Age for Girls
5. Weight for Age for Girls: Birth to 2 years
6. Length for Age for Girls: Birth to 2 years
7. Weight for Length for Girls: Birth to 2 years
8. BMI for Age for Girls: Birth to 2 years
9. Weight for Age for Girls: 2 to 5 years
10. Height for Age for Girls: 2 to 5 years
11. Weight for Height for Girls: 2 to 5 years
12. BMI for Age for Girls: 2 to 5 years
13. Weight for Age for Girls: 5 to 10 years
14. Height for Age for Girls: 5 to 19 years
15. BMI for Age for Girls: 5 to 19 years
16. Head Circumference for Age for Boys
17. Weight for Age for Boys: Birth to 2 years
18. Length for Age for Boys: Birth to 2 years
19. Weight for Length for Boys: Birth to 2 years
20. BMI for Age for Boys: Birth to 2 years
21. Weight for Age for Boys: 2 to 5 years
22. Height for Age for Boys: 2 to 5 years
23. Weight for Height for Boys: 2 to 5 years
24. BMI for Age for Boys: 2 to 5 years
25. Weight for Age for Boys: 5 to 10 years
26. Height for Age for Boys: 5 to 19 years
27. BMI for Age for Boys: 5 to 19 years
28. US CDC-NCHS Growth Chart for Boys
29. US CDC-NCHS Growth Chart for Girls
30. BP Levels for Boys by Age and Height Percentile
31. BP Levels for Girls by Age and Height Percentile
32. Food Pyramid
33. Immunization Table 2018
34. Immunization of Teens and Pre-Teens

Windows of Achievement



Figure 1. Windows of Achievement



FIGURES

Figure 2. Developmental Milestones of Early Literacy

Developmental Milestones of Early Literacy		
MOTOR:	COGNITIVE:	WHAT PARENTS CAN DO:
6-12 months		
- reaches for book - book to mouth - sits in lap, head steady - turns pages with adult help	- looks at pictures - vocalizes, pats pictures - prefers pictures of faces	- hold child comfortably; face-to-face gaze - follow baby's cues for "more" and "stop" - point and name pictures
12-18 months		
- sits without support - may carry book - holds book with help - turns board pages, several at a time	- no longer mouths right away - points at pictures with one finger - may make same sound for particular picture (labels) - points when asked, "where's...?" - turns book right side up - gives book to adult to read	- respond to child's prompting to read - let the child control the book - be comfortable with toddler's short attention span - ask "where's the...?" and let child point
18-24 months		
- turns board book pages easily, one at a time - carries book around the house - may use book as transitional object	- names familiar pictures - fills in words in familiar stories "reads" to dolls or stuffed animals - recites parts of well-known stories - attention span highly variable	- relate books to child's experiences - use books in routines, bedtimes - ask "what's that?" and give child time to answer - pause and let child complete the sentence
24-36 months		
- learns to handle paper pages - goes back and forth in - books to find favorite pictures	- recites whole phrases, sometimes whole stories - coordinates text with picture - protests when adult gets word wrong in a familiar story - reads familiar book to self	- keep using books in routines - read at bedtime - be willing to read the same story over and over - ask "what's that?" - relate books to child's experiences - provide crayons and paper
3 years and up		
- competent book handling - turns paper pages one at a time	- listens to longer stories - can retell familiar story - understands what text is - moves finger along text "writes" name - moves toward letter recognition	- ask "what's happening?" - encourage writing and drawing - let the child tell the story



Reach Out and Read National Center • 56 Roland Street, Suite 100D, Boston, MA 02129
 Voice: 617-455-0600 • Fax: 617-455-0601 • Email: info@reachoutandread.org • www.reachoutandread.org
 BOSTON UNIVERSITY SCHOOL OF MEDICINE • BOSTON MEDICAL CENTER • DEPARTMENT OF PEDIATRICS



FIGURES

Figure 3. Z- SCORE INTERPRETATION

Compare the points plotted on the child's growth charts with the z-score lines to determine whether they indicate a growth problem. Measurements in the shaded boxes are in the normal range.

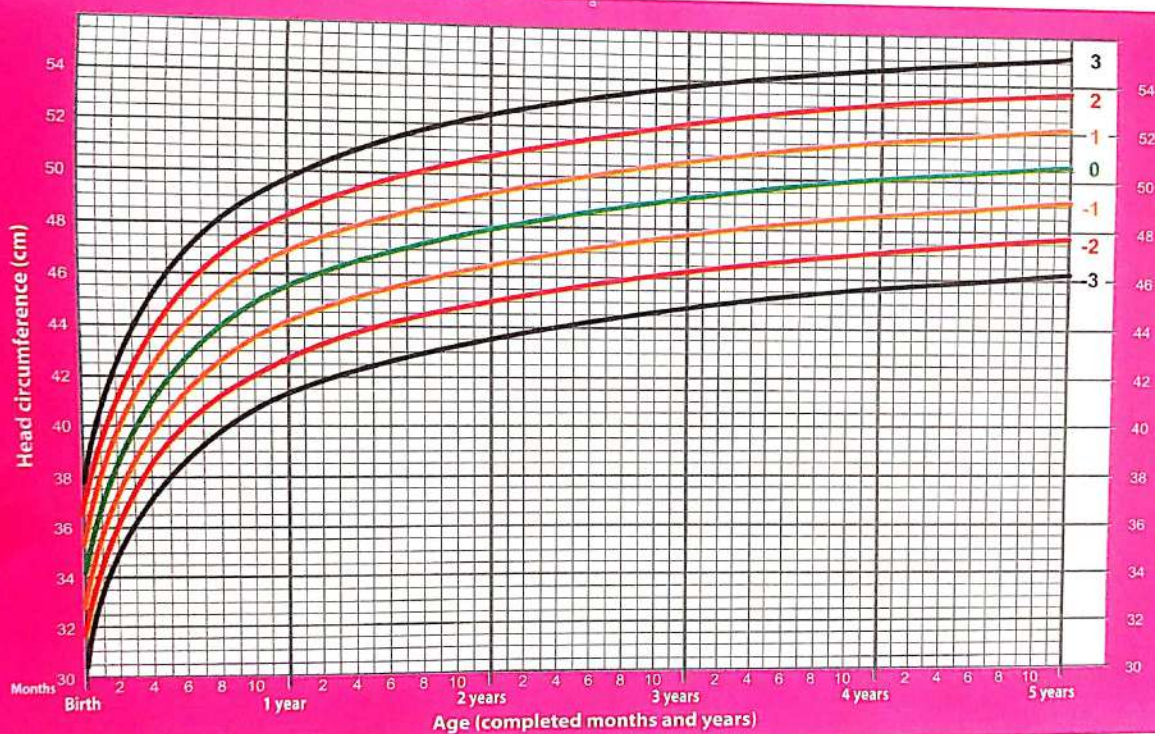
Z-SCORE	GROWTH INDICATORS			
	Length/Height - for - Age	Weight - for - Age	Weight - for - Length/Height	BMI - for - Age
Above 3	See note 1	See note 2	Obese	Obese
Above 2			Overweight	Overweight
Above 1			Possible risk of overweight (See note 3)	Possible risk of overweight (See note 3)
0 (Median)				
Below - 1				
Below - 2	Stunted (See note 4)	Underweight	Wasted	Wasted
Below - 3	Severely stunted (See note 4)	Severely Underweight (See note 5)	Severely wasted	Severely wasted

Notes:

1. A child in this range is very tall. Tallness is rarely a problem, unless it is so excessive that it may indicate an endocrine disorder such as a growth-hormone-producing tumor. Refer a child in this range of assessment if you suspect an endocrine disorder (e.g. if parents of normal height have a child who is excessively tall for his or her age).
2. A child whose weight-for-age falls in this range may have a growth problem, but this is better assessed from weight-for-length/height or BMI-for-age.
3. A plotted point above 1 shows possible risk. A trend towards the 2 z-score line shows definite risk.
4. It is possible for a stunted or severely stunted child to become overweight.
5. This is referred to as very low weight in IMCI training modules. (Integrated Management of Childhood Illness, In-service training. WHO, Geneva, 1977.)

Head circumference-for-age GIRLS

Birth to 5 years (z-scores)



WHO Child Growth Standards

Figure 4. Head Circumference for Age for Girls

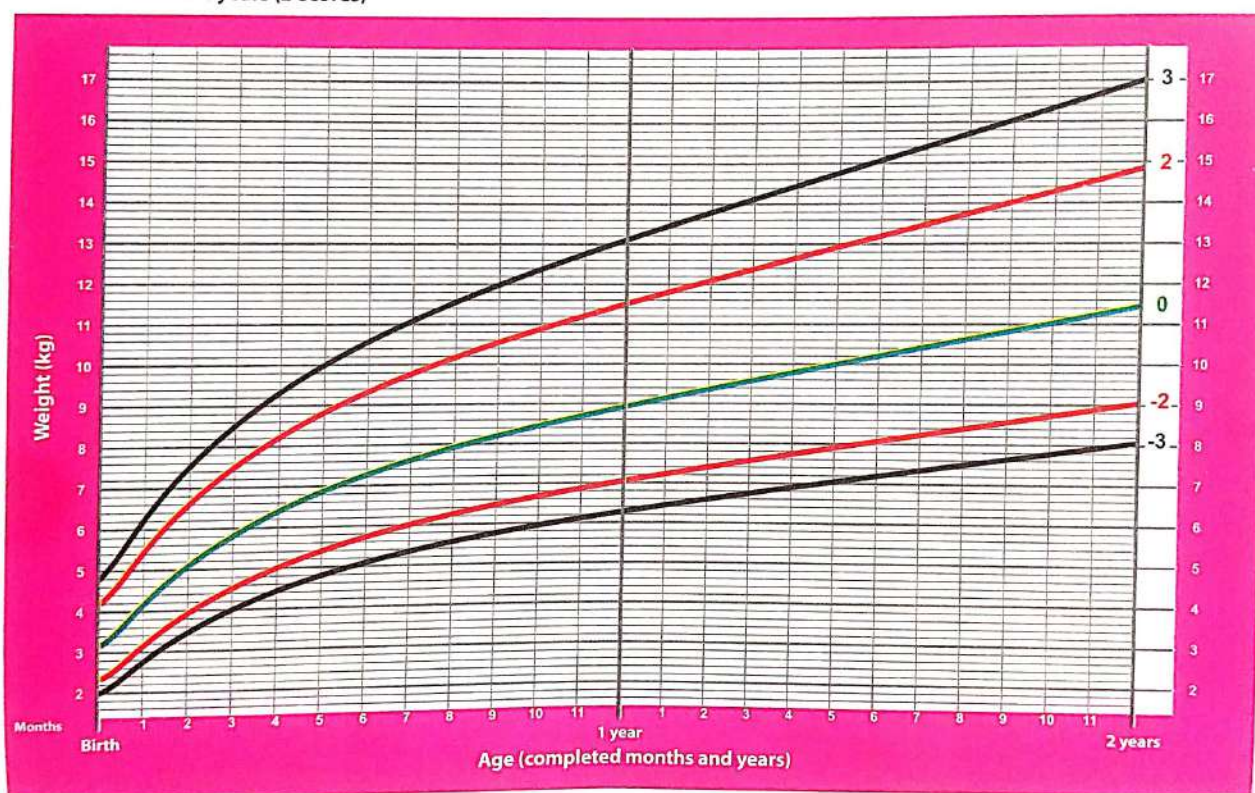
FIGURES

Weight-for-age GIRLS

Birth to 2 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 5. Weight for Age for Girls: Birth to 2 years

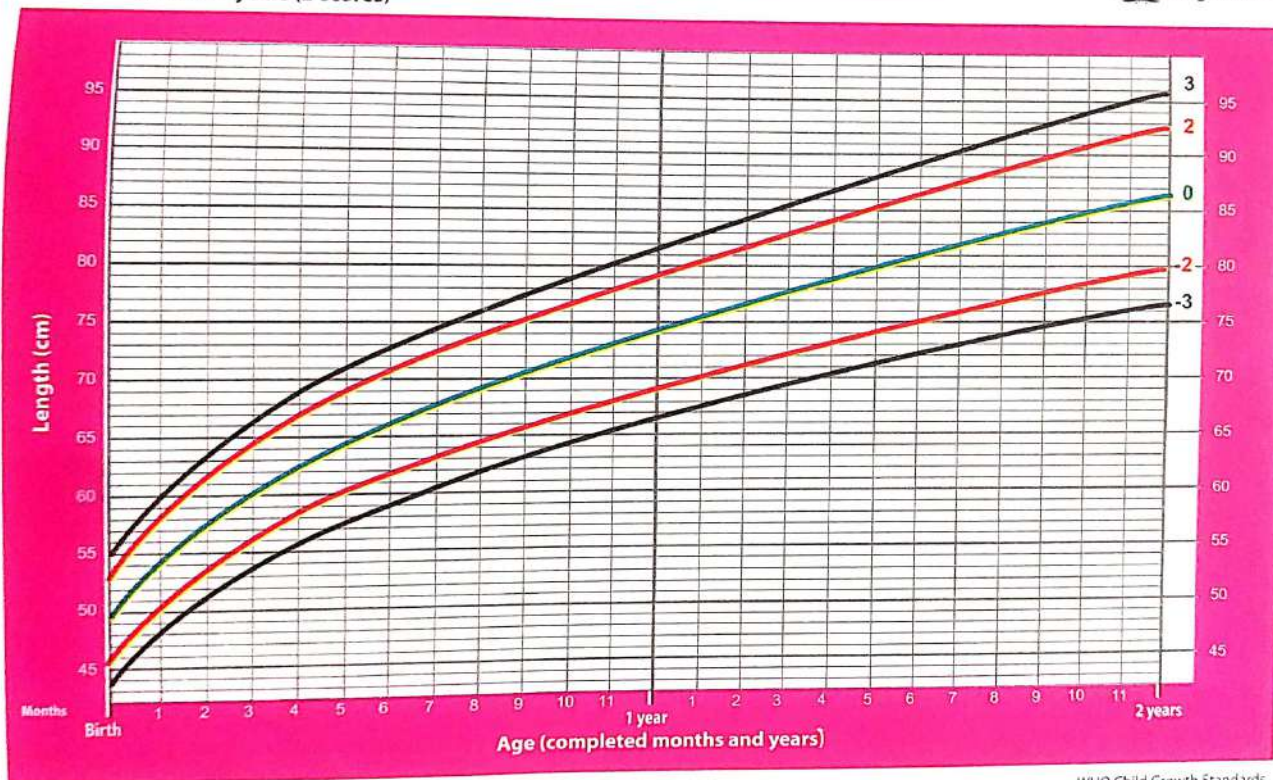
FIGURES

Length-for-age GIRLS

Birth to 2 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 6. Length for Age for Girls: Birth to 2 years

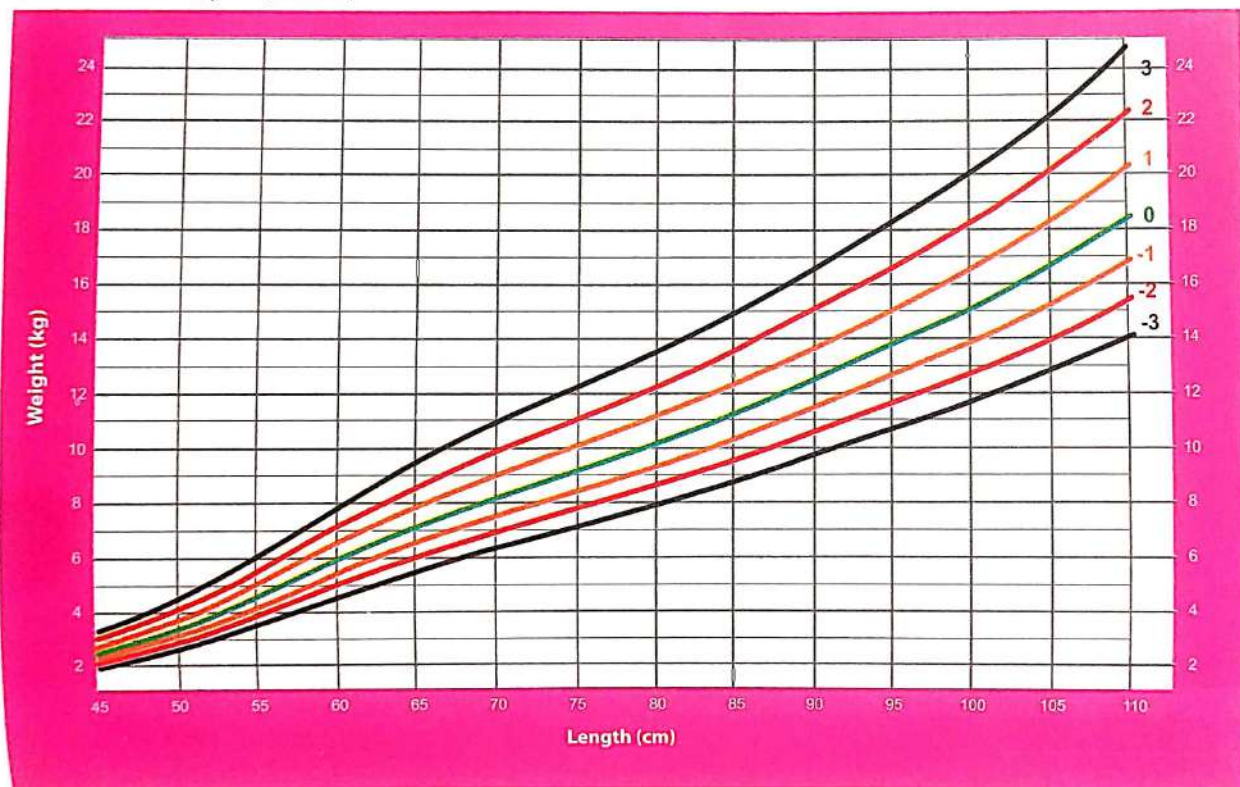
FIGURES

Weight-for-length GIRLS

Birth to 2 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 7. Weight for Length for Girls: Birth to 2 years

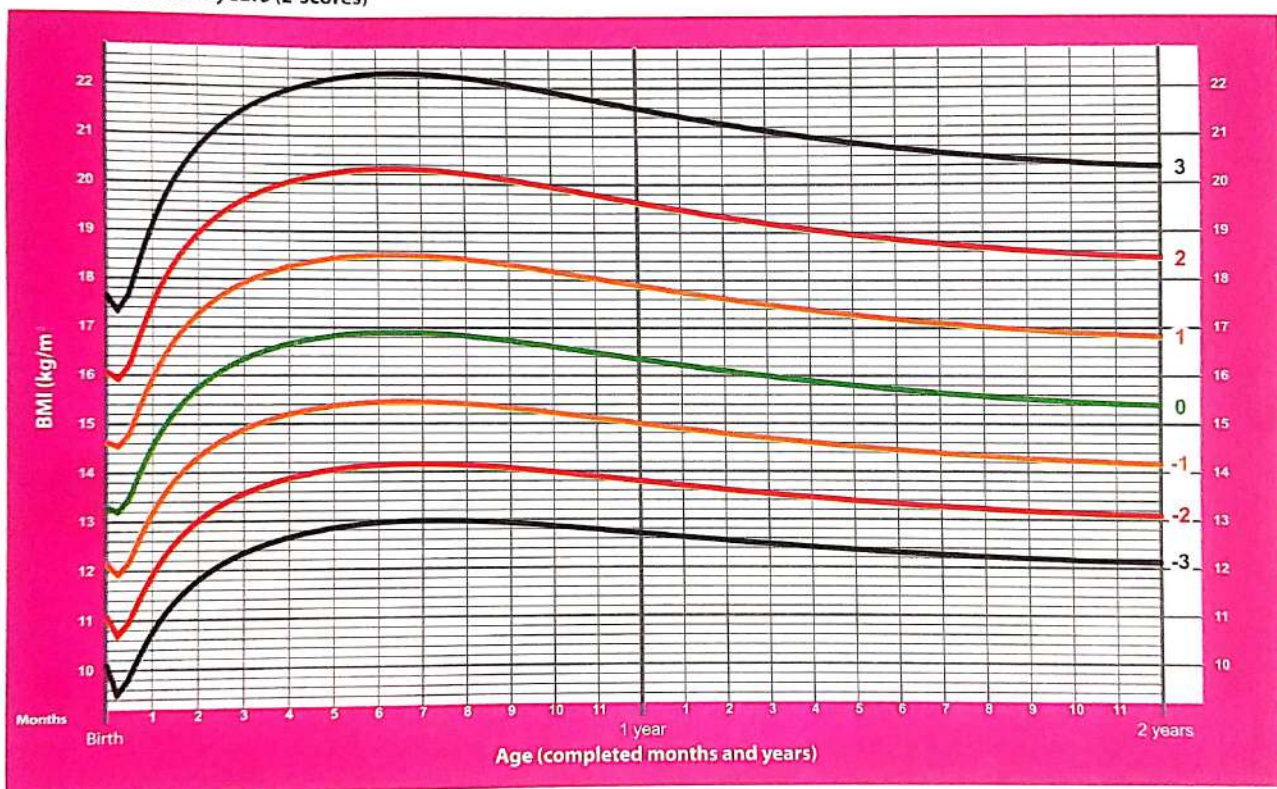
FIGURES

BMI-for-age GIRLS

Birth to 2 years (z-scores)



World Health Organization



WHO Child Growth Standards

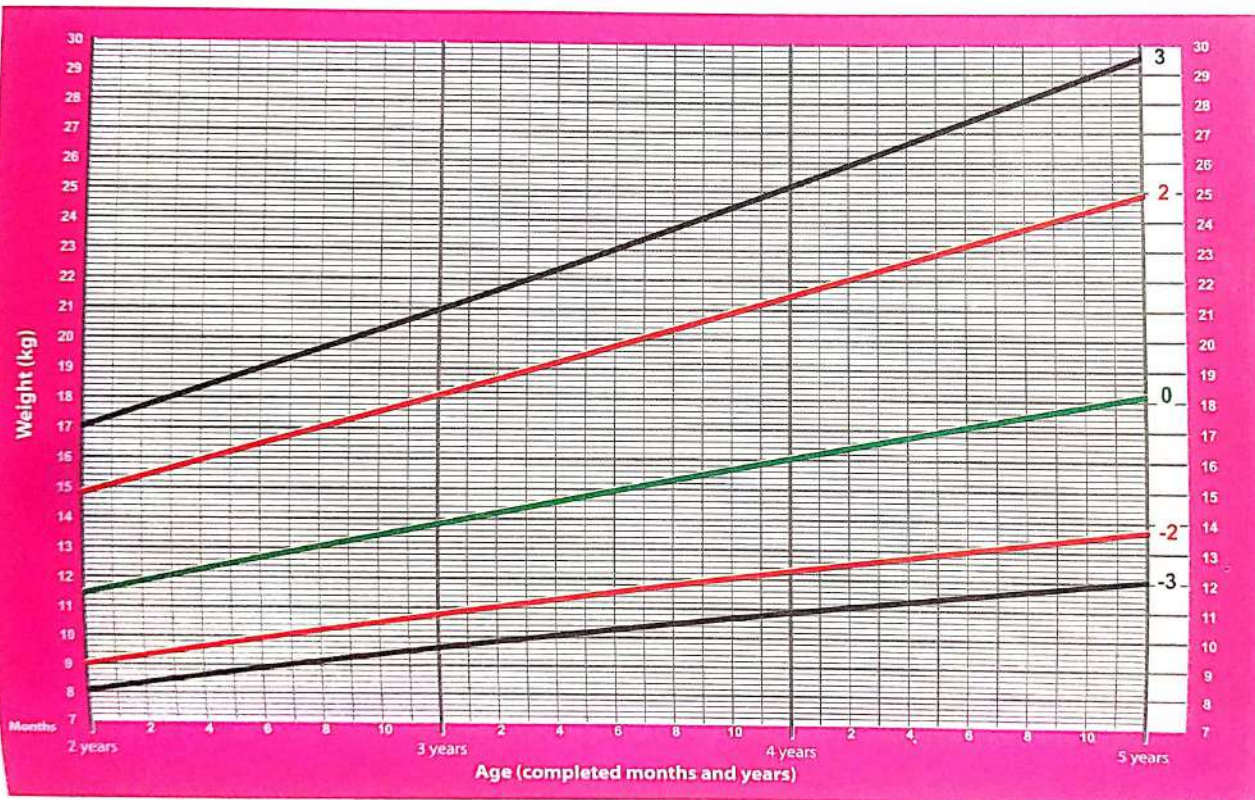
Figure 8. BMI for Age for Girls: Birth to 2 years

Weight-for-age GIRLS

2 to 5 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 9. Weight for Age for Girls: 2 to 5 years

FIGURES

Height-for-age GIRLS

2 to 5 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 10. Height for Age for Girls: 2 to 5 years

FIGURES



Philippine Pediatric Society, Inc.

Recommendations for Preventive Pediatric Health Care 2018

N.B. The recommendations in this table and optional procedures are described in detail in the ANNOTATIONS.

	INFANCY ¹																	EARLY CHILDHOOD ¹			MIDDLE CHILDHOOD ¹		ADOLESCENCE ¹									
	Pre Natal ¹	at birth ²	2-4 days ³	1 mo	2 mos	4 mos	6 mos	9 mos	12 mos	15 mos	18 mos	24 mos	3Y	4Y	5Y	6Y	8Y	10Y	11Y	12Y	13Y	14Y	15Y	16Y	17Y	18Y	19Y					
PRENATAL VISIT & COUNSELING	●																															
HISTORY: INITIAL/INTERVAL ⁴	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
DEVELOPMENTAL SURVEILLANCE ⁵	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
SCREENING FOR ATOPY ⁶	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
PHYSICAL EXAMINATION ⁴	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
MEASUREMENTS																																
A. WEIGHT & LENGTH/ HEIGHT ¹¹	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
B. HEAD CIRCUMFERENCE ¹²	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
C. BLOOD PRESSURE ¹³		★	★	★	★	★	★	★	★	★	★	★																				
GENERAL PROCEDURES ¹⁴																																
A. SCREENING FOR INBORN ERRORS OF METABOLISM ¹⁵		●																														
B. SCREENING FOR HEARING IMPAIRMENT ¹⁶		●	→																													
C. SCREENING FOR EYE/ VISUAL DEFECTS ¹⁷		●					●						●	→																		
D. IMMUNIZATION ¹⁸	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
E. IRON SUPPLEMENTATION ¹⁹							←	→	→	→	→	→			→			←	←	←	←	←	←	←	←	←	←	←	←			
F. VITAMIN A SUPPLEMENTATION ²⁰							←	→	→	→	→	→			→																	
G. DEWORMING ²¹									●									●														
ANTICIPATORY GUIDANCE ²²																																
A. FIRST DENTAL VISIT ²³							←	→	→	→	→	→																				
B. NUTRITION COUNSELING ²⁴	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
C. PHYSICAL ACTIVITY ²⁵	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
D. INJURY & POISONING PREVENTION ²⁶	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
E. PREVENTION OF CHILD MALTREATMENT ²⁷	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
F. COUNSELING ON EXPOSURE ON LEAD AND OTHER TOXICANTS ²⁸	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●			
PROCEDURES FOR PATIENTS AT RISK																																
A. CBC ²⁹							←	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→	→			
B. URINALYSIS ³⁰																																
C. WORK-UP FOR SEXUALLY ACTIVE ADOLESCENTS ³¹																																
D. MANTOUX TEST ³²																																

●

DONE AT EVERY VISIT OR AT RECOMMENDED INTERVALS
FOLLOW-UPS DURING THE 1ST YEAR OF LIFE DONE ON A MONTHLY BASIS

→

AGE RANGE A PROCEDURE IS DONE AT LEAST ONCE OR AT RECOMMENDED INTERVALS

→●→

AGE RANGE A PROCEDURE IS DONE AT LEAST ONCE FOR ALL CHILDREN
THE DOT INDICATES THE PREFERRED AGE

★

DONE FOR THOSE AT HIGH RISK ONLY

↔

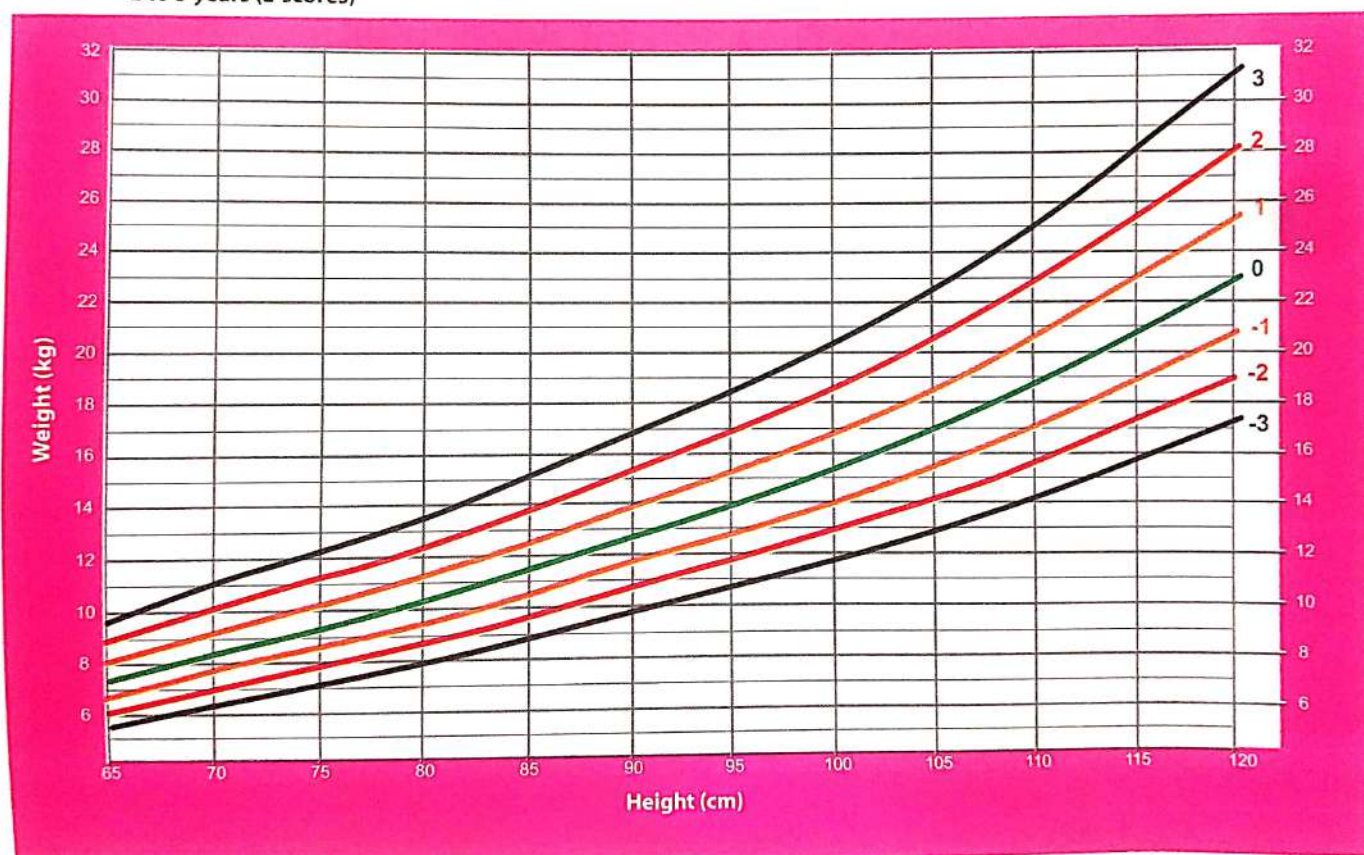
AGE RANGE A PROCEDURE IS DONE AT LEAST ONCE FOR THOSE AT HIGH RISK

● DONE AT EVERY VISIT OR AT RECOMMENDED INTERVALS
FOLLOW-UPS DURING THE 1ST YEAR OF LIFE DONE ON A MONTHLY BASIS
←→ AGE RANGE A PROCEDURE IS DONE AT LEAST ONCE OR AT RECOMMENDED INTERVALS
←→ AGE RANGE A PROCEDURE IS DONE AT LEAST ONCE FOR ALL CHILDREN
★ DONE FOR THOSE AT HIGH RISK ONLY
←→★ AGE RANGE A PROCEDURE IS DONE AT LEAST ONCE FOR THOSE AT HIGH RISK

Disclaimer: The recommendations contained in this document are intended to GUIDE practitioners in the conduct of anticipatory care and guidance and periodic health examinations of infants, children and adolescents. In no way should the recommendations be regarded as absolute rules, since nuances and peculiarities in individual cases or particular communities may entail differences in the specific approach. In the end, the recommendations should supplement and not replace sound clinical judgement made on a case to case basis.

Weight-for-height GIRLS

2 to 5 years (z-scores)



WHO Child Growth Standards

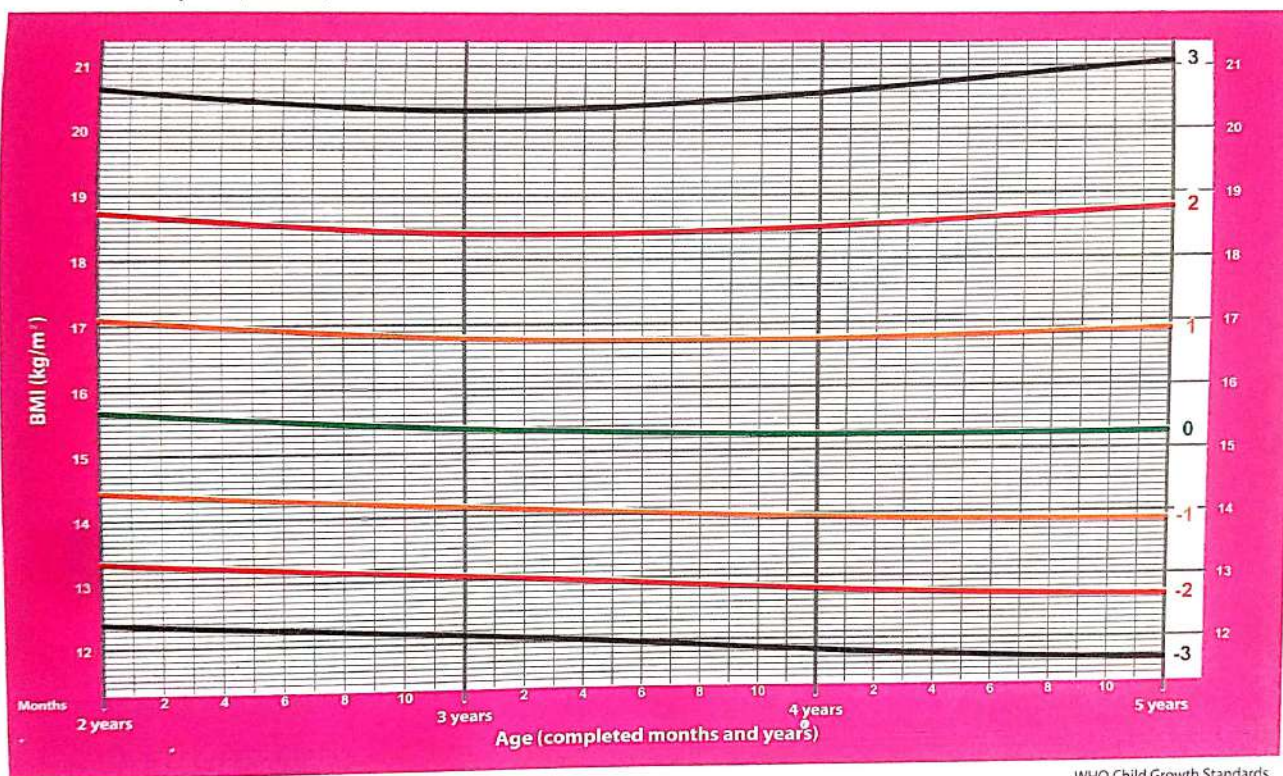
Figure 11. Weight for Height for Girls: 2 to 5 years

BMI-for-age GIRLS

2 to 5 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 12. BMI for Age for Girls: 2 to 5 years

FIGURES

Weight-for-age GIRLS

5 to 10 years (z-scores)



2007 WHO Reference

Figure 13. Weight for Age for Girls: 5 to 10 years

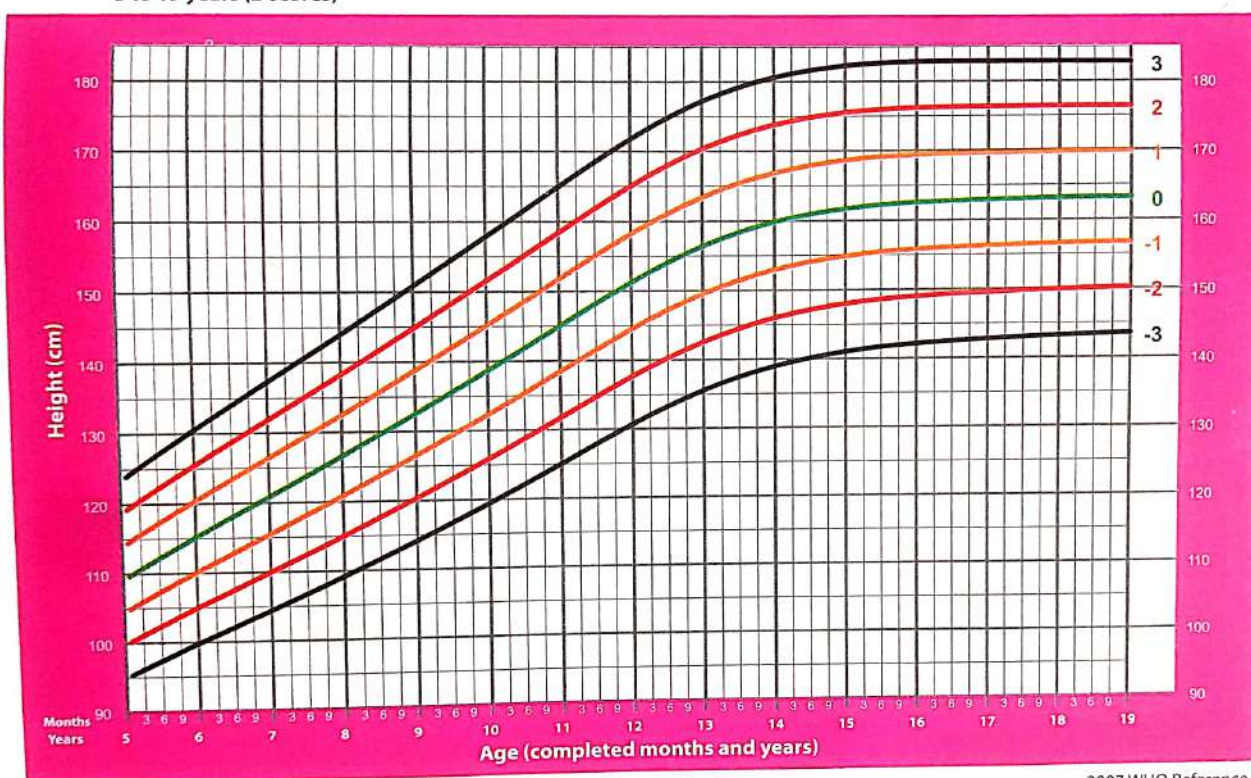
FIGURES

Height-for-age GIRLS

5 to 19 years (z-scores)



World Health Organization



2007 WHO Reference

Figure 14. Height for Age for Girls: 5 to 19 years

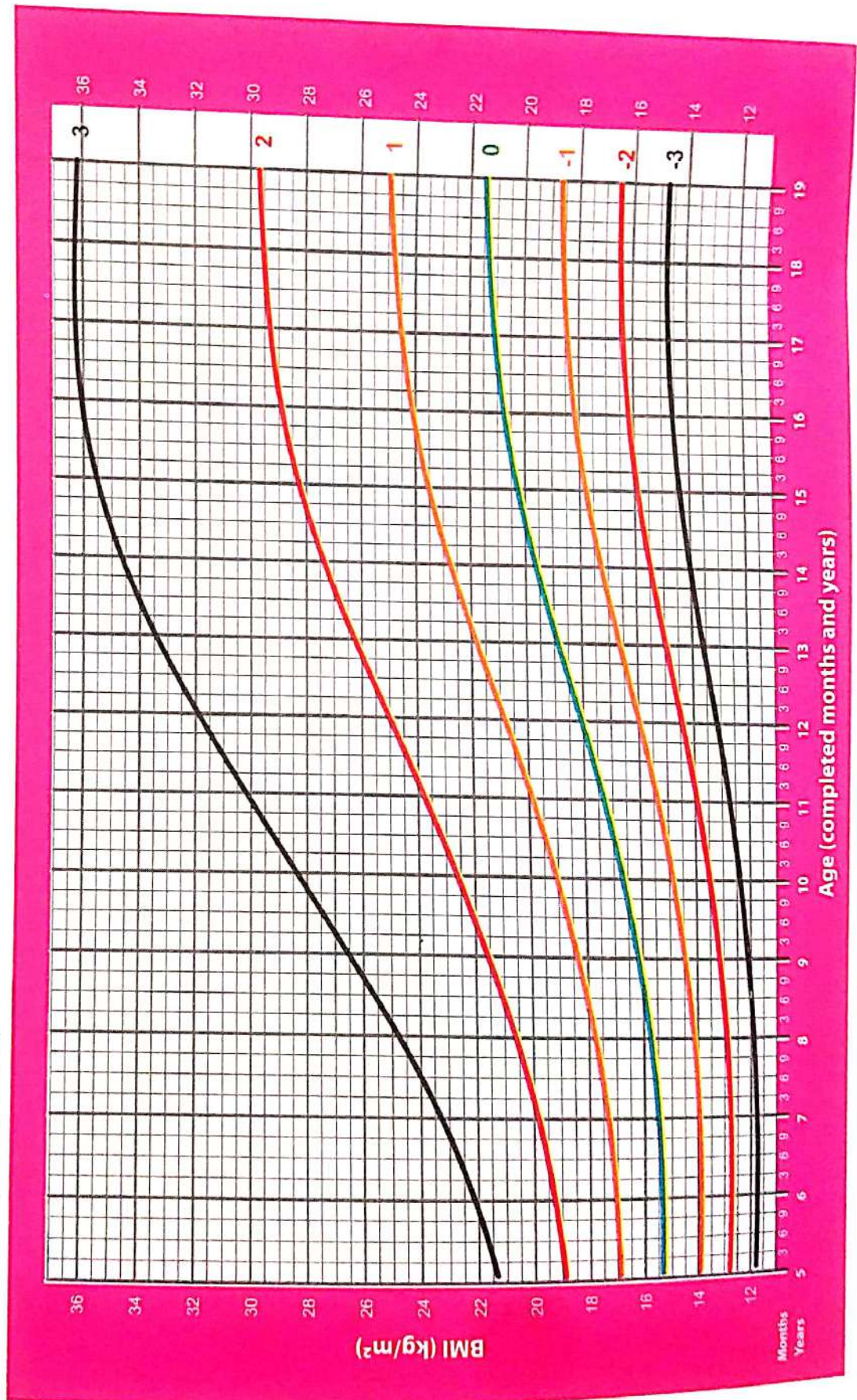
FIGURES

FIGURES

Figure 15. BMI for Age for Girls: 5 to 19 years

BMI-for-age GIRLS

5 to 19 years (z-scores)



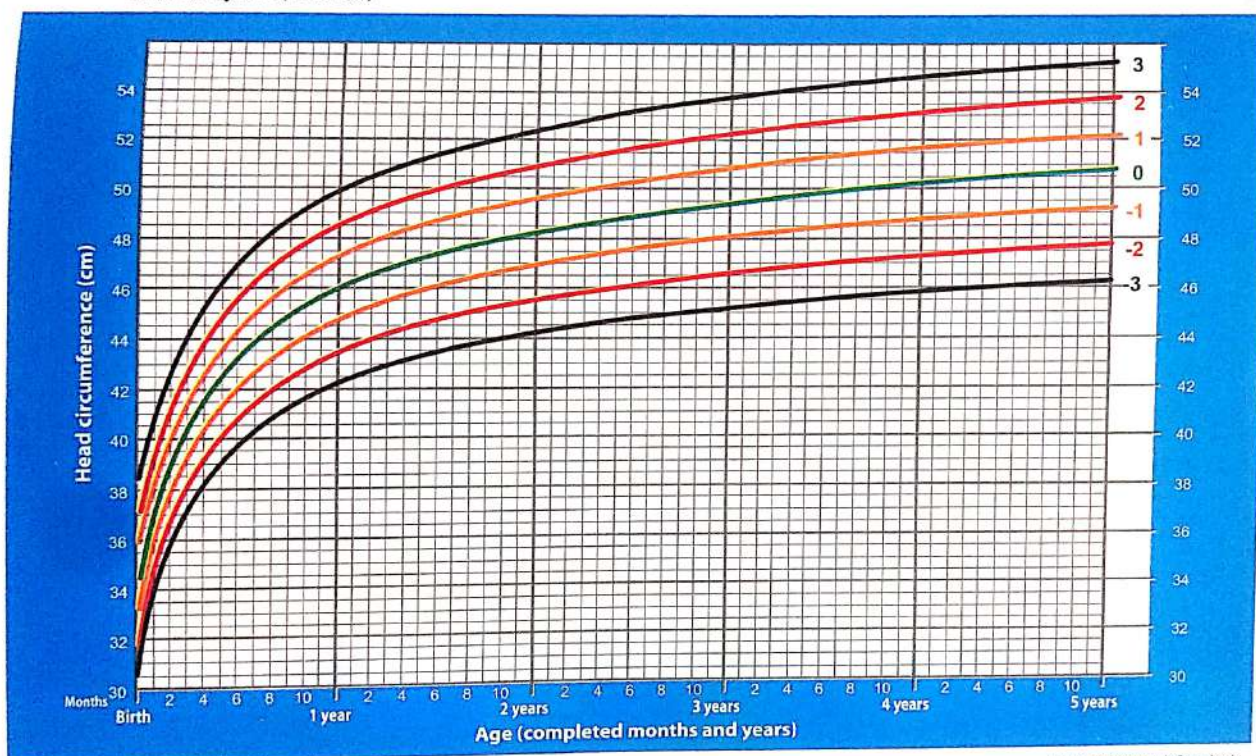
2007 WHO Reference

Head circumference-for-age BOYS

Birth to 5 years (z-scores)



World Health Organization



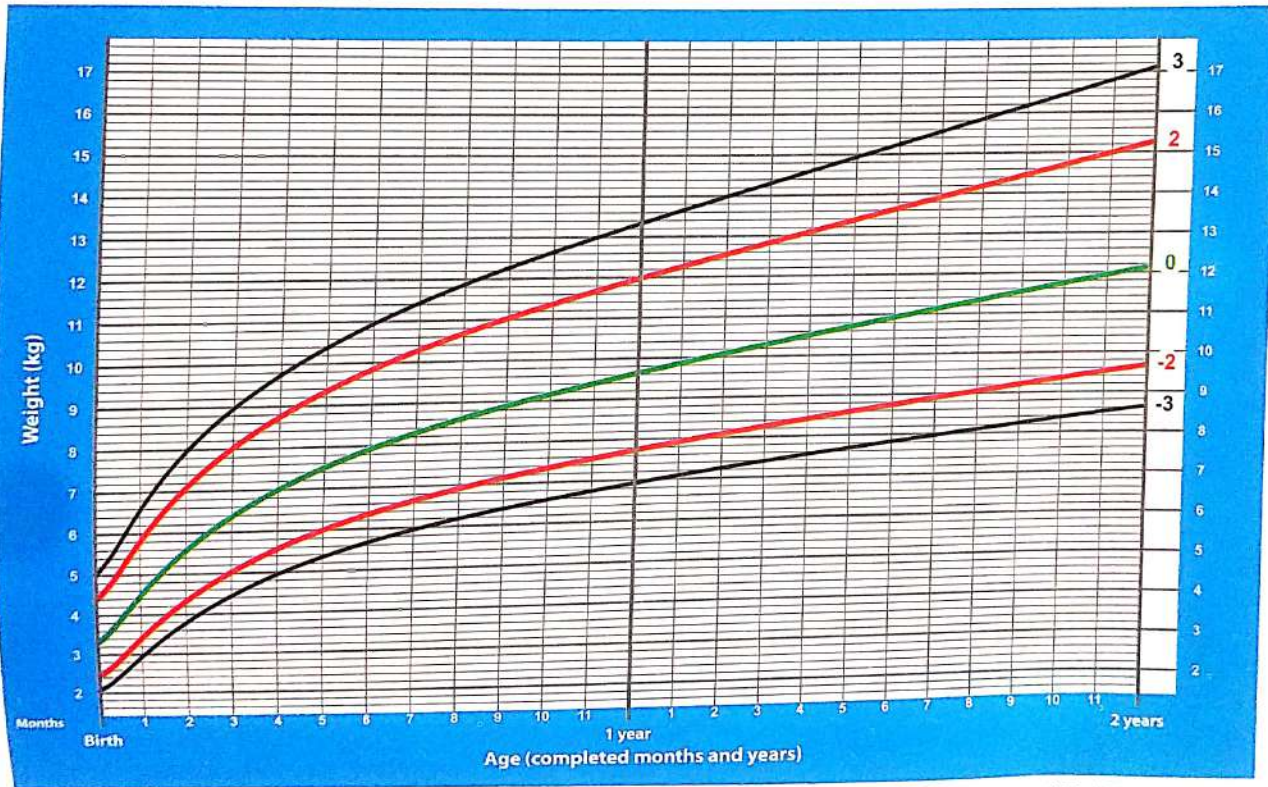
WHO Child Growth Standards

Figure 16. Head Circumference for Age for Boys

FIGURES

Weight-for-age BOYS

Birth to 2 years (z-scores)



WHO Child Growth Standards

Figure 17. Weight for Age for Boys: Birth to 2 years

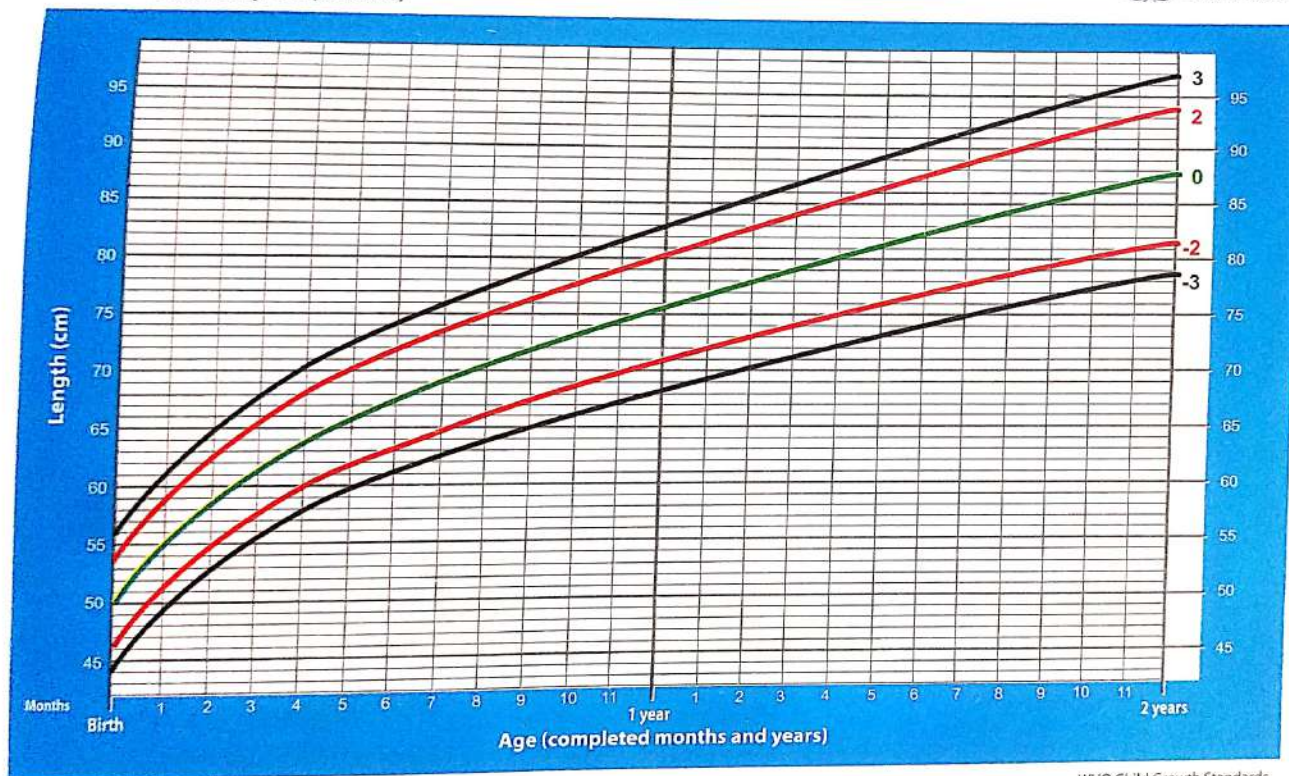
FIGURES

Length-for-age BOYS

Birth to 2 years (z-scores)



World Health Organization

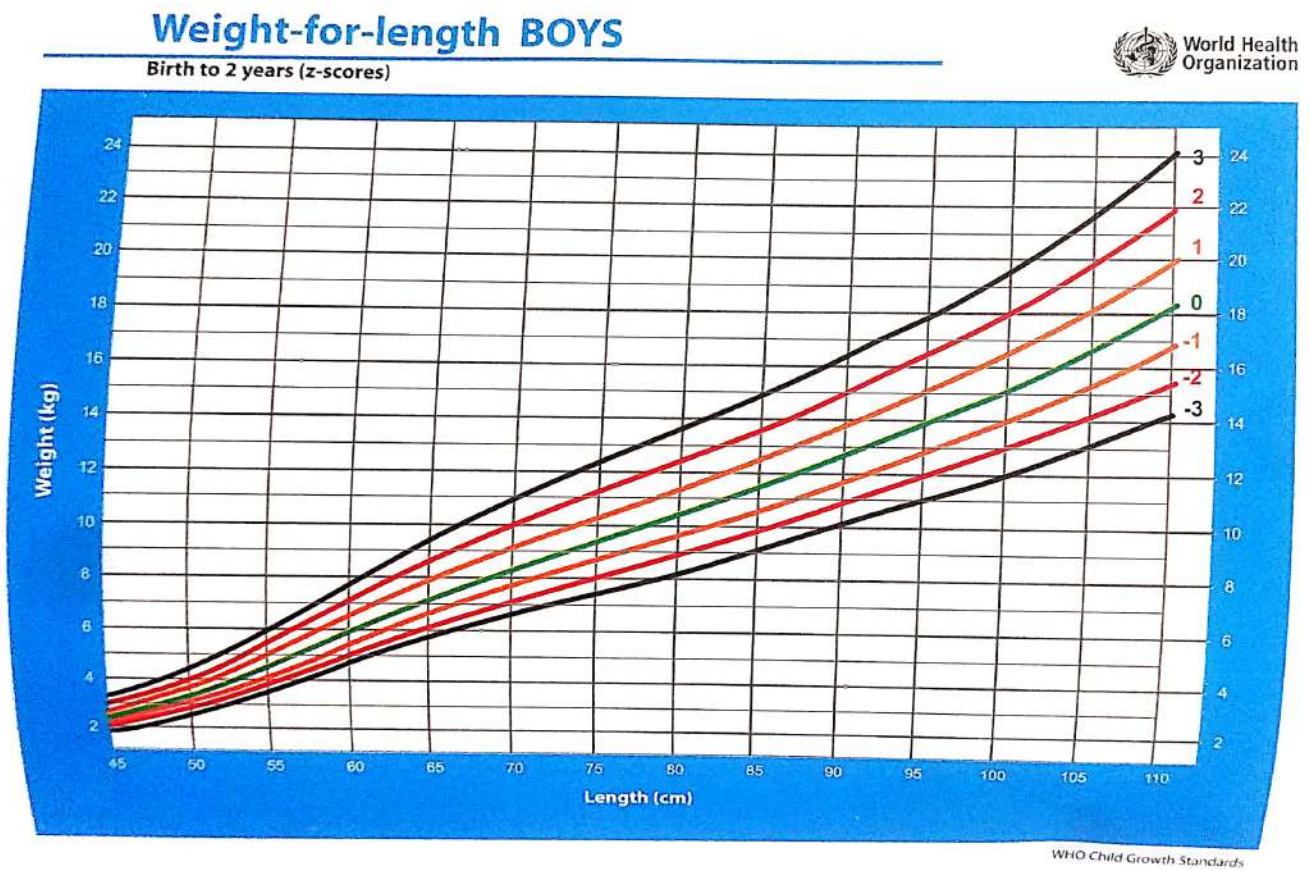


WHO Child Growth Standards

Figure 18. Length for Age for Boys: Birth to 2 years

FIGURES

Figure 19. Weight for Length for Boys: Birth to 2 years

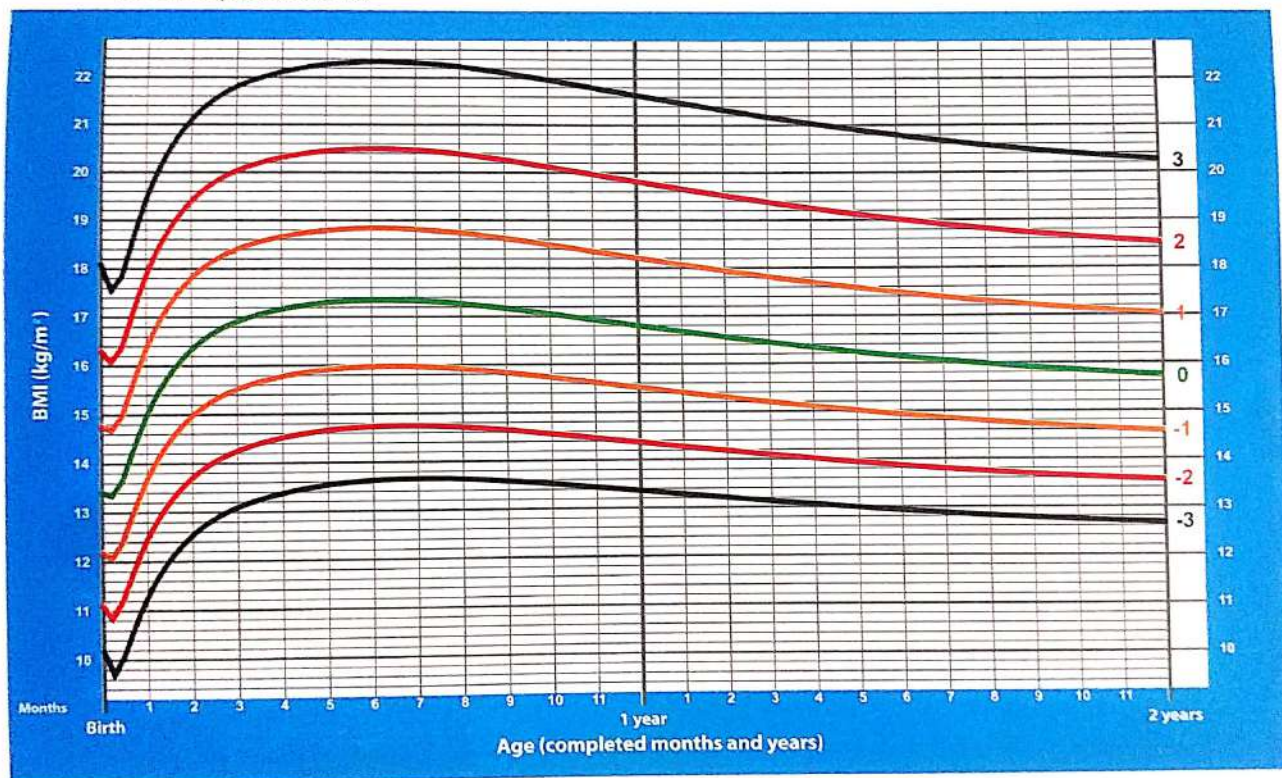


BMI-for-age BOYS

Birth to 2 years (z-scores)



World Health Organization



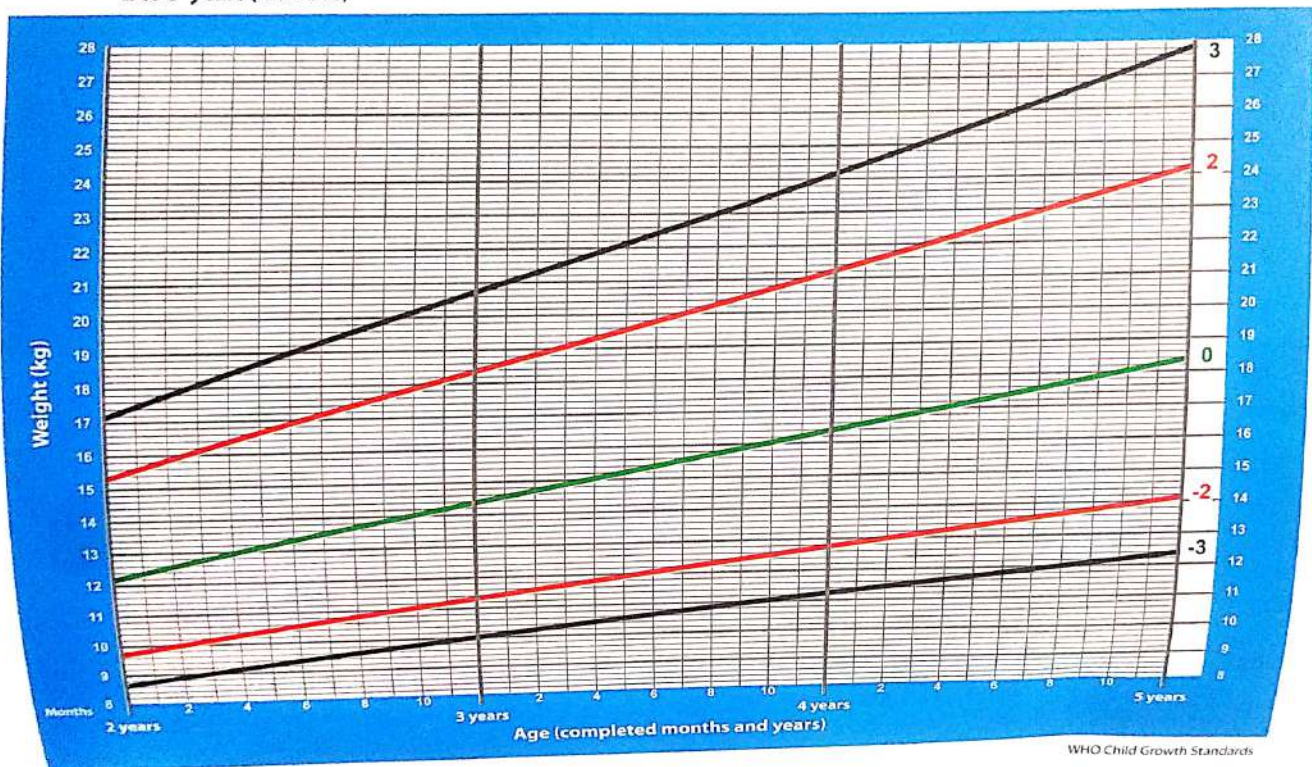
WHO Child Growth Standards

Figure 20. BMI for Age for Boys: Birth to 2 years

FIGURES

Weight-for-age BOYS

2 to 5 years (z-scores)



WHO Child Growth Standards

Figure 21. Weight for Age for Boys: 2 to 5 years

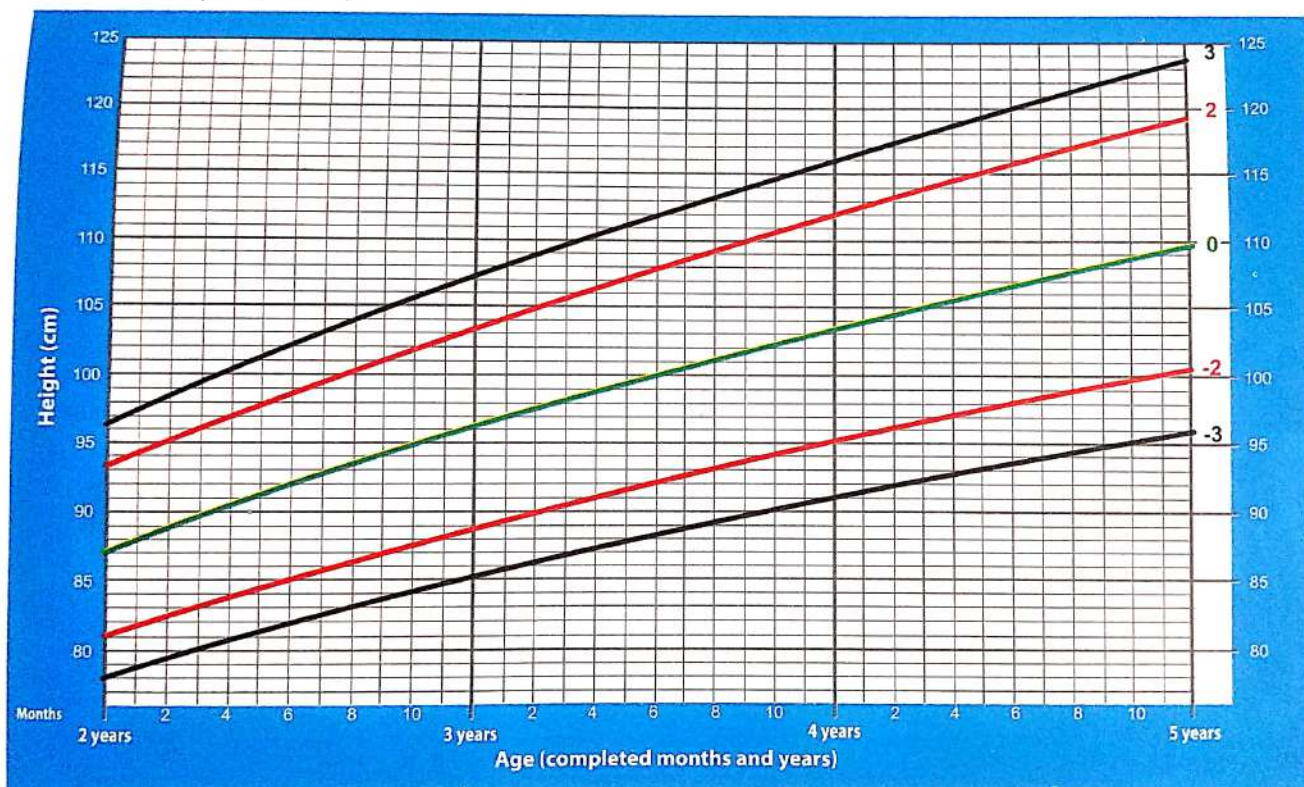
FIGURES

Height-for-age BOYS

2 to 5 years (z-scores)



World Health Organization



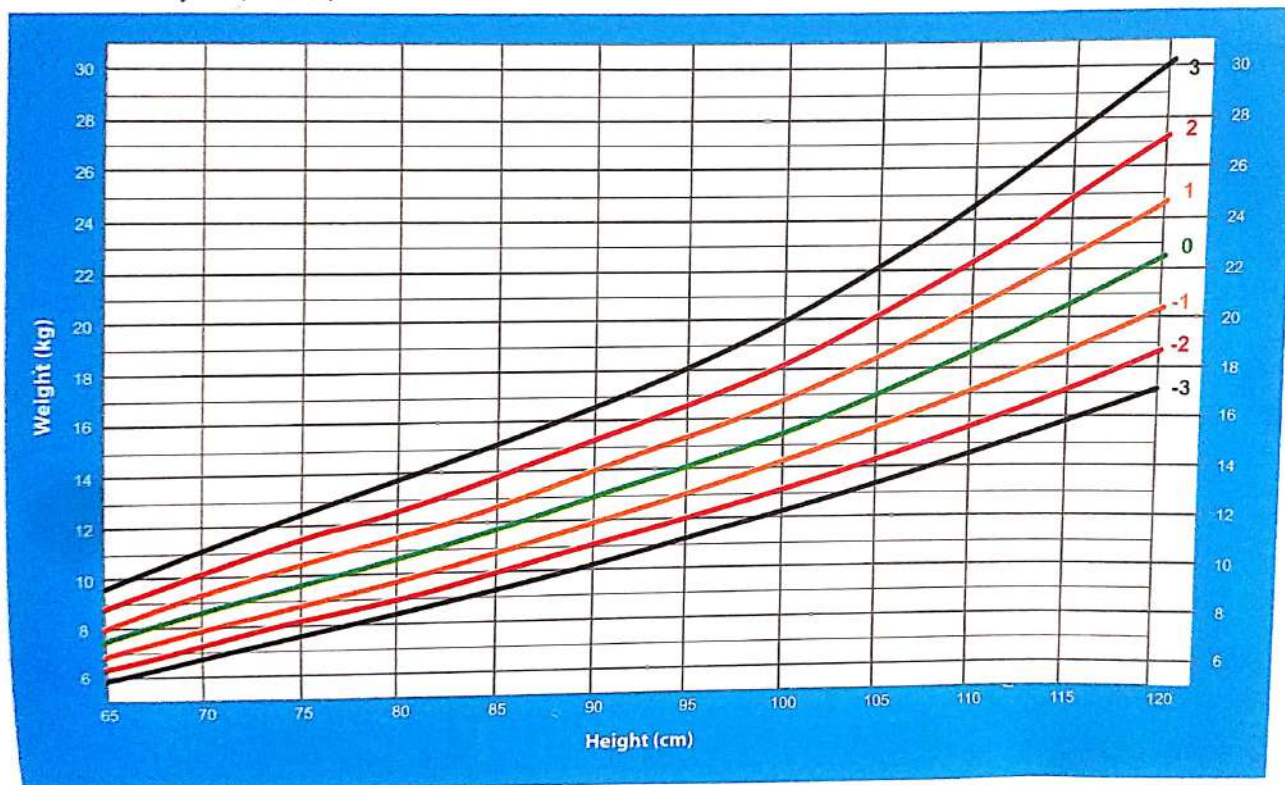
WHO Child Growth Standards

Figure 22. Height for Age for Boys: 2 to 5 years

FIGURES

Weight-for-height BOYS

2 to 5 years (z-scores)



WHO Child Growth Standards

Figure 23. Weight for Height for Boys: 2 to 5 years

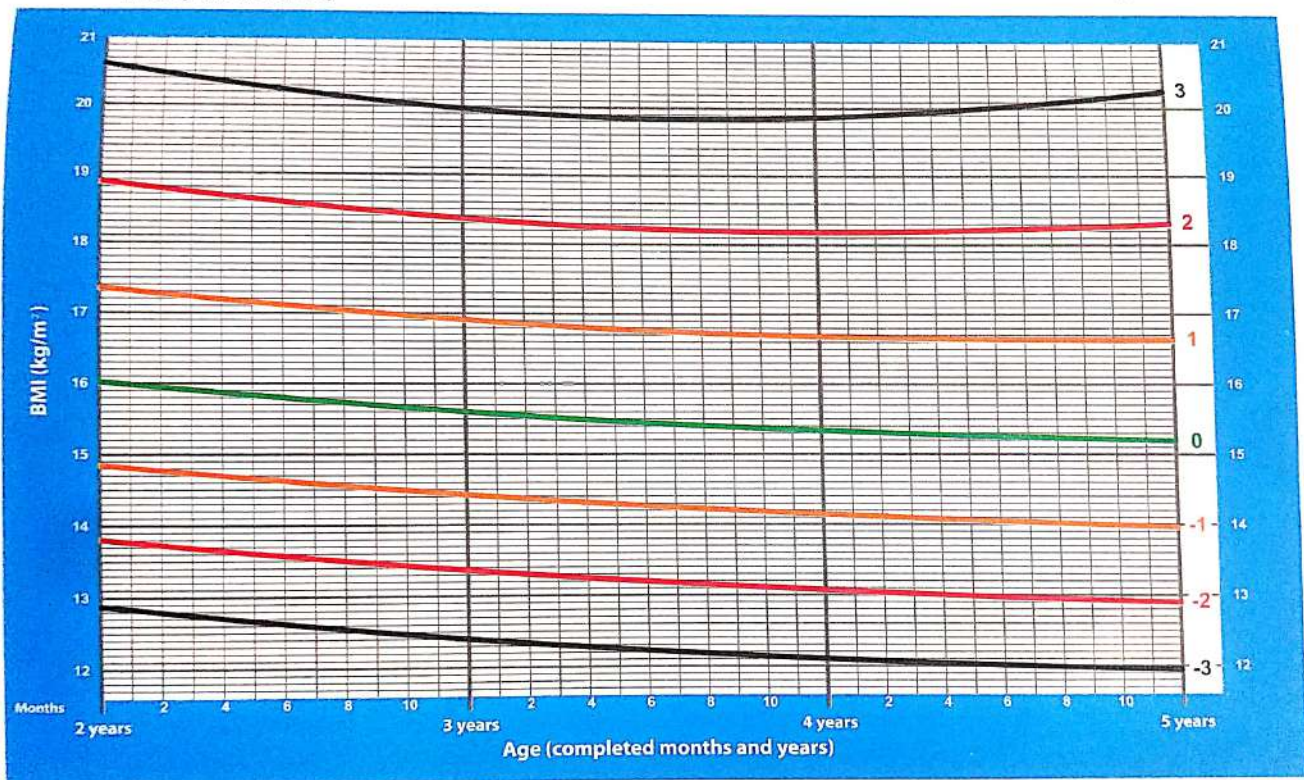
FIGURES

BMI-for-age BOYS

2 to 5 years (z-scores)



World Health Organization



WHO Child Growth Standards

Figure 24. BMI for Age for Boys: 2 to 5 years

FIGURES

Weight-for-age BOYS

5 to 10 years (z-scores)



World Health Organization

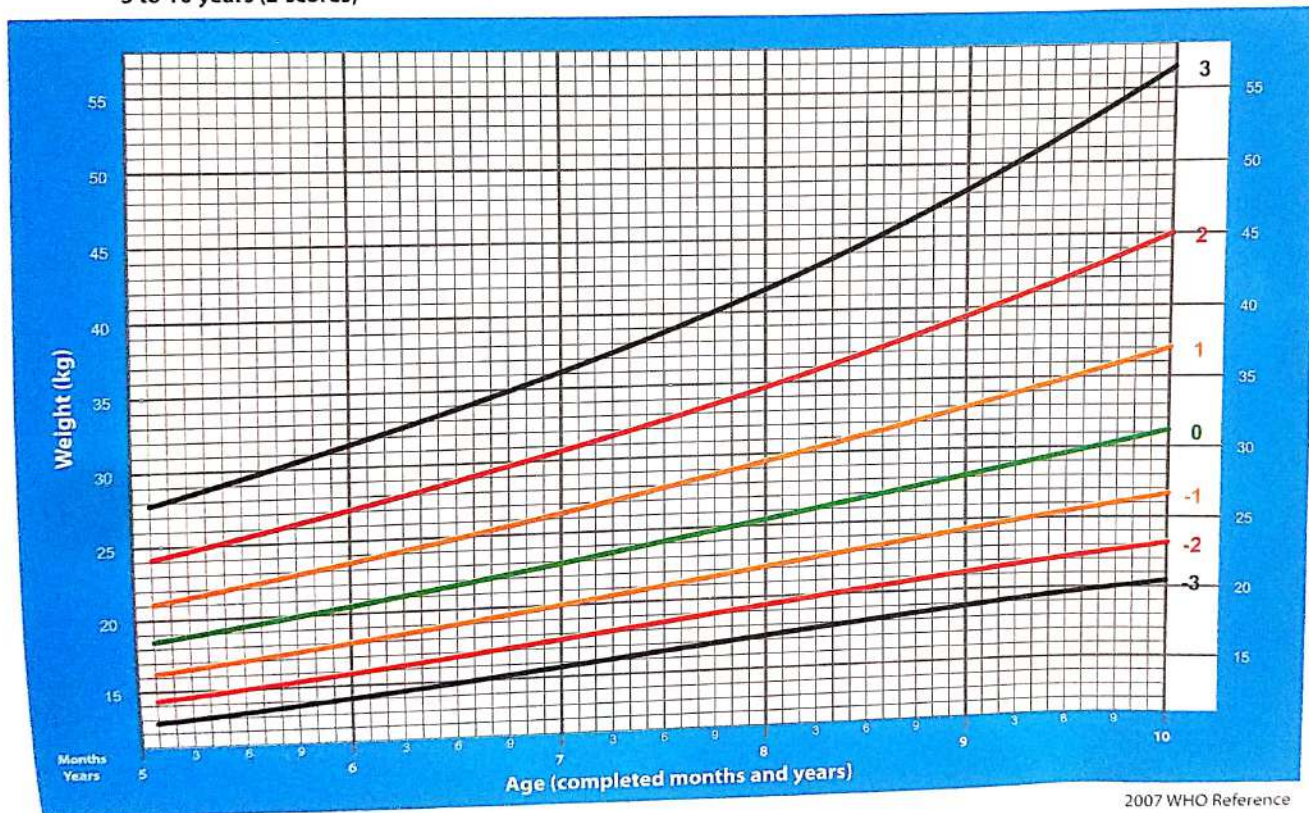


Figure 25. Weight for Age for Boys: 5 to 10 years

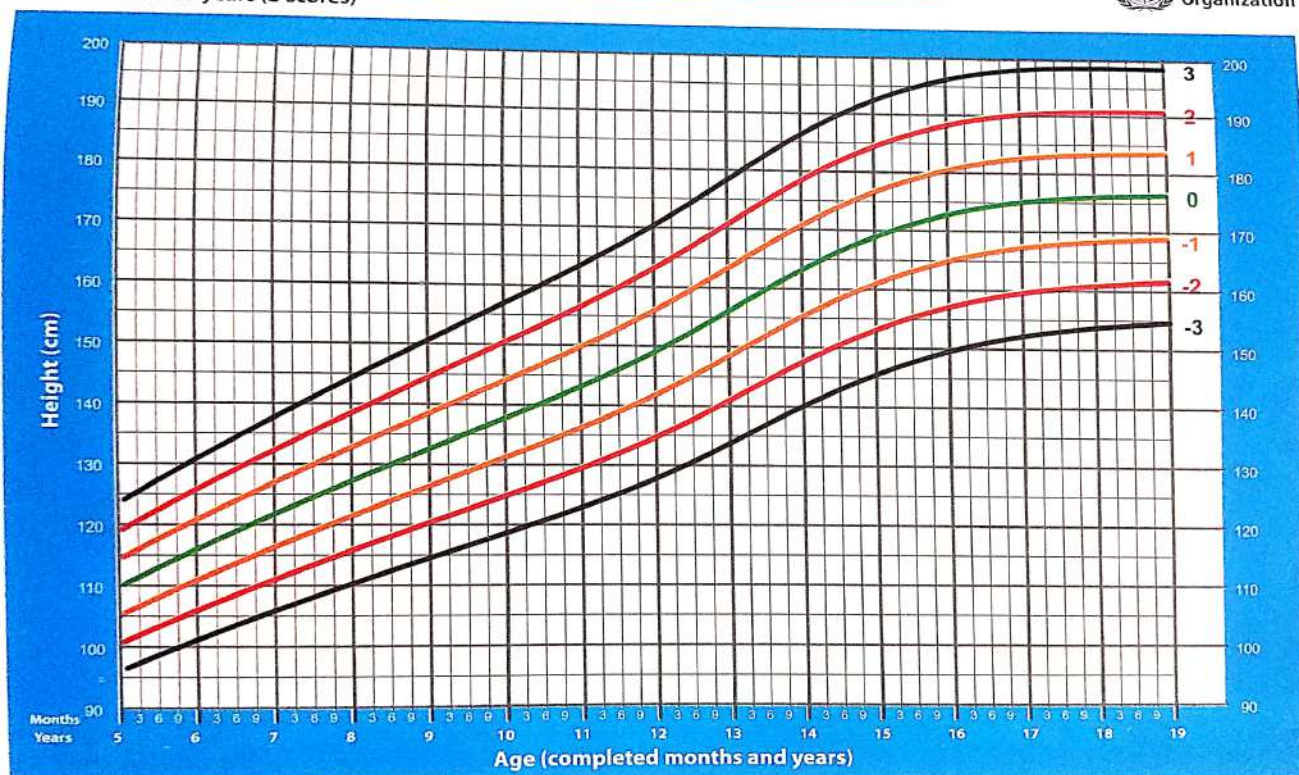
FIGURES

Height-for-age BOYS

5 to 19 years (z-scores)



World Health Organization



2007 WHO Reference

Figure 26. Height for Age for Boys: 5 to 19 years

FIGURES

BMI-for-age BOYS

5 to 19 years (z-scores)



World Health Organization

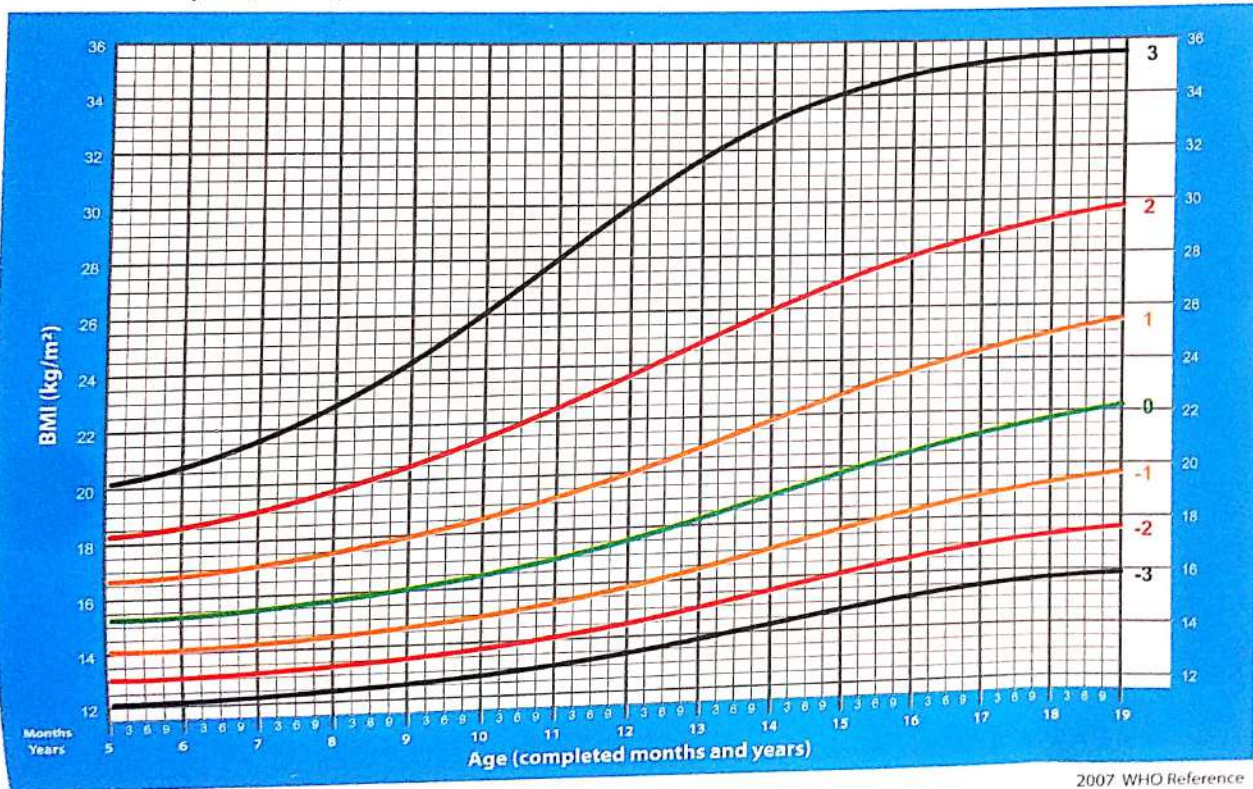


Figure 27. BMI for Age for Boys: 5 to 19 years

FIGURES

FIGURES

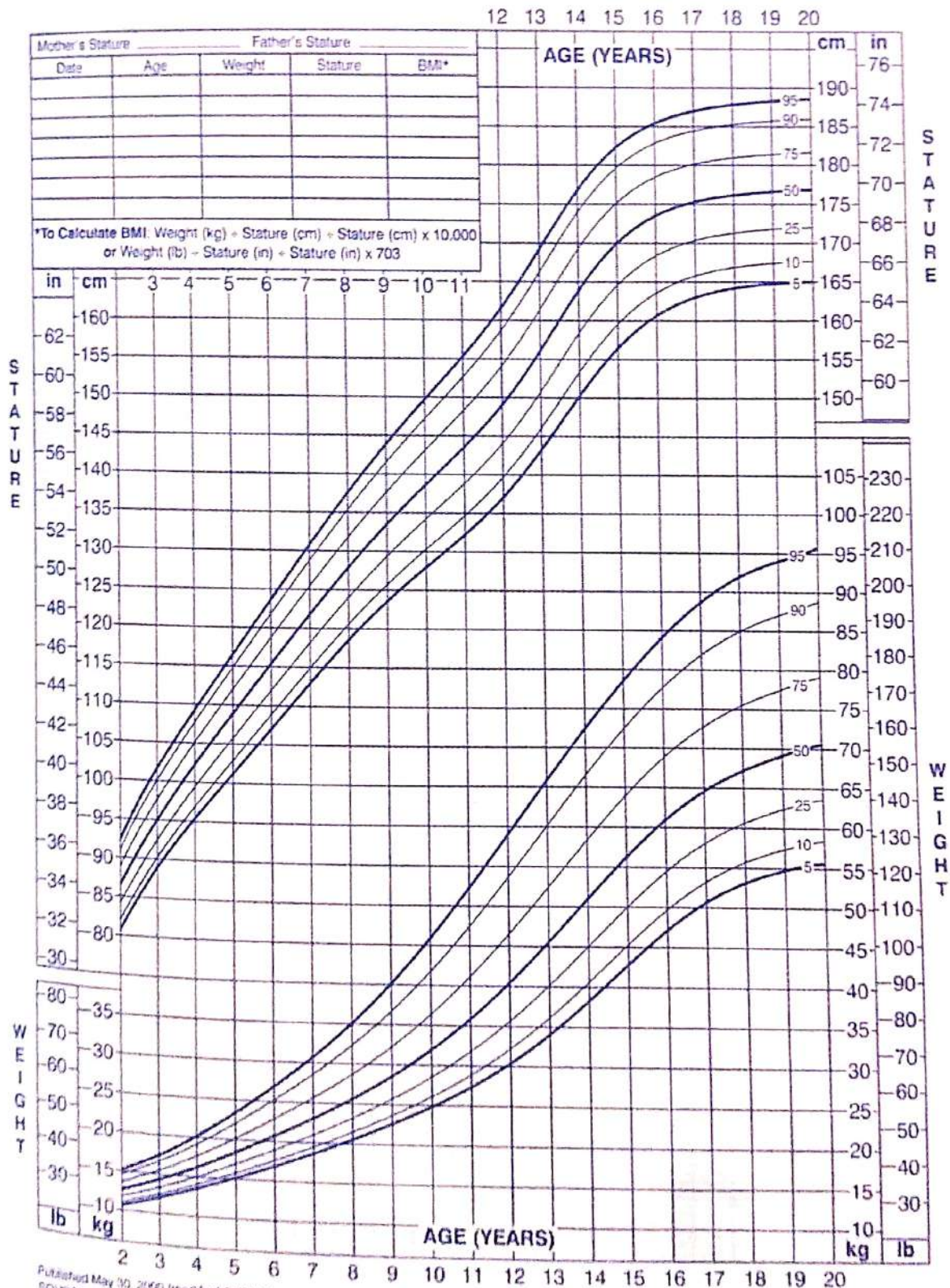
Figure 28. US CDC-NCHS Growth Chart for Boys (Stature of Age: to be used in locating Blood Pressure Percentile of children and adolescents)

2 to 20 years: Boys

Stature-for-age and Weight-for-age percentiles

NAME _____

RECORD # _____



Published May 30, 2000 (modified 11/21/00)
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000)
<http://www.cdc.gov/growthcharts>



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FIGURES

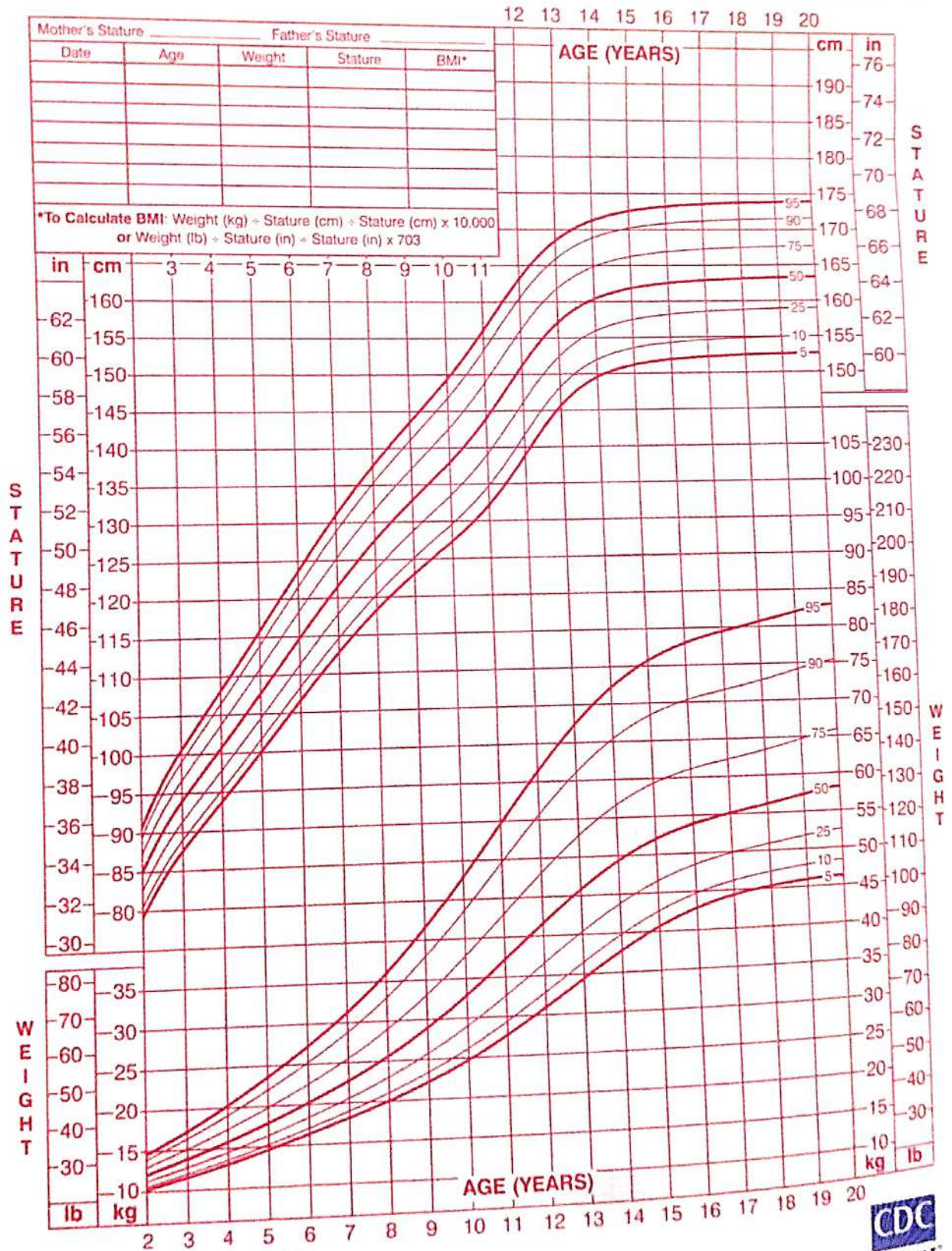
Figure 29. US CDC-NCHS Growth Chart for Girls (Stature of Age: to be used in locating Blood Pressure Percentile of children and adolescents)

2 to 20 years: Girls

Stature-for-age and Weight-for-age percentiles

NAME _____

RECORD # _____



Published May 30, 2000 (modified 11/21/00)
SOURCE: Developed by the National Center for Health Statistics in collaboration with
the National Center for Chronic Disease Prevention and Health Promotion (2000)
<http://www.cdc.gov/growthcharts>

CDC
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FIGURES

Figure 30. BP Levels for Boys by Age and Height Percentile

Age (y)	BP Percentile	SBP (mmHg)							DBP (mmHg)						
		Height Percentile or Measured Height							Height Percentile or Measured Height						
		5%	10%	25%	50%	75%	90%	95%	5%	10%	25%	50%	75%	90%	95%
1	Height (in)	30.4	30.8	31.6	32.4	33.3	34.1	34.6	30.4	30.8	31.6	32.4	33.3	34.1	34.6
	Height (cm)	77.2	78.3	80.2	82.4	84.6	86.7	87.9	77.2	78.3	80.2	82.4	84.6	86.7	87.9
	50th	85	85	86	86	87	88	88	40	40	40	41	41	42	42
	90th	98	99	99	100	100	101	101	52	52	53	53	54	54	54
	95th	102	102	103	103	104	105	105	54	54	55	55	56	57	57
	95th + 12 mmHg	114	114	115	115	116	117	117	66	66	67	67	68	69	69
2	Height (in)	33.9	34.4	35.3	36.3	37.3	38.2	38.8	33.9	34.4	35.3	36.3	37.3	38.2	38.8
	Height (cm)	86.1	87.4	89.6	92.1	94.7	97.1	98.5	86.1	87.4	89.6	92.1	94.7	97.1	98.5
	50th	87	87	88	89	89	90	91	43	43	44	44	45	46	46
	90th	100	100	101	102	103	103	104	55	55	56	56	57	58	58
	95th	104	105	105	106	107	107	108	57	58	58	59	60	61	61
	95th + 12 mmHg	116	117	117	118	119	119	120	69	70	70	71	72	73	73
3	Height (in)	36.4	37	37.9	39	40.1	41.1	41.7	36.4	37	37.9	39	40.1	41.1	41.7
	Height (cm)	92.5	93.9	96.3	99	101.8	104.3	105.8	92.5	93.9	96.3	99	101.8	104.3	105.8
	50th	88	89	89	90	91	92	92	45	46	46	47	48	49	49
	90th	101	102	102	103	104	105	105	58	58	59	59	60	61	61
	95th	106	106	107	107	108	109	109	60	61	61	62	63	64	64
	95th + 12 mmHg	118	118	119	119	120	121	121	72	73	73	74	75	76	76
4	Height (in)	38.8	39.4	40.5	41.7	42.9	43.9	44.5	38.8	39.4	40.5	41.7	42.9	43.9	44.5
	Height (cm)	98.5	100.2	102.9	105.9	108.9	111.5	113.2	98.5	100.2	102.9	105.9	108.9	111.5	113.2
	50th	90	90	91	92	93	94	94	48	49	49	50	51	52	52
	90th	102	103	104	105	105	106	107	60	61	62	62	63	64	64
	95th	107	107	108	108	109	110	110	63	64	65	66	67	67	68
	95th + 12 mmHg	119	119	120	120	121	122	122	75	76	77	78	79	79	80
5	Height (in)	41.1	41.8	43.0	44.3	45.5	46.7	47.4	41.1	41.8	43.0	44.3	45.5	46.7	47.4
	Height (cm)	104.4	106.2	109.1	112.4	115.7	118.6	120.3	104.4	106.2	109.1	112.4	115.7	118.6	120.3
	50th	91	92	93	94	95	96	96	51	51	52	53	54	55	55
	90th	103	104	105	106	107	108	108	63	64	65	65	66	67	67
	95th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
	95th + 12 mmHg	119	120	121	121	122	123	124	78	79	80	81	82	82	83
6	Height (in)	43.4	44.2	45.4	46.8	48.2	49.4	50.2	43.4	44.2	45.4	46.8	48.2	49.4	50.2
	Height (cm)	110.3	112.2	115.3	118.9	122.4	125.6	127.5	110.3	112.2	115.3	118.9	122.4	125.6	127.5
	50th	93	93	94	95	96	97	98	54	54	55	56	57	57	58
	90th	105	105	106	107	109	110	110	66	66	67	68	68	69	69
	95th	108	109	110	111	112	113	114	69	70	70	71	72	72	73
	95th + 12 mmHg	120	121	122	123	124	125	126	81	82	82	83	84	84	85
7	Height (in)	45.7	46.5	47.8	49.3	50.8	52.1	52.9	45.7	46.5	47.8	49.3	50.8	52.1	52.9
	Height (cm)	116.1	118	121.4	125.1	128.9	132.4	134.5	116.1	118	121.4	125.1	128.9	132.4	134.5
	50th	94	94	95	97	98	98	99	56	56	57	58	58	59	59
	90th	106	107	108	109	110	111	111	68	68	69	70	70	71	71
	95th	110	110	111	112	114	115	116	71	71	72	73	73	74	74
	95th + 12 mmHg	122	122	123	124	126	127	128	83	83	84	85	85	86	86
8	Height (in)	47.8	48.6	50	51.6	53.2	54.6	55.5	47.8	48.6	50	51.6	53.2	54.6	55.5
	Height (cm)	121.4	123.5	127	131	135.1	138.8	141	121.4	123.5	127	131	135.1	138.8	141
	50th	95	96	97	98	99	99	100	57	57	58	59	59	60	60
	90th	107	108	109	110	111	112	112	69	70	70	71	72	72	73
	95th	111	112	112	114	115	116	117	72	73	73	74	75	75	75
	95th + 12 mmHg	123	124	124	126	127	128	129	84	85	85	86	87	87	87
9	Height (in)	49.6	50.5	52	53.7	55.4	56.9	57.9	49.6	50.5	52	53.7	55.4	56.9	57.9
	Height (cm)	126	128.3	132.1	136.3	140.7	144.7	147.1	126	128.3	132.1	136.3	140.7	144.7	147.1
	50th	96	97	98	99	100	101	101	57	58	59	60	61	62	62
	90th	107	108	109	110	112	113	114	70	71	72	73	74	74	74
	95th	112	112	113	115	116	118	119	74	74	75	76	76	77	77
	95th + 12 mmHg	124	124	125	127	128	130	131	86	86	87	88	88	89	89

FIGURES

BP Levels for Boys by Age and Height Percentile (continued)

Age (y)	BP Percentile	SBP (mmHg)							DBP (mmHg)						
		Height Percentile or Measured Height							Height Percentile or Measured Height						
		5%	10%	25%	50%	75%	90%	95%	5%	10%	25%	50%	75%	90%	95%
10	Height (in)	51.3	52.2	53.8	55.6	57.4	59.1	60.1	51.3	52.2	53.8	55.6	57.4	59.1	60.1
	Height (cm)	130.2	132.7	136.7	141.3	145.9	150.1	152.7	130.2	132.7	136.7	141.3	145.9	150.1	152.7
	50th	97	98	99	100	101	102	103	59	60	61	62	63	63	64
	90th	108	109	111	112	113	115	116	72	73	74	74	75	75	76
	95th	112	113	114	116	118	120	121	76	76	77	77	78	78	78
	95th + 12 mmHg	124	125	126	128	130	132	133	88	88	89	89	90	90	90
11	Height (in)	53	54	55.7	57.6	59.6	61.3	62.4	53	54	55.7	57.6	59.6	61.3	62.4
	Height (cm)	134.7	137.3	141.5	146.4	151.3	155.8	158.6	134.7	137.3	141.5	146.4	151.3	155.8	158.6
	50th	99	99	101	102	103	104	106	61	61	62	63	63	63	63
	90th	110	111	112	114	116	117	118	74	74	75	75	75	76	76
	95th	114	114	116	118	120	123	124	77	78	78	78	78	78	78
	95th + 12 mmHg	126	126	128	130	132	135	136	89	90	90	90	90	90	90
12	Height (in)	55.2	56.3	58.1	60.1	62.2	64	65.2	55.2	56.3	58.1	60.1	62.2	64	65.2
	Height (cm)	140.3	143	147.5	152.7	157.9	162.6	165.5	140.3	143	147.5	152.7	157.9	162.6	165.5
	50th	101	101	102	104	106	108	109	61	62	62	62	62	63	63
	90th	113	114	115	117	119	121	122	75	75	75	75	75	76	76
	95th	116	117	118	121	124	126	128	78	78	78	78	78	79	79
	95th + 12 mmHg	128	129	130	133	136	138	140	90	90	90	90	90	91	91
13	Height (in)	57.9	59.1	61	63.1	65.2	67.1	68.3	57.9	59.1	61	63.1	65.2	67.1	68.3
	Height (cm)	147	150	154.9	160.3	165.7	170.5	173.4	147	150	154.9	160.3	165.7	170.5	173.4
	50th	103	104	105	108	110	111	112	61	60	61	62	63	64	65
	90th	115	116	118	121	124	126	126	74	74	74	75	76	77	77
	95th	119	120	122	125	128	130	131	78	78	78	78	80	81	81
	95th and 12 mmHg	131	132	134	137	140	142	143	90	90	90	90	92	93	93
14	Height (in)	60.6	61.8	63.8	65.9	68.0	69.8	70.9	60.6	61.8	63.8	65.9	68.0	69.8	70.9
	Height (cm)	153.8	156.9	162	167.5	172.7	177.4	180.1	153.8	156.9	162	167.5	172.7	177.4	180.1
	50th	105	106	109	111	112	113	113	60	60	62	64	65	66	67
	90th	119	120	123	126	127	128	129	74	74	75	77	78	79	80
	95th	123	125	127	130	132	133	134	77	78	79	81	82	83	84
	95th and 12 mmHg	135	137	139	142	144	145	146	89	90	91	93	94	95	96
15	Height (in)	62.6	63.8	65.7	67.8	69.8	71.5	72.5	62.6	63.8	65.7	67.8	69.8	71.5	72.5
	Height (cm)	159	162	166.9	172.2	177.2	181.6	184.2	159	162	166.9	172.2	177.2	181.6	184.2
	50th	108	110	112	113	114	114	114	61	62	64	65	66	67	68
	90th	123	124	126	128	129	130	130	75	76	78	79	80	81	81
	95th	127	129	131	132	134	135	135	78	79	81	83	84	85	85
	95th and 12 mmHg	139	141	143	144	146	147	147	90	91	93	95	96	97	97
16	Height (in)	63.8	64.9	66.8	68.8	70.7	72.4	73.4	63.8	64.9	66.8	68.8	70.7	72.4	73.4
	Height (cm)	162.1	165	169.6	174.6	179.5	183.8	186.4	162.1	165	169.6	174.6	179.5	183.8	186.4
	50th	111	112	114	115	115	116	116	63	64	66	67	68	69	69
	90th	126	127	128	129	131	131	132	77	78	79	80	81	82	82
	95th	130	131	133	134	135	136	137	80	81	83	84	85	86	86
	95th and 12 mmHg	142	143	145	146	147	148	149	92	93	95	96	97	98	98
17	Height (in)	64.5	65.5	67.3	69.2	71.1	72.8	73.8	64.5	65.5	67.3	69.2	71.1	72.8	73.8
	Height (cm)	163.8	166.5	170.9	175.8	180.7	184.9	187.5	163.8	166.5	170.9	175.8	180.7	184.9	187.5
	50th	114	115	116	117	117	118	118	65	66	67	68	69	70	70
	90th	128	129	130	131	132	133	134	78	79	80	81	82	82	83
	95th	132	133	134	135	137	138	138	81	82	84	85	86	86	87
	95th and 12 mmHg	144	145	146	147	149	150	150	93	94	96	97	98	98	99

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FIGURES

Figure 31. BP Levels for Girls by Age and Height Percentile

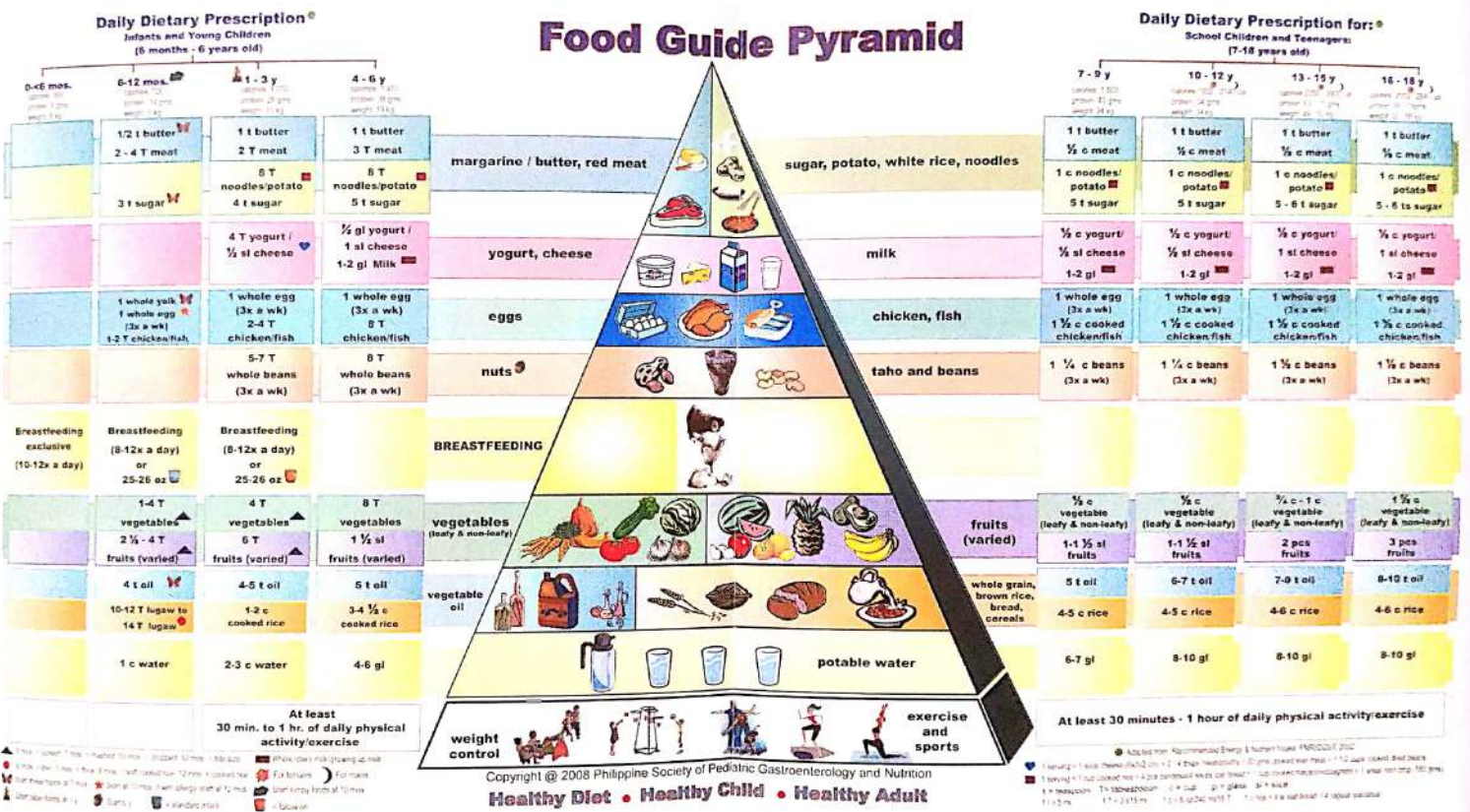
Age (y)	BP Percentile	SBP (mm Hg)							DBP (mm Hg)						
		Height Percentile or Measured Height							Height Percentile or Measured Height						
		5%	10%	25%	50%	75%	90%	95%	5%	10%	25%	50%	75%	90%	95%
1	Height (in)	29.7	30.2	30.9	31.8	32.7	33.4	33.9	29.7	30.2	30.9	31.8	32.7	33.4	33.9
	Height (cm)	75.4	76.6	78.6	80.8	83	84.9	86.1	75.4	76.6	78.6	80.8	83	84.9	86.1
	50th	84	85	86	86	87	88	88	41	42	42	43	44	45	46
	90th	98	99	99	100	101	102	102	54	55	56	56	57	58	58
	95th	101	102	102	103	104	105	105	59	59	60	60	61	62	62
	95th + 12 mm Hg	113	114	114	115	116	117	117	71	71	72	72	73	74	74
2	Height (in)	33.4	34	34.9	35.9	36.9	37.8	38.4	33.4	34	34.9	35.9	36.9	37.8	38.4
	Height (cm)	84.9	86.3	88.6	91.1	93.7	96	97.4	84.9	86.3	88.6	91.1	93.7	96	97.4
	50th	87	87	88	89	90	91	91	45	46	47	48	49	50	51
	90th	101	101	102	103	104	105	106	58	58	59	60	61	62	62
	95th	104	105	106	106	107	108	109	62	63	63	64	65	66	66
	95th + 12 mm Hg	116	117	118	118	119	120	121	74	75	75	76	77	78	78
3	Height (in)	35.8	36.4	37.3	38.4	39.6	40.6	41.2	35.8	36.4	37.3	38.4	39.6	40.6	41.2
	Height (cm)	91	92.4	94.9	97.6	100.5	103.1	104.6	91	92.4	94.9	97.6	100.5	103.1	104.6
	50th	88	89	89	90	91	92	93	48	48	49	50	51	53	53
	90th	102	103	104	104	105	106	107	60	61	61	62	63	64	65
	95th	106	106	107	108	109	110	110	64	65	65	66	67	68	69
	95th + 12 mm Hg	118	118	119	120	121	122	122	76	77	77	78	79	80	81
4	Height (in)	38.3	38.9	39.9	41.1	42.4	43.5	44.2	38.3	38.9	39.9	41.1	42.4	43.5	44.2
	Height (cm)	97.2	98.8	101.4	104.5	107.6	110.5	112.2	97.2	98.8	101.4	104.5	107.6	110.5	112.2
	50th	89	90	91	92	93	94	94	50	51	51	53	54	55	55
	90th	103	104	105	106	107	108	108	62	63	64	65	66	67	67
	95th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
	95th + 12 mm Hg	119	120	121	121	122	123	124	78	79	80	81	82	82	83
5	Height (in)	40.8	41.5	42.6	43.9	45.2	46.5	47.3	40.8	41.5	42.6	43.9	45.2	46.5	47.3
	Height (cm)	103.6	105.3	108.2	111.5	114.9	118.1	120	103.6	105.3	108.2	111.5	114.9	118.1	120
	50th	90	91	92	93	94	95	96	52	52	53	55	56	57	57
	90th	104	105	106	107	108	109	110	64	65	66	67	68	69	70
	95th	108	109	109	110	111	112	113	68	69	70	71	72	73	73
	95th + 12 mm Hg	120	121	121	122	123	124	125	80	81	82	83	84	85	85
6	Height (in)	43.3	44	45.2	46.6	48.1	49.4	50.3	43.3	44	45.2	46.6	48.1	49.4	50.3
	Height (cm)	110	111.8	114.9	118.4	122.1	125.6	127.7	110	111.8	114.9	118.4	122.1	125.6	127.7
	50th	92	92	93	94	96	97	97	54	54	55	56	57	58	59
	90th	105	106	107	108	109	110	111	67	67	68	69	70	71	71
	95th	109	109	110	111	112	113	114	70	71	72	72	73	74	74
	95th + 12 mm Hg	121	121	122	123	124	125	126	82	83	84	84	85	86	86
7	Height (in)	45.6	46.4	47.7	49.2	50.7	52.1	53	45.6	46.4	47.7	49.2	50.7	52.1	53
	Height (cm)	115.9	117.8	121.1	124.9	128.8	132.5	134.7	115.9	117.8	121.1	124.9	128.8	132.5	134.7
	50th	92	93	94	95	97	98	99	55	55	56	57	58	59	60
	90th	106	106	107	109	110	111	112	68	68	69	70	71	72	72
	95th	109	110	111	112	113	114	115	72	72	73	73	74	74	75
	95th + 12 mm Hg	121	122	123	124	125	126	127	84	84	85	85	86	86	87
8	Height (in)	47.6	48.4	49.8	51.4	53	54.5	55.5	47.6	48.4	49.8	51.4	53	54.5	55.5
	Height (cm)	121	123	126.5	130.6	134.7	138.5	140.9	121	123	126.5	130.6	134.7	138.5	140.9
	50th	93	94	95	97	98	99	100	56	56	57	59	60	61	61
	90th	107	107	108	110	111	112	113	69	70	71	72	72	73	73
	95th	110	111	112	113	115	116	117	72	73	74	74	75	75	75
	95th + 12 mm Hg	122	123	124	125	127	128	129	84	85	86	86	87	87	87
9	Height (in)	49.3	50.2	51.7	53.4	55.1	56.7	57.7	49.3	50.2	51.7	53.4	55.1	56.7	57.7
	Height (cm)	125.3	127.6	131.3	135.6	140.1	144.1	146.6	125.3	127.6	131.3	135.6	140.1	144.1	146.6
	50th	95	95	97	98	99	100	101	57	58	59	60	60	61	61
	90th	108	108	109	111	112	113	114	71	71	72	73	73	73	73
	95th	112	112	113	114	116	117	118	74	74	75	75	75	75	75
	95th + 12 mm Hg	124	124	125	126	128	129	130	86	86	87	87	87	87	87

FIGURES

BP Levels for Girls by Age and Height Percentile (Continued)

Age (y)	BP Percentile	SBP (mmHg)							DBP (mmHg)						
		Height Percentile or Measured Height							Height Percentile or Measured Height						
		5%	10%	25%	50%	75%	90%	95%	5%	10%	25%	50%	75%	90%	95%
10	Height (in)	51.1	52	53.7	55.5	57.4	59.1	60.2	51.1	52	53.7	55.5	57.4	59.1	60.2
	Height (cm)	129.7	132.2	136.3	141	145.8	150.2	152.8	129.7	132.2	136.3	141	145.8	150.2	152.8
	50th	96	97	98	99	101	102	103	58	59	59	60	61	61	62
	90th	109	110	111	112	113	115	116	72	73	73	73	73	73	73
	95th	113	114	114	116	117	119	120	75	75	76	76	76	76	76
	95th + 12 mm Hg	125	126	126	128	129	131	132	87	87	88	88	88	88	88
11	Height (in)	53.4	54.5	56.2	58.2	60.2	61.9	63	53.4	54.5	56.2	58.2	60.2	61.9	63
	Height (cm)	135.6	138.3	142.8	147.8	152.8	157.3	160	135.6	138.3	142.8	147.8	152.8	157.3	160
	50th	98	99	101	102	104	105	106	60	60	60	61	62	63	64
	90th	111	112	113	114	116	118	120	74	74	74	74	74	75	75
	95th	115	116	117	118	120	123	124	76	77	77	77	77	77	77
	95th + 12 mm Hg	127	128	129	130	132	135	136	88	89	89	89	89	89	89
12	Height (in)	56.2	57.3	59	60.9	62.8	64.5	65.5	56.2	57.3	59	60.9	62.8	64.5	65.5
	Height (cm)	142.8	145.5	149.9	154.8	159.6	163.8	166.4	142.8	145.5	149.9	154.8	159.6	163.8	166.4
	50th	102	102	104	105	107	108	108	61	61	61	62	64	65	65
	90th	114	115	116	118	120	122	122	75	75	75	75	76	76	76
	95th	118	119	120	122	124	125	126	78	78	78	78	79	79	79
	95th and 12 mm Hg	130	131	132	134	136	137	138	90	90	90	90	91	91	91
13	Height (in)	58.3	59.3	60.9	62.7	64.5	66.1	67	58.3	59.3	60.9	62.7	64.5	66.1	67
	Height (cm)	148.1	150.6	154.7	159.2	163.7	167.8	170.2	148.1	150.6	154.7	159.2	163.7	167.8	170.2
	50th	104	105	106	107	108	108	109	62	62	63	64	65	65	66
	90th	116	117	119	121	122	123	123	75	75	75	76	76	76	76
	95th	121	122	123	124	126	126	127	79	79	79	79	80	80	81
	95th + 12 mm Hg	133	134	135	136	138	138	139	91	91	91	91	92	92	93
14	Height (in)	59.3	60.2	61.8	63.5	65.2	66.8	67.7	59.3	60.2	61.8	63.5	65.2	66.8	67.7
	Height (cm)	150.6	153	156.9	161.3	165.7	169.7	172.1	150.6	153	156.9	161.3	165.7	169.7	172.1
	50th	105	106	107	108	109	109	109	63	63	64	65	66	66	66
	90th	118	118	120	122	123	123	123	76	76	76	76	77	77	77
	95th	123	123	124	125	126	127	127	80	80	80	80	81	81	82
	95th + 12 mm Hg	135	135	136	137	138	139	139	92	92	92	92	93	93	94
15	Height (in)	59.7	60.6	62.2	63.9	65.6	67.2	68.1	59.7	60.6	62.2	63.9	65.6	67.2	68.1
	Height (cm)	151.7	154	157.9	162.3	166.7	170.6	173	151.7	154	157.9	162.3	166.7	170.6	173
	50th	105	106	107	108	109	109	109	64	64	64	65	66	67	67
	90th	118	119	121	122	123	123	124	76	76	76	77	77	78	78
	95th	124	124	125	126	127	127	128	80	80	80	81	82	82	82
	95th + 12 mm Hg	136	136	137	138	139	139	140	92	92	92	93	94	94	94
16	Height (in)	59.9	60.8	62.4	64.1	65.8	67.3	68.3	59.9	60.8	62.4	64.1	65.8	67.3	68.3
	Height (cm)	152.1	154.5	158.4	162.8	167.1	171.1	173.4	152.1	154.5	158.4	162.8	167.1	171.1	173.4
	50th	106	107	108	109	109	110	110	64	64	65	66	66	67	67
	90th	119	120	122	123	124	124	124	76	76	76	77	78	78	78
	95th	124	125	125	127	127	128	128	80	80	80	81	82	82	82
	95th + 12 mm Hg	136	137	137	139	139	140	140	92	92	92	93	94	94	94
17	Height (in)	60.0	60.9	62.5	64.2	65.9	67.4	68.4	60.0	60.9	62.5	64.2	65.9	67.4	68.4
	Height (cm)	152.4	154.7	158.7	163.0	167.4	171.3	173.7	152.4	154.7	158.7	163.0	167.4	171.3	173.7
	50th	107	108	109	110	110	110	111	64	64	65	66	66	66	67
	90th	120	121	123	124	124	125	125	76	76	77	77	78	78	78
	95th	125	125	126	127	128	128	128	80	80	80	81	82	82	82
	95th + 12 mm Hg	137	137	138	139	140	140	140	92	92	92	93	94	94	94

Downloaded from:
 Flynn, J., Kaelber, D., Baker-Smith, C, et al; Subcommittee on Screening and Management of High Blood Pressure in Children.
 Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents. Pediatrics 2017;
 140(3):e20171904. DOI: 10.1542/peds.2017-1904

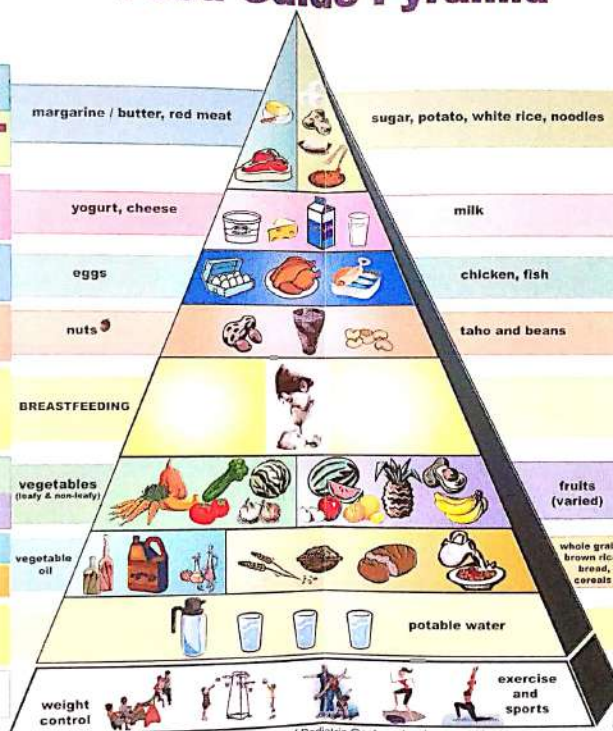


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Daily Dietary Prescription® Infants and Young Children (6 months - 6 years old)

0-6 mos.	6-12 mos.	1-3 y	4-6 y
breastfeed protein 8 gms weight 14 kg	breastfeed protein 18 gms weight 18 kg	protein 28 gms weight 22 kg	protein 38 gms weight 28 kg
	1/2 t butter 2-4 T meat	1 t butter 2 T meat	1 t butter 3 T meat
	3 t sugar	8 T noodles/potato 4 t sugar	8 T noodles/potato 5 t sugar
		4 T yogurt / 1/2 sl cheese	1/2 gl yogurt / 1 sl cheese 1-2 gl Milk
	1 whole yolk 1 whole egg (3x a wk) 1-2 T chicken/fish	1 whole egg (3x a wk) 2-4 T chicken/fish	1 whole egg (3x a wk) 8 T chicken/fish
		5-7 T whole beans (3x a wk)	8 T whole beans (3x a wk)
Breastfeeding exclusive (10-12x a day)	Breastfeeding (8-12x a day) or 25-26 oz	Breastfeeding (8-12x a day) or 25-26 oz	
	1-4 T vegetables 2 1/2 - 4 T fruits (varied)	4 T vegetables 6 T fruits (varied)	8 T vegetables 1 1/2 sl fruits (varied)
	4 t oil	4-5 t oil	5 t oil
	10-12 T sugaw to 14 T sugaw	1-2 c cooked rice	3-4 1/2 c cooked rice
	1 c water	2-3 c water	4-6 gl
			At least 30 min. to 1 hr. of daily physical activity/exercise

Food Guide Pyramid



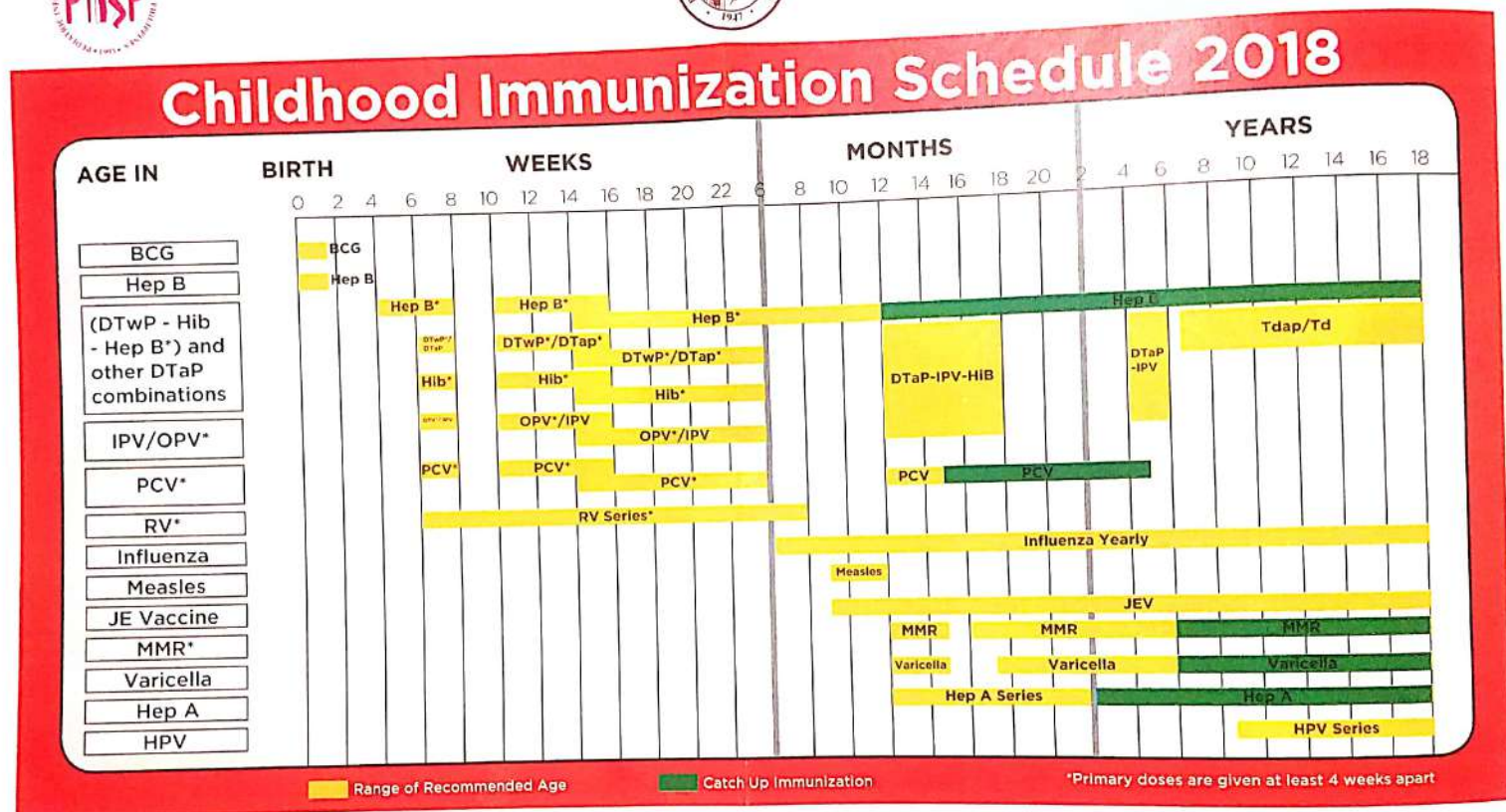
Daily Dietary Prescription for: School Children and Teenagers: (7-18 years old)

7-9 y	10-12 y	13-15 y	16-18 y
protein 18 gms weight 24 kg	protein 18 gms weight 34 kg	protein 28 gms weight 45 kg	protein 38 gms weight 55 kg
1 t butter 1/2 c meat	1 t butter 1/2 c meat	1 t butter 1/2 c meat	1 t butter 1/2 c meat
1 c noodles/ potato 5 t sugar	1 c noodles/ potato 5 t sugar	1 c noodles/ potato 5-6 t sugar	1 c noodles/ potato 5-6 ts sugar
1/2 c yogurt/ 1/2 sl cheese 1-2 gl	1/2 c yogurt/ 1/2 sl cheese 1-2 gl	1/2 c yogurt/ 1 sl cheese 1-2 gl	1/2 c yogurt/ 1 sl cheese 1-2 gl
1 whole egg (3x a wk) 1 1/2 c cooked chicken/fish	1 whole egg (3x a wk) 1 1/2 c cooked chicken/fish	1 whole egg (3x a wk) 1 1/2 c cooked chicken/fish	1 whole egg (3x a wk) 1 1/2 c cooked chicken/fish
1 1/4 c beans (3x a wk)	1 1/4 c beans (3x a wk)	1 1/4 c beans (3x a wk)	1 1/4 c beans (3x a wk)
1/2 c vegetable (leafy & non-leafy) 1-1 1/2 sl fruits	1/2 c vegetable (leafy & non-leafy) 1-1 1/2 sl fruits	1/2 c - 1 c vegetable (leafy & non-leafy) 2 pcs fruits	1 1/2 c vegetable (leafy & non-leafy) 3 pcs fruits
5 t oil	6-7 t oil	7-9 t oil	8-10 t oil
4-5 c rice	4-5 c rice	4-6 c rice	4-6 c rice
6-7 gl	8-10 gl	8-10 gl	8-10 gl
			At least 30 minutes - 1 hour of daily physical activity/exercise

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Healthy Diet • Healthy Child • Healthy Adult

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Figure 33. Immunization Table 2018



DISCLAIMER:
The Childhood Immunization Schedule presents recommendations for immunization for children and adolescents based on updated literature review, experience and premises current at the time of publication. The PPS, PIDSP and PFV acknowledge that individual circumstances may warrant a decision differing from the recommendations given here. Physicians must regularly update their knowledge about specific vaccines and their use because information about safety and efficacy of vaccines and recommendations relative to their administration continue to develop after a vaccine is licensed.

Vaccines in the Philippine National Immunization Program (NIP)

The following vaccines are in the 2018 NIP

- BCG, monovalent Hep B, Pentavalent vaccine (DTwP-Hib-HepB), bivalent OPV, IPV, PCV, MMR, MR, Td and HPV.

Recommended Vaccines

These are vaccines not included in the NIP which are recommended by the Philippines Pediatric Society (PPS), Pediatric Infectious Disease Society of the Philippines (PIDSP) and the Philippine Foundation for Vaccination (PFV)

IMMUNIZATION ANNOTATIONS

Bacille Calmette-Guérin (BCG)

- Given intradermally (ID)
- The dose of BCG is 0.05 ml for children < 12 months of age and 0.1 ml for children > 12 months of age
- Given at the earliest possible age after birth preferably within the first 2 months of life
- For healthy infants and children > 2 months who were not given BCG at birth, PPD prior to BCG vaccination is not necessary. However, PPD is recommended prior to BCG vaccination if any of the following is present:
 - Congenital TB
 - History of close contact to known or suspected infectious TB cases
 - Clinical findings suggestive of TB and/or chest x-ray suggestive of TB In the presence of any of these conditions, an induration of ≥ 5 mm is considered positive and BCG is no longer recommended.

Hepatitis B Vaccine (HBV)

- Given intramuscularly (IM)
- Administer the first dose of monovalent HBV to all newborns >2kgs within 24 hours of life.
- A 2nd dose is given 1-2 months after the birth dose
- The final dose is administered not earlier than 24 weeks of age. Another dose is needed if the last dose was given at age <24 weeks.

For infants born to HBsAg (+) mothers:

- Administer HBV and HBIG (0.5ml) within 12 hours of life. HBIG should be administered not later than 7 days of age if not immediately available.

For infants born to mothers with unknown HBsAg status:

- with birth weight >2kgs, administer HBV within 12 hours of birth and determine the mother's HBsAg as soon as possible. If HBsAg (+), administer also HBIG not later than 7 days of age.
- with birth weight <2kgs, administer HBIG in addition to HBV within 12 hours of life.

For preterm infants:

- If born to HBsAg (-) mothers and medically stable, the 1st dose of HBV may be given at 30 days of chronological age regardless of weight, and this can be counted as part of the 3-dose primary series.
- For those <2 kgs, the 1st dose received at birth is not counted as part of the vaccine series. Additional 3 HBV doses are needed.

Haemophilus influenzae Type b Conjugate Vaccine (Hib)

- Given intramuscularly (IM)
 - Given as a 3-dose primary series with a minimum age of 6 weeks and a minimum interval of 4 weeks
 - A booster dose is given between 12-15 months of age with an interval of 6 months from the 3rd dose
- Refer to Vaccines for Special Groups for Hib recommendation in high risk children*

Diphtheria and Tetanus Toxoid and Pertussis Vaccine (DTP)

- Given intramuscularly (IM)
- Given at a minimum age of 6 weeks with a minimum interval of 4 weeks
- Complete a 5-dose series at ages 2, 4, 6, 15 through 18 months, and 4 through 6 years. The recommended interval between the 3rd and 4th dose is 6 months, but a minimum interval of 4 months is valid
- The 5th dose of DTaP vaccine may not be given if the 4th dose was administered at age 4 years or older.

Inactivated Poliovirus Vaccine (IPV)

- Given intramuscularly (IM)
- Usually given in combination with DTaP and Hib, with or without Hep B
- Given at a minimum age of 6 weeks with a minimum interval of 4 weeks
- The primary series consists of 3 doses
- A booster dose should be given on or after the 4th birthday and at least 6 months from the previous dose

Rotavirus Vaccine (RV)

- Given per os (PO)
- Given at a minimum age of 6 weeks with a minimum interval of 4 weeks between doses. The last dose should be administered not later than 32 weeks of age.
- The monovalent human rotavirus vaccine (RV1) is given as a 2-dose series and the pentavalent human bovine rotavirus vaccine (RV5) is given as a 3-dose series.

Pneumococcal Conjugate Vaccines (PCV)

- Given intramuscularly (IM)
 - Given at a minimum age of 6 weeks for PCV10 and PCV 13
 - Primary vaccination consists of 3 doses with an interval of at least 4 weeks between doses plus a booster dose given 6 months after the 3rd dose.
 - Healthy children 2 to 5 years old who do not have previous PCV vaccination may be given 1 dose of PCV 13, or 2 doses of PCV 10 at least 8 weeks apart
- Refer to Vaccines for Special Groups for Pneumococcal Vaccine recommendation in high-risk children.*

Influenza Vaccine (Trivalent/Quadrivalent Influenza Vaccine)

- Trivalent influenza vaccine (TIV) given intramuscularly (IM) or subcutaneously (SC)
- Quadrivalent influenza vaccine (QIV) given intramuscularly (IM)
- Given at a minimum age of 6 months
- The dose of influenza vaccine is 0.25 ml for children 6 months to 35 months and 0.5 ml for children 36 months to 18 years

IMMUNIZATION ANNOTATIONS con't.

- Children 6 months to 8 years receiving influenza vaccine for the 1st time should receive 2 doses separated by at least 4 weeks
- If only one dose was given during the previous influenza season, give 2 doses of the vaccine then one dose yearly thereafter
- Children aged 9 to 18 years should receive one dose of the vaccine yearly
- Annual vaccination should begin in February but may be given throughout the year

Measles Vaccine

- Given subcutaneously (SC)
- Given at the age of 9 months, but may be given as early as 6 months of age in cases of outbreaks as declared by public health authorities
- If monovalent measles is not available, MMR may be given

Japanese Encephalitis Vaccine (JE)

- Given subcutaneously (SC)
- Given at a minimum age of 9 months
- Children 9 months to 17 years of age should receive one primary dose followed by a booster dose 12-24 months after the primary dose
- Individuals 18 years and older should receive a single dose only

Measles-Mumps-Rubella (MMR) Vaccine

- Given subcutaneously (SC)
- Given at a minimum age of 12 months
- 2 doses of MMR vaccine are recommended
- The 2nd dose is usually given from 4-6 years of age but may be given at an earlier age with a minimum of 4 weeks interval between doses.

Varicella Vaccine

- Given subcutaneously (SC)
- Given at a minimum age of 12 months
- 2 doses of varicella vaccine are recommended
- The 2nd dose is usually given at 4-6 years of age, but may be given earlier at an interval of 3 months from the first dose.
- If the 2nd dose was given 4 weeks from the first dose, it is considered valid.
- For children 13 years and above, the recommended minimum interval between doses is 4 weeks.

Hepatitis A Vaccine (HAV)

- Given intramuscularly (IM)
- Given at a minimum age of 12 months
- 2 doses of the vaccine are recommended
- The 2nd dose is given at least 6 months from the 1st dose

Measles-Mumps-Rubella-Varicella Vaccine (MMRV)

- Given subcutaneously (SC)
- Given at a minimum age of 12 months
- MMRV may be given as an alternative to separately administered MMR and Varicella vaccines

- The maximum age is 12 years
- The recommended minimum interval between doses is 3 months

Human Papillomavirus Vaccine (HPV)

- Given intramuscularly (IM)
- For ages 9-14 years, a 2-dose series is recommended
 - Bivalent HPV (2vHPV), quadrivalent (4vHPV) or nonavalent (9vHPV) given at 0 and 6 months
 - If the interval between the 1st and 2nd dose is less than 6 months a 3rd dose is needed. The minimum interval between the 2nd and 3rd dose is 3 months.
- For ages 15 years and older, a 3-dose series is recommended.
 - Bivalent HPV (2vHPV), quadrivalent (4vHPV) or nonavalent (9vHPV) at 0, 2 and 6 months.
 - The minimum interval between the 1st and the 2nd dose is 1 month and the minimum interval between the 2nd and 3rd dose is 3 months. The 3rd dose should be given at least 6 months from the 1st dose.

For males 9-18 years of age, a 4vHPV and 9vHPV can be given for the prevention of anogenital warts and anal cancer

Tetanus and Diphtheria Toxoid (Td)/ Tetanus and Diphtheria Toxoid and Acellular Pertussis Vaccine (Tdap)

- Given intramuscularly (IM)
- For children who are fully immunized, Td booster doses should be given every 10 years.
- For children aged > 7 years old, a single dose of Tdap can be given and can replace due Td. It can be administered regardless of the interval since the last tetanus and diphtheria toxoid containing vaccine. Subsequent doses are given as Td

For pregnant adolescents

- Fully immunized:
 - administer 1 dose of Tdap vaccine during 27 to 36 wks AOG regardless of previous Td or Tdap vaccination
- Unimmunized:
 - administer a 3 dose tetanus-diphtheria containing vaccine (Td) following a 0-1-6 month schedule. Tdap should replace one dose of Td given during 27 to 36 wks AOG

Pneumococcal Conjugate Vaccine (PCV)/ Pneumococcal Polysaccharide Vaccine (PPSV23)

- Given intramuscularly (IM)
 - All recommended PCV doses should be given prior to PPSV23 if possible. The two vaccines should not be co-administered. If a dose of PPSV23 is inadvertently given earlier than the recommended interval, the dose need not be repeated.

IMMUNIZATION ANNOTATIONS con't.

- The following tables summarize the indication and schedule of PCV/PPSV23 administration to children with high risk conditions according to age group:

SCHEDULE OF PCV 13-PPSV23 VACCINATION SEQUENCE	INDICATION
Age: 24 mos to 5 years - Administer 1 dose of PCV13 if any incomplete schedule of 3 doses of PCV13 was received previously. - Administer 2 doses of PCV13 at least 8 weeks apart if unvaccinated or any incomplete schedule of fewer than 3 doses of PCV13 was received previously. - The minimum interval between doses of PCV13 is 8 weeks. - For children with no history of PPSV23 vaccination, administer PPSV23 at least 8 weeks after the most recent dose of PCV13.	- Chronic heart disease, particularly cyanotic congenital heart disease and cardiac failure - Chronic lung disease, including asthma if treated with high-dose oral corticosteroid therapy - Diabetes mellitus - Cerebrospinal fluid leaks - Cochlear implant(s) - Sickle cell disease and other hemoglobinopathies - Congenital or acquired asplenia, or splenic dysfunction - HIV infection - Chronic renal failure and nephrotic syndrome - Diseases associated with treatment with immunosuppressive drugs or radiation therapy, including malignant neoplasms, leukemias, lymphomas, and Hodgkin disease; or solid organ transplantation - Congenital immunodeficiency (includes B-(humoral) or T-lymphocyte deficiency, complement deficiencies (particularly C1, C2, C3, and C4 deficiencies), and phagocytic disorders (excluding chronic granulomatous disease)
Age: 6 yrs to 18 years Administer 1 dose of PCV13 if they have not previously received this vaccine, regardless of whether the previous vaccine received was PCV7 or PPSV 23 Children aged 2 years to 64 years old, with any 1 of the listed chronic medical conditions should get 1 dose of PPSV23	- Chronic heart disease, including heart failure and cardiomyopathies - Chronic lung disease, including chronic obstructive pulmonary disease, emphysema, and asthma - Diabetes mellitus - Cerebrospinal fluid leaks - Cochlear implant(s) - Alcoholism - Chronic liver disease

NO. OF DOSES OF PPSV 23 ACCORDING TO INDICATION

Children aged 2 years to 64 years old, with any 1 of the listed chronic medical conditions should get 1 dose of PPSV23	- Chronic heart disease, including heart failure and cardiomyopathies - Chronic lung disease, including chronic obstructive pulmonary disease, emphysema, and asthma - Diabetes mellitus - Cerebrospinal fluid leaks - Cochlear implant(s) - Alcoholism - Chronic liver disease
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Children 2 years to 64 years old, with any 1 of the listed immunocompromising conditions should get 2 doses of PPSV23, 5 years apart

Congenital or acquired immunodeficiencies includes B- (humoral) or T-lymphocyte deficiency, complement deficiencies (particularly C1, C2, C3, and C4 deficiencies), and phagocytic disorders (excluding chronic granulomatous disease)

- HIV infection
- Chronic renal failure or nephrotic syndrome
- Leukemia or lymphoma
- Hodgkin's disease
- Generalized malignancy
- Iatrogenic immunosuppression (diseases requiring treatment with immunosuppressive drugs, including long-term systemic corticosteroids and radiation therapy)
- Solid organ transplant
- Multiple myeloma

Hemophilus influenzae Type b Conjugate Vaccine (Hib)

- Given intramuscularly (IM)
 - Indications for children with the following high risk conditions:
 - Chemotherapy recipients, anatomic functional asplenia including sickle cell disease, HIV infection, immunoglobulin or early component complement deficiency
 - Children aged 12-59 months:
 - Unimmunized* or with one Hib vaccine dose received before age 12 months, give 2 additional doses 8 weeks apart
 - With ≥ 2 Hib vaccine doses received before age 12 months, give 1 additional dose
 - For children ≤ 5 years old who received a Hib vaccine dose(s) during or within 14 days of starting chemotherapy or radiation treatment, repeat the dose(s) of Hib vaccine at least 3 months after completion of therapy
 - For children who are hematopoietic stem cell transplant recipients, revaccination with 3 doses of Hib vaccine given 4 weeks apart, starting 6-12 months after transplant, is recommended regardless of vaccination history.
 - Unimmunized* children ≥ 15 months of age and undergoing elective splenectomy should be given 1 dose of Hib-containing vaccine at least 14 days before the procedure
 - Unimmunized* children 5-18 years old and with either anatomic or functional asplenia (including sickle cell disease) or HIV infection, should be given 1 dose of Hib vaccine
- *Unimmunized children are those without a primary series and booster dose or those without at least one dose of the vaccine after 14 months of age*

*Fully immunized is defined as 5 doses of DTP or 4 doses of DTP if the 4th dose was given on or after the 4th birthday.

Meningococcal Vaccines

- Tetravalent meningococcal (ACYW-135) conjugate vaccine MCV4-D, MCV4-TT, MCV4-CRM given intramuscularly (IM)
- Indicated for those at high risk for invasive disease:
Persistent complement component deficiencies (including those with inherited or chronic deficiencies in C3, C5-9, properdin, factor D, factor H), anatomic/functional asplenia (including sickle cell disease), HIV, travelers to or resident of areas where meningococcal disease is hyperendemic or epidemic, including countries in the African meningitis belt or the Hajj, or belonging to a defined risk group during a community or institutional meningococcal outbreak

Dosing schedule:

- MCV4-D: minimum age is 9 months. For children 9-23 months give 2 doses 3 months apart. For children 2 years and above give one dose, except in cases of asplenia, HIV and persistent complement component deficiency where 2 doses, 8 weeks apart are recommended.
- MCV4-TT given to children 12 months and above as a single dose
- MCV4-CRM given to children 2 years and above as a single dose
- Revaccinate with a MCV4 vaccine every 5 years as long as the person remains at increased risk of infection
- Co-administration of MCV4 and other vaccines MCV4-D and PCV13
- If MCV4-D is administered to a child with asplenia (including sickle cell disease) or HIV infection, do not administer MCV4-D until age 2 years and at least 4 weeks after the completion of all PCV13 doses MCV4-D and Tdap
- If MCV4-D is to be administered to a child at high risk for meningococcal disease, it is recommended that MCV4-D be given either before or at the same time as DTaP. MCV4-TT with Tetanus toxoid (TT) containing vaccines
- Whenever feasible, MCV4-TT should be co-administered with TT-containing vaccines, or administer MCV4-TT 1 month before the other TT-containing vaccines

Rabies Vaccine

- Given intramuscularly (IM) or intradermally (ID)
- Recommended regimens for pre-exposure prophylaxis:
 - Intramuscular regimen (IM):
Purified Vero Cell Rabies (PVRV) 0.5 ml OR
Purified Chick Embryo Cell Vaccine (PCECV) 1 ml given on days 0, 7, 21 or 28
 - Intradermal regimen (ID): PVRV or PCECV 0.1 ml given on days 0, 7, 21 or 28
- A repeat dose should be given if the vaccine is inadvertently given subcutaneously.

- Rabies vaccine should never be given in the gluteal area since absorption is unpredictable.
- In the event of subsequent exposures, those who have completed 3 doses of pre-exposure prophylaxis regardless of the interval between exposure and last dose of the vaccine will require only booster doses given on day 0 and 3. Booster doses may be given IM (0.5 ml PVRV or 1 ml PCECV) or ID (0.1 ml of PVRV or PCECV). There is no need to give rabies immune globulin.

Typhoid Vaccine

- Given intramuscularly (IM)
- Given at a minimum age of 2 years old with revaccination every 2–3 years
- Recommended for travelers to areas where there is a risk for exposure and for outbreak situations as declared by public health authorities

Cholera Vaccine

- Given per os (PO)
- Given at a minimum age of 12 months as a 2-dose series two weeks apart.
- Recommended for outbreak situations and natural disasters as declared by public health authorities

Hepatitis A Vaccine

- Given intramuscularly (IM)
- Administer 2 doses of Hepatitis A vaccine at least 6 months apart to unvaccinated individuals who are at increased risk for infection:
- Travelers to or are working in countries with intermediate or high endemicity of infection.
- Men having sex with men (MSM)
- Users of injection and non-injection illicit drugs.
- Working with HAV infected primates or with HAV in research laboratories.
- With clotting factor disorders and chronic liver disease

Human Papillomavirus Vaccine (HPV)

- Given intramuscularly (IM)
- Give 3 doses of HPV vaccine following the 0, 1-2, and 6 month schedule, regardless of age at vaccine initiation to the following:
 - Children with history of sexual abuse or assault starting at age 9 years
 - Immunocompromised children including those with HIV infection
- HPV vaccination is not recommended during pregnancy. If HPV vaccine is inadvertently given during pregnancy, delay the remaining doses until after pregnancy. Pregnancy testing is not necessary before initiating HPV vaccination.

Dengue Vaccine

Recommendation under review, pending re-labelling of the product

**SUMMARY TABLE 2018: Immunization of
Pre-Adolescents and Adolescents (7 to 18 yrs.old)**

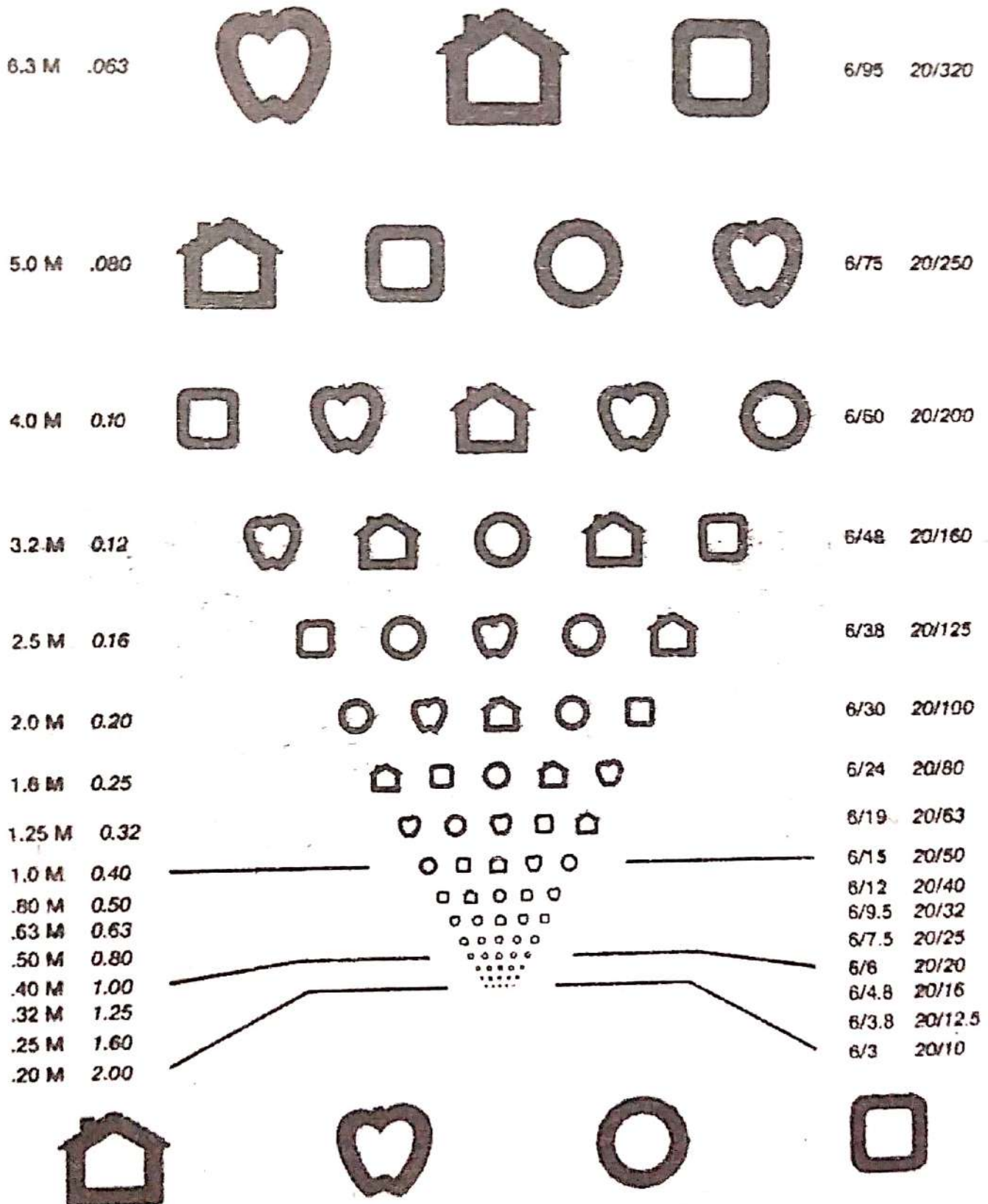
Vaccine	Range of Recommended Age	Dose(s) Needed	Schedule of Immunization	Route of Administration	Precautions & Contraindication
Hep B Vaccine	Unvaccinated 7-18 yrs. old	3	0,1,6 months	IM	- Severe allergic reaction to vaccine component - Moderate to severe illness
Hep A Vaccine	Unvaccinated 7-18 yrs. old	2	2nd dose given at least 6 months from the 1st dose	IM	- Severe allergic reaction to vaccine component - Moderate to severe illness
MMR	Unvaccinated 7-18 yrs. old	2	4 weeks interval between doses	SC	- Severe allergic reaction to vaccine component - Pregnancy - Immunosuppression - Recent receipt of blood products - Moderate to severe illness
	Incompletely vaccinated 7-18 yrs. old	1	2nd dose given anytime but at least 4 weeks from 1st dose		
Varicella	Unvaccinated 7-12 yrs. old	2	Minimum interval between doses is 3 months	SC	- Severe allergic reaction to vaccine component - Pregnancy - Immunosuppression - Recent receipt of blood products - Moderate to severe illness
	Unvaccinated ≥ 13 yrs. old	2	Minimum interval between doses is one month		
	Incompletely vaccinated 7-18 yrs. old	1	Given anytime 7-12 yrs. old at least 3 mos. from the 1st dose, 13 yrs old at least 1 month from the 1st dose		
Influenza Vaccine	9-18 yrs. old	1	Give annually beginning February	IM/SC	- Severe allergic reaction to vaccine component - Moderate to severe illness - History of Guillain-Barre syndrome following a previous dose
Td/Tdap	Unvaccinated 7-18 yrs. old	3	0,1 and 6 months Tdap preferably as the 1st dose then Td for the remaining doses	IM	- Severe allergic reaction to vaccine component - Moderate to severe illness
	Incompletely vaccinated 7-18 yrs. old	1-2	One dose Tdap then Td for remaining dose		
	Fully vaccinated 7-18 yrs. old (Fully vaccinated defined as 5 doses of DTaP or 4 doses of Tdap if the 4th dose was administered on or after the 4th birthday)	1	1 dose Tdap then Td every 10 years		
HPV: Bivalent HPV (2vHPV)	Females: 15-18 yrs. old	3	0,1,6 months	IM	- Severe allergic reaction to vaccine component - Moderate to severe illness - If found to be pregnant after starting immunization, delay remaining doses until completion of pregnancy
Quadrivalent HPV (4vHPV)/Nonavalent HPV (9vHPV)	Females: 15-18 yrs. old Males: 15-18 yrs. old	3	0,2 and 6 months		
For Females: (2vHPV)/ Quadrivalent HPV (4vHPV)/ Nonavalent HPV (9vHPV) For Males: 4vHPV/9vHPV	9-14 yrs. old	2	0,6-12 months	IM	- Severe allergic reaction to vaccine component - Moderate to severe illness - If found to be pregnant after starting immunization, delay remaining doses until completion of pregnancy

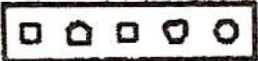
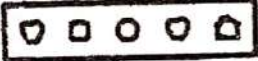
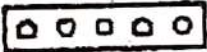
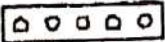
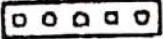
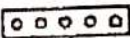
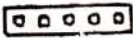
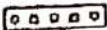
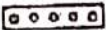
LOGARITHMIC VISUAL ACUITY CHART

LETTER
SIZE DECIMAL

FOR TESTING AT 40CM (16 INCHES)
SIZE INCREMENTS = 0.1 LOG UNITS

SNELLEN DISTANCE
EQUIVALENT
METER FOOT



LETTER SIZE	DECIMAL	<i>Mass</i> VAT® LOGARITHMIC VISUAL ACUITY CHART		SNELLEN DISTANCE EQUIVALENT	
		FOR TESTING AT 40CM (16 INCHES)		METER FOOT	
		SIZE INCREMENTS = 0.1 LOG UNITS			
1.6 M	0.25			6/24	20/80
1.25 M	0.32			6/19	20/63
1.0 M	0.40			6/15	20/50
.80 M	0.50			6/12	20/40
.63 M	0.63			6/9.5	20/32
.50 M	0.80			6/7.5	20/25
.40 M	1.00			6/6	20/20
.32 M	1.25			6/4.8	20/16

